

Transportation Infrastructure: The Challenges of Rebuilding America



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Transportation Infrastructure

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Contributors



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Petra Todorovich is the director of America 2050, a national planning initiative to develop infrastructure and growth strategy for the United States. She has convened some of the brightest thinkers in the field of transportation, land-use planning, social equity, and urban development to help create the vision for America’s future growth. That vision includes using infrastructure investments and sustainable transportation, the energy grid, and greener water infrastructure to shape quality communities, create jobs, and make America more competitive and livable in the 21st century.

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CHAPTER 1

Transportation Planning: Crisis and Opportunity

Marlon G. Boarnet



Two themes dominate current transportation discussions: one, we hope, will be transitory, but the other will define the field for a generation or more. The transportation profession, which for decades has focused on enhancing mobility, must now grapple with short-term job creation and long-term greenhouse-gas emissions reduction.

Politicians, economists, and policy analysts have suggested, sensibly so, that transportation investment can be an important part of economic stimulus packages. Never before in the interstate highway era have transportation projects been so explicitly valued for their short-term job-generating abilities. This landmark moment will likely be short-lived, however, and the best uses of today's funds will both generate jobs and start to build the transportation systems we will require for the decades ahead.

That brings us to the more long-lived challenge in transportation policy, which is not merely how to build and maintain our systems but how to ensure that we as a culture reduce greenhouse-gas emissions. Adapting to and mitigating the impacts of climate change promise to be the defining environmental challenges for the next generation, and ones that transportation will play a significant role in addressing. The pressures on our policies and institutions can be understood, in part, with some historical perspective.

Contemporary transportation planning was born with the dawn of the interstate highway system, and the institutions, financing arrangements, and modeling techniques familiar to us today were developed to address a singular project: the task of building a national highway system. That was a relatively brief era—from the late 1950s to the early 1970s—and since then transportation policy has struggled to find a similarly unifying mission and to adapt legal, regulatory, financial, institutional, and analytic frameworks shaped during the interstate era to more recent challenges. The initial, almost exclusive, focus on mobility has given way to include broader concerns about how road, rail, and sidewalk infrastructure shapes the fabric and vitality of towns and cities. In addition, environmental issues have loomed increasingly large, forcing us to assess transportation not in isolation but as part of complex social and natural systems. The role of travel continues to change, while sustainability and quality-of-life goals make it all the more important to understand travel behavior and its relationship to communities. And yet, we have not had the resources to fully support the new thinking that is required, since funding mechanisms, which are tied to gasoline taxes, have for decades failed to keep pace with increases in driving. In short, the transportation profession is rooted in an earlier time and has long been challenged to update itself.¹ Rebuilding America—and changing the ground rules for so much of our thinking about transportation—is a challenge that will require responses on many scales, using a variety of new tools and mechanisms.

This PAS Report was compiled with that context in mind—and with an eye to the urgency and severity of the challenges that we now face. We have canvassed some of the best scholars and practitioners in transportation planning, providing a forum for both best practices and visionary ideas. Much of the work included here is fresh. Without sacrificing quality, we have worked to move cutting-edge ideas into print and into the public debate. The pieces in this report include overviews from leading researchers and practitioners and expert opinion, compiled in formats that combine the goals of communicating well-documented research and moving ideas into print now, when the policy makers can make the most of this information.

A key message that emerges from this work is that planning has much to offer to transportation policy. Planners bring a unique ability to view projects within a broader context, adapting goals to the situation and navigating a complex web of actors and interests. Planners have pioneered efforts to view transportation infrastructure not just as mobility tools but as projects that must exist in concert with the community and the natural environment. They are therefore well situated to invigorate the current policy discussions across levels of government and across disciplines. Planners have much to offer to debates about reauthorization of the federal surface transportation programs and the future of transportation more generally, and we expect that the ideas in these pages will be useful to planners and policy makers at all levels of government, as well as to citizens in their engagement with community issues.



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The Challenges of Rebuilding America through Transportation Infrastructure

Transportation investment is essential. In many metropolitan areas, population growth will necessitate additional capacity. Maintenance and refurbishment needs have reached crisis proportions in some cases, as illustrated by high-profile events such as the collapse of the I-35W bridge in Minneapolis in 2007. The dollars at stake are tremendous—the U.S. Department of Transportation spends approximately 50 percent more than the Department of Education, and the Department of Transportation's budget is approximately the size of the budgets for the Departments of Housing and Urban Development, Commerce, and Interior combined—increasing the political pressure that is inherent in the transportation field.² While funding is vital, dollars alone will not move us to our goals. The chapters here articulate key areas for reform.

First, the transportation mission needs to be redefined. Since the effective completion of the national highway system almost four decades ago, entities designed to build infrastructure have had mixed success in adapting to a multifaceted world in which enhancing mobility is only one of many goals for the transportation system. The most notable step in changing the transportation paradigm was the passage of the Intermodal Surface Transportation Efficiency Act (ISTEA) in 1991. ISTEA dramatically raised the profile of metropolitan planning organizations (MPOs), strengthened requirements for conformity between transportation investment plans and federal air-quality standards, and acknowledged the need to balance funding across modes and between new investments and maintenance. The ISTEA reforms were important, but they were only a start. The key question of transportation's mission has remained unanswered, especially at the federal level. As a result, the field has become highly susceptible to political pressure, as projects are increasingly determined more by horse-trading and funding availability than by thoughtful long-term strategizing.

Because our cities' transportation infrastructures create networks of accessibility that influence growth patterns, economic activity, environmental impacts, and quality of life, the new mission for transportation must enable the full

*The Arthur Ravenel Jr.
Bridge over the Cooper River,
Charleston, South Carolina.*

integration of more holistic planning into the policy process. This will require elevating the role of metropolitan areas in passenger-transportation decision making. Having built a national highway network almost half a century ago, we now find that our passenger transportation challenges are largely confined within and defined by the context of metropolitan areas. We must rewire our institutions, funding mechanisms, and planning approaches to allow more proactive and systematic planning and policy making at the metropolitan level. There are a few examples in the United States of how this might be done.

For more than three decades, Portland, Oregon, has engaged in an ambitious coordinated program of growth management and transportation planning. While transit mode splits in Portland were, as of 2000, similar to the national average, there are some hints that over time that city's efforts might be making a difference.³ The number of vehicle miles of travel (VMT) per capita in the urbanized area of Portland was similar to the national average in the first half of the 1990s, yet from 1996 to 2007 Portland's VMT per capita fell by 7.8 percent, from 21.7 daily miles to 20.0, while the national average rose by 8.3 percent during the same period.⁴ The Rosslyn-Ballston corridor in Arlington County, Virginia (discussed by Robert Cervero in his chapter here), is another place that has had success with long-range land use-transportation planning. The point is not that every metropolitan area should closely follow those examples—far from it. But transportation needs to be more effectively immersed in planning at the metropolitan level, and federal efforts should work to empower metropolitan areas.

Federal funds should be allocated more directly to MPOs when there is a clear metropolitan purpose. For example, federal funds could be used as incentive grants, to reward states with MPOs that more effectively unify funding, planning, and implementation of passenger transport with the pursuit of long-range planning and environmental goals. Places could be incentivized by matching grants to tax themselves at the metropolitan level. Better yet, locations (ideally MPOs) that pursue coordinated strategies such as congestion tolling, parking prices, and user fees that manage and mitigate environmental impacts could be rewarded with even larger grants. More generally, the allocation of federal funds could be reformed to reflect the need to better price our system, including pricing of environmental and air-quality impacts, and to coordinate those efforts with land-use planning. Such a reform could start small, with matching funds as targeted incentives to reward forward-looking MPOs, on top of existing funding allocations that flow to states.

This implies shifting the locus of transportation policy from the federal to the metropolitan level and from infrastructure construction to coordinated pricing, investment, maintenance, and sustainability. That in turn will require changes in data collection and analytic methods. Most important, transportation needs to build a tradition of evaluation and adaptation. We rarely gather evidence on the impacts of transportation projects. Even simple ideas, like cross-checking forecasts with actual experience, are implemented rarely in research and almost never in practice. Cost-benefit analysis is almost completely absent in U.S. transportation-policy decisions. The difficulty is twofold. First, transportation has long focused on forecasting impacts, to the detriment of retrospective evaluations of projects. Second, and more seriously, there has been resistance to evaluating transportation projects, partly from a misguided belief that policy pressures make evaluation impossible or not useful.

Transportation planners will not agree on goals, measurement techniques, or tools, but certainly we can and should agree that it is important to evaluate our efforts and build a knowledge base for future action. Other highly politicized policy realms, such as health or education, have robust evaluation traditions. In those sectors, disagreements about goals or about the quality or interpretation of a particular evaluation do not lead to the conclusion that work should not be evaluated. The transportation profession's lack of

a robust tradition of retrospective evaluation is a relic from before the advent of modern data and analysis capabilities. The transportation mission, going forward, cannot be separated from efforts to consistently develop measures to track the efficiency of the system, to evaluate environmental, land use, community, and mobility impacts, and to make those data widely available both for third-party analysis and as a tool for policy decisions.

In short, our transportation policy needs to realign the federal role with the growing need to empower metropolitan areas, broadening to link to questions of sustainability that span from the local to the global, while using modern information technology to foster an open, publicly accessible, data-driven evaluation tradition.

Economic Impacts of Transportation Infrastructure

Economic impacts of transportation infrastructure can be separated into two parts: construction jobs and productivity gains from the infrastructure. Traditionally academics have looked only at productivity gains, making the assumption that transportation should be viewed not as a jobs program per se but as an infrastructure program. And although construction jobs are now a short-term policy goal, evidence on the link between infrastructure and long-term economic productivity remains important. We know that the economic impacts of modern transportation projects in developed countries are localized, and that transportation often shifts economic activity from one location to another. One place may gain jobs while another location loses them. The long-run impact is often not an aggregate increase in productivity or economic activity but a series of shifts in jobs and competitive position across the landscape.⁵

Because of their focus on land use, planners know that the economic impacts from transportation projects are land-use impacts, and planning's attention to questions about where growth occurs provides a useful framework. In this report, Reid Ewing canvasses the evidence on land-use impacts of highway projects, noting that highways typically change the location of urban growth rather than generating new growth. In his related chapter, Robert Cervero adds an important point, discussing the ways that infrastructure investment should be coordinated with land-use planning.

Implications for Transportation Policy

Because transportation shapes the economic landscape of metropolitan areas, creating patterns of economic gains and losses within those areas, planners have argued for some time that transportation governance, planning, and financing should be more explicitly metropolitan focused. Here, Robert Puentes, Petra Todorovich, and Mariia Zimmerman all advocate for a stronger metropolitan role in transportation. Puentes notes that federal transportation policy is often misaligned—something that the changing nature of our national transportation efforts illustrates. Transportation policies and institutions still reflect the idea that infrastructure investment is a national goal—that we are building a national highway system. Yet traffic congestion, environmental impacts, and even maintenance and incremental investment decisions are largely confined within metropolitan areas. The transportation challenges of New York or Chicago differ from the problems in Des Moines or Peoria. Federal policy should better foster a robust metropolitan role that can coordinate transportation funding, planning, and implementation with the local context, needs, and impacts.

While the locus of passenger movement planning should shift more to metropolitan areas, freight movement requires federal leadership that to date has been missing. The explosion in the volume of international trade over the past two decades has heightened the importance and the impacts of goods movement to and from ports. Large ports are a national resource, but the environmental impacts from goods-movement activities tend to be localized within metropolitan areas, though at times those impacts extend much farther

than the immediate area around the ports. While policy solutions should be developed with attention to local context, questions of interstate commerce and international treaties will require federal action to enable, assist, and encourage effective metropolitan policy responses. The case of the Ports of Los Angeles and Long Beach is especially instructive, and three chapters by Michael Armstrong, Thomas O'Brien, and Marlon G. Boarnet, Lindell Marsh, Chris Lunghino, and Lucilla Olmos provide an in-depth discussion of the Southern California situation. While federal action is needed in goods movement, planners' ability to draw deeply on local knowledge and context is vital, and Boarnet et al. discuss the opportunities for building partnerships that can leverage that planning perspective. The same can be said, on a larger scale, for investments in commercial waterways, which Noël P. Comeaux discusses as alternatives to congested highway and rail routes.

On all scales, infrastructure, land use, and planning must be integrated so that we can build livable, sustainable cities that respond to the environmental, economic, and social challenges of our time, while continuing to abet mobility. Ellen Greenberg describes the emerging practice of sustainable streets and illustrates ways that practice is sensitive to local context, though its widespread implementation would have larger-scale positive impacts. On a larger canvas, transportation must be central to efforts to reduce greenhouse-gas emissions. Daniel Sperling, Deborah Salon, and Nic Lutsey provide an overview of issues and policy options in that realm. The metropolitan impacts of all these considerations and more are discussed here by a panel of planning directors that convened at the American Planning Association's National Planning Conference in April 2009. That panel included San Diego planning director William R. Anderson, Minneapolis planning director Barbara L. Sporlein, and the Washington, D.C., director of planning, Harriet Tregoning, as well as Armando Carbonell of the Lincoln Institute of Land Policy and APA's director of research, William R. Klein. Last and most helpfully, Robert E. Paaswell offers a broad overview, placing transportation in historical context and suggesting reforms to our mechanisms and our ways of thinking.

Rebuilding America by Reforming Transportation

Planners have for years argued that transportation is an intimate part of our cities and towns, grounded in context. Transportation is not something to be programmed from above but instead must be planned in concert with the broader goals of communities. Now, in a time of economic and environmental challenge, transportation must respond by adapting. That will require sustained efforts to learn from practice and research, to foster innovation, and to create exemplar projects and programs. The contributions in this book represent some of the best that planning has to offer, illustrating thinking about how transportation planning can help rebuild America, now and for generations to come.

ENDNOTES

1. Robert E. Paaswell, in his chapter herein, expands on many of these ideas.
2. Budget data are from www.gpoaccess.gov/usbudget/fy10/browse.html.
3. Data for journeys to work from the 2000 census show that 5.7 percent of commute trips in Portland were by public transportation, compared to 4.7 percent nationally.
4. Data are from Portland Metro, www.oregonmetro.gov/files/maps/02b_portland-dmvt-us-avrg_1990-2007.pdf.
5. See, e.g., Marlon G. Boarnet, "Spillovers and the Locational Effects of Public Infrastructure," *Journal of Regional Science* 38(3) (1998): 381–400, and Saksith Chalermpong, "Empirical Study of Economic Spillovers of Interstate 105 in Los Angeles," *Transportation Research Record: Journal of the Transportation Research Board* 1864 (2004): 94–102.

CHAPTER 2.1**The Federal Agenda
for Rebuilding America***Petra Todorovich*

I personally believe that the choices we make today as a nation about how we invest in infrastructure, how we plan our settlements, and how aggressively we address climate change in both these areas will determine our country's future prosperity and the health of our planet.*

* This is an edited and condensed transcript of a presentation given at the 101st National Planning Conference in Minneapolis, April 26, 2009.

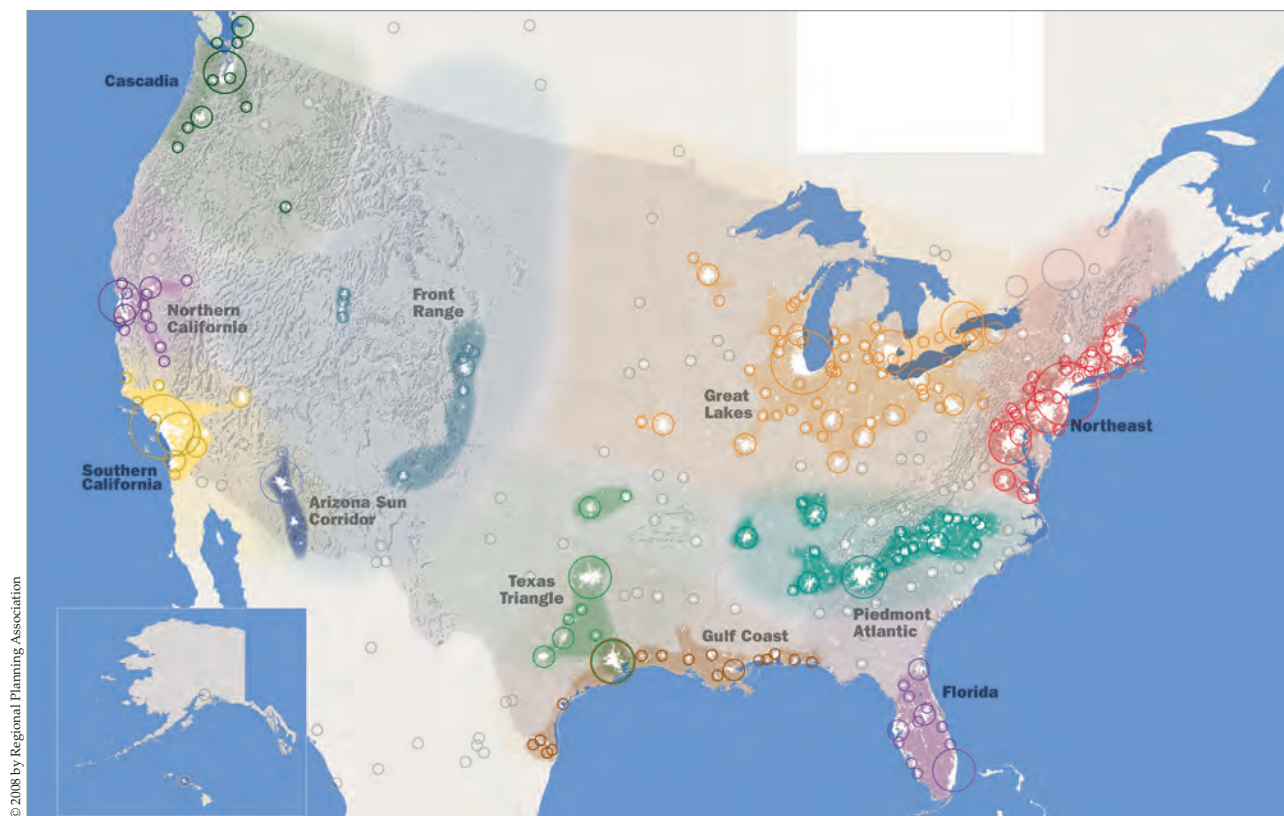
Megaregions are networks of metropolitan areas connected by overlapping commuting patterns, business travel, and economic links, and by large environmental resources, such as watersheds and large landscapes, as well as shared history and culture.

I work for the Regional Plan Association (RPA), an 80-year old civic group in the New York metropolitan region, which promotes long-range planning, infrastructure investment, environmental sustainability, and quality urban design. America 2050 is a national urban planning initiative based at RPA to develop a framework for the nation's future growth. We have advanced key concepts such as megaregions. We advocate for changes in federal policy, such as the surface transportation bill, which will begin the reauthorizing process at the end of this year. And we do some national planning, looking at population projections, where people are migrating, and what type of transportation, energy, and water infrastructure investments are needed to support sustainable urban development patterns and a competitive economy in this coming century. We get our message out in a variety of ways. We have a national committee for America 2050 of business, civic and government leaders; our website, America2050.org; and frequent research seminars, public forums, and conferences to raise awareness on these issues. This type of planning for the country has never been more urgently needed, as we face the challenge of rebuilding our economy in the short and the long terms.

RPA launched America 2050 about four years ago to respond to a handful of key challenges that we face as a nation: the projected 40 percent population growth by 2050 and demographic shifts that will make for an older and more racially and ethnically diverse nation; threats to our position in the global economy; our dangerous dependence on foreign oil and the need to cut carbon emissions 80 percent by 2050; and the growing disparities and the loss of opportunity in communities and regions that have failed to gain a foothold in the changing global economy.

A major focus of America 2050 is the changing land-use patterns in the United States and the emergence of 11 megaregions, where more than 70 percent of the population and economic growth will be focused by 2050. Megaregions are networks of metropolitan areas connected by overlapping

Figure 2.1.1. The 11 megaregions.



commuting patterns, business travel, and economic links, and by large environmental resources, such as watersheds and large landscapes, as well as shared history and culture. In the context of national planning, these are the spaces that reflect the travel patterns, business relationships, and manufacturing chains of the global economy. Our competitors in Asia and Europe are making very ambitious investments at this scale—in intercity rail and in goods movement. The trans-European network connects the European Union with continent-wide infrastructure investments. In China’s Pearl River delta, which is a megaregion the size of our Northeast megaregion, they’re investing the equivalent of \$50 billion in a high-speed rail system that they’re going to build in five years.

The building blocks of a national infrastructure plan can be built from the ground up in the megaregions, with some guidance and policy goals set by Washington, because megaregions are the places with the greatest population and economic growth pressures. And because most of them encompass multiple states or multiple regions—in the Northeast, in the Texas Triangle, in the Piedmont-Atlantic megaregion centered around Atlanta, for example—if they can come together and set priorities for infrastructure investment, then they can act as more effective partners to the federal government in advancing national-scale investments. This national infrastructure plan would include investments in passenger and goods movement, in energy infrastructure, in transmission and the smart grid, and in water infrastructure and natural-systems approaches to water-resources management.

To facilitate planning these investments, America 2050 has been holding a series of forums around the country, working with APA, Transportation for America, and other local partners. The purpose of the forums is to build support for the idea of a national infrastructure plan, an ambitious long-range framework for investments for the country’s growth, and to identify priorities for megaregion-scale collaboration and investments in energy, transportation, and water infrastructure.

In the meantime, the RPA has been convening the Business Alliance for Northeast Mobility, which includes more than 30 chambers of commerce, economic policy groups, and civic organizations, to advocate for transportation investments at the Northeast megaregion scale. For the first two years of our advocacy, we focused on the passage of the Passenger Rail Investment and Improvement Act, the five-year authorizing legislation for intercity passenger rail, which passed last October and was signed by former President Bush. We felt that the Northeast Corridor intercity rail line was a vital transportation asset and that we needed to have all of the players from Boston down to Washington speaking as one, working with our senators to advance this critical piece of legislation. Moving forward, this coalition will be planning and advocating for more comprehensive investments in transportation infrastructure in the Northeast.

National planning may seem like something that we don’t do that often in the United States. We know that local planning is the most dominant form of planning, and there is some resistance to the idea of land-use decisions coming from the federal government. Historian Robert Fishman writes that “national planning is often viewed as an exercise in bureaucratic hubris, best left to the French” (Fishman 2007). But he goes on to argue that in fact this is one of the oldest traditions in American history. In 1808, President Thomas Jefferson’s secretary of the treasury, Albert Gallatin, created a national plan of ports and roads and inland waterways to encourage settlement of our new country and to connect independent farmers to market. (See Figure 2.1.2.) The Gallatin Plan wasn’t implemented as intended—the construction of the Erie Canal shifted trade patterns—but many of these routes were used during the construction of the transcontinental railway.

Figure 2.1.2. The Gallatin Plan, created in 1808, developed a national plan of ports and roads and inland waterways to encourage settlement and connect independent farmers to market.

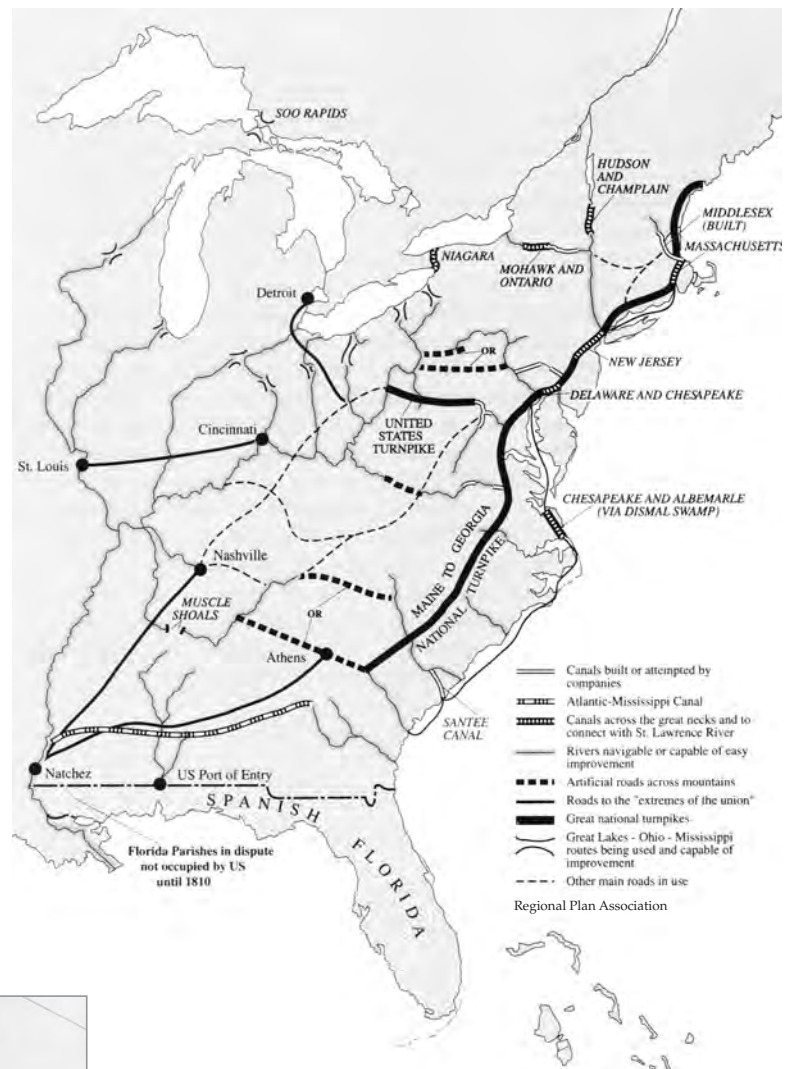


Figure 2.1.3. President Theodore Roosevelt and Forest Service chief Gifford Pinchot.



Then in 1908, President Theodore Roosevelt convened a conference of governors at the White House to launch a series of conservation and power-generation projects that were drafted by the Inland Waterways Commission, which was led by the Forest Service chief, Gifford Pinchot. These included projects such as the Roosevelt Dam in Phoenix, the Colorado Compact, and the Boulder Dam. This series of environmental restoration and economic-development/power-generation projects was aimed at developing the southern and western parts of the country; previously, most of the wealth and economic activity had been concentrated in the Northeast and the Midwest because of the railroads and railroad ownership in those areas. So this was a concerted national effort to promote economic opportunity and development nationwide.

In the 1930s, President Franklin Roosevelt created the National Resources Planning Board during the New Deal to plan infrastructure investments and to help rebuild the economy. In 1939, the NRPB, working with the Bureau of Public Roads, developed this map of toll roads and free



Figure 2.1.4. The template for the Interstate Highway System in 1956 shows existing routes tentatively selected for a proposed interregional highway system.

roads, which became the template for the Interstate Highway System, which was legislated in 1956, during the Eisenhower administration, with the Interstate Highway Defense Act. And we all know that the Interstate Highway System provided the template for growth and fueled development in the metropolitan regions over the last 50 years, with both good and negative effects for our country and for our regions.

So there is a historical basis for national planning today, as we think about the elements of a 21st-century infrastructure plan to prepare for the nation's growth, our changing demographics, our economic challenges, and the need to respond to climate change.

One such national plan is the Obama administration's strategic plan for high-speed rail, which will harness the \$8 billion appropriation in the American Recovery and Reinvestment Act and another \$5 billion proposed in the president's budget. Figure 2.1.5 shows the federal government's 10 designated high-speed-rail corridors, where states can apply



Figure 2.1.5. President Obama's strategic plan for high-speed rail.

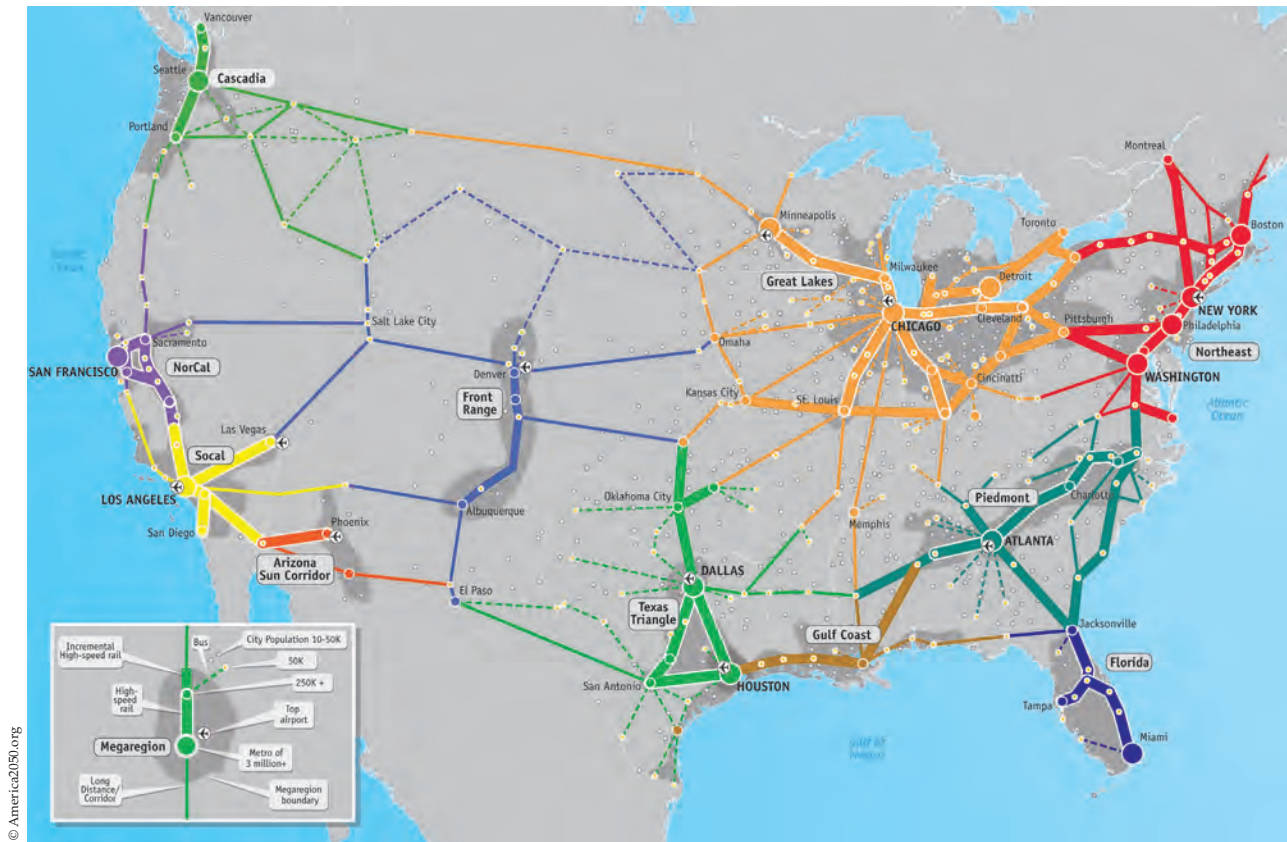


Figure 2.1.6. Proposed trans-America passenger network.

America needs a vision for transportation in this country as ambitious as the Interstate Highway System in the 1950s. But this time, it should not just be about roads.

for funding, made available through the stimulus bill and a competitive grant program. And the federal government will also be making awards to states for corridor-development projects and planning grants as part of this program. But as President Obama said, these investments are just the beginning of building a smart transportation system equal to the needs of the 21st century.

America needs a vision for transportation in this country as ambitious as the Interstate Highway System in the 1950s. But this time, it should not just be about roads. We need a trans-American network of intercity passenger rail, of goods movement, of metropolitan mobility systems to carry ourselves forward in the next 50 years and guide development of the country. Figure 2.1.6, a map of a proposed trans-American passenger network, is the result of RPA's analysis of 250 city pairs, to determine those best suited for high-speed rail and incremental high-speed-rail investment. This passenger network is built around the megaregions, where high-speed rail has the best chance to work well because these are cities that are located between 100 and 500 miles apart. They are the places with the most economic and population growth. We also looked at cities that have existing regional and local transit networks—so that when you take a high-speed train and arrive at your destination, you don't then need to rent a car but can make use of the existing urban transit options. This map shows three categories: high-speed rail, incremental high-speed rail, and existing corridors and long-distance services.

We are also developing recommendations for a trans-American freight network, which would alleviate bottlenecks on highways and rail corridors; electrify freight rail corridors; green the nation's seaports by reducing small-particulate-matter pollution, which has serious health impacts; and

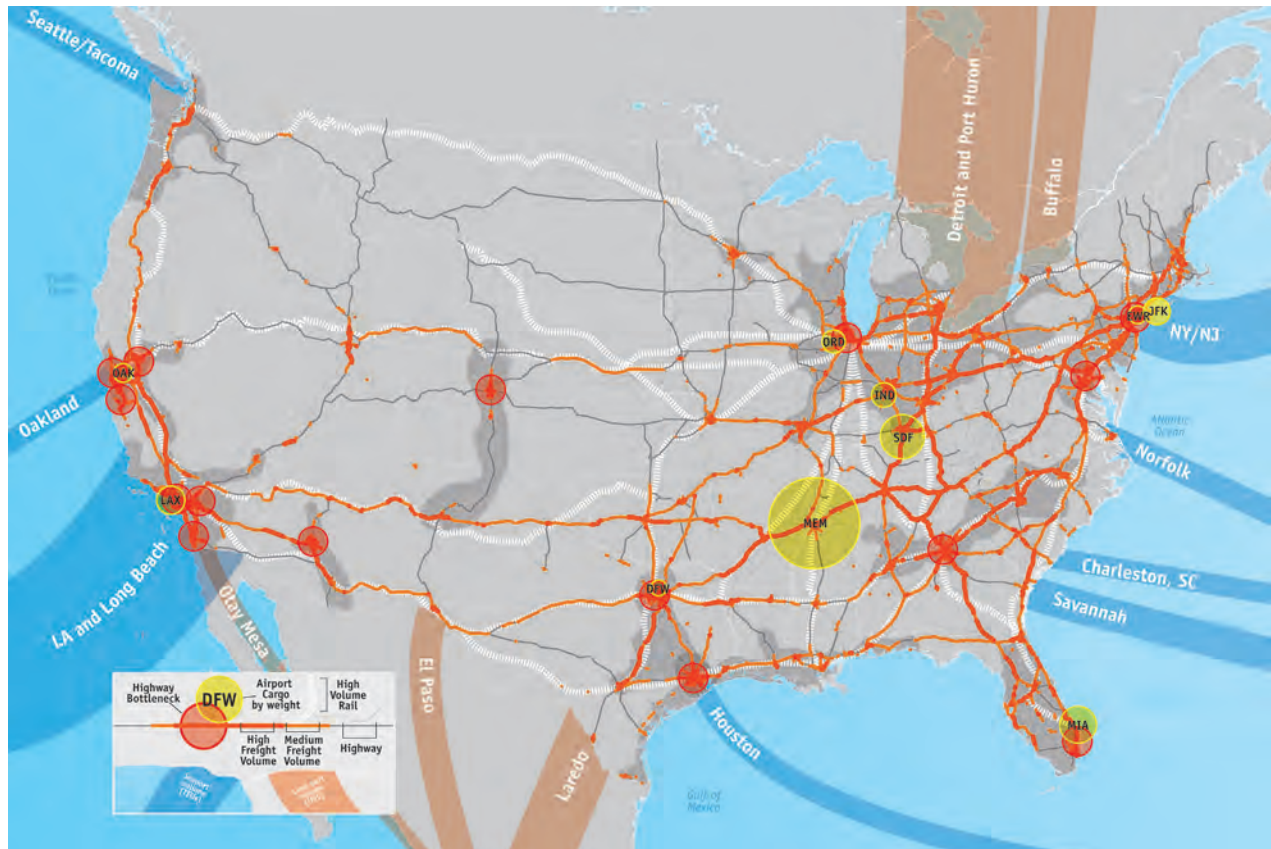
improve intermodal freight connections. Figure 2.1.7 shows goods movement flows, the major bottlenecks, and our most congested airports. And these national investments in intercity travel and freight movement are just part of the equation.

In the megaregions, the federal government should provide incentives and remove obstacles to multistate or multiregion planning and financing. We need stronger regional planning as well, since at that scale we have the best opportunities to coordinate transportation and land-use decisions. We agree with Transportation for America's proposals for a federal metropolitan blueprint program. At the local level, where the foot meets the sidewalk, transportation investments must consider the needs of all users by planning complete streets equally accessible by cars, pedestrians, bicyclists, transit, the young, the old, and so on.

Of course, transportation is just one piece of rebuilding America. We face immense challenges in rebuilding our water and energy infrastructure systems for a postcarbon economy. Questions of water supply and energy are inseparable from transportation.

For example, thermoelectric power generation accounts for 39 percent of our nation's freshwater withdrawals (U.S. Geological Survey 2000). Even in the water-rich regions like the Midwest and the Northeast, we're facing serious challenges, as the rate of urbanization is outpacing the rate of population growth and putting pressures on the quality of our water supplies. New approaches to water-resources management have implications for our transportation infrastructure, from hard engineering—gray infrastructure techniques—to the interaction of the built environment with natural systems. At the local level, porous pavement, vegetated swales, disconnected downspouts, and other practices reduce stormwater runoff and allow stormwater

Figure 2.1.7. Proposed trans-America freight network.



to filtrate through the natural environment and provide more efficient, less costly municipal sewer systems. At the regional scale—and at the megaregional scale—we need investments in land preservation, wetland restoration, and growth management to protect our natural drinking watersheds. At the federal level, we need policies that promote and finance green infrastructure solutions on a level playing field with the frequently very expensive gray infrastructure solutions, which are really the status quo of how we manage water infrastructure.

With energy, we have an enormous opportunity in the coming years to rebuild our economy in the process of phasing out our use of fossil fuels, dramatically increasing the efficiency of our built environment, transportation, and industry, and developing new renewable-energy technologies. As planners, energy efficiency should be our watchword. It is by far the most cost-effective strategy and should be thought of as a source of energy. Many cities—New York, Seattle, Chicago—are leading the way with energy-efficiency efforts by promoting greater efficiency of their buildings, greater use of public transit, and changes in their vehicle fleets.

In conclusion, I urge everyone to get involved in the important federal policy decisions that are before us in the coming years; to demand a surface-transportation bill that provides an ambitious framework and national investments that level the playing field for road building and public transit and other modes; to put a price on carbon so we can begin the transition to a low-carbon economy; and to demand water-infrastructure policies that promote efficiency, green infrastructure solutions, and natural-systems approaches.

At risk of overusing this quote, Rahm Emanuel said at the beginning of Obama's presidency, "A crisis is a terrible thing to waste." The crisis is here, and the time to act is now.

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CHAPTER 2.2

A Bridge to Somewhere: A New Transportation Program for the 21st Century

Robert Puentes



There has been an explosion of interest in federal transportation policy. While this could not come at a better time, we must recognize that our institutions, methods, and programs for transportation are in many ways a half century old and are tailored to the challenges of a bygone era. In this piece, I suggest a reform program designed to deliver transportation investments that meet 21st-century opportunities and challenges.

Pedestrians enjoying one of America's great bridges, the Golden Gate Bridge, in San Francisco.



Tyler Barton

In transportation policy, the federal government is often absent when it should be present. The federal transportation system lacks any overarching vision, goals, or guidance. It is no wonder, then, that the U.S. Government Accountability Office recently referred to transportation as a “cash transfer, general purpose grant program” (U.S. GAO 2004, p. 5).

In addition, as a program with its roots in the 1950s, the federal surface transportation program is woefully outdated. Federal transportation policy has only haltingly recognized the centrality of metropolitan areas to transportation outcomes and continues to favor roads over transit and other nonmotorized alternatives to traditional highway building. This is despite the fact that our top 100 metropolitan areas account for 65 percent of our population, 68 percent of jobs, and 75 percent of the nation’s economic output. This is in part due to their high concentrations of the nation’s key economic assets, such as infrastructure. These largest metro areas handle 72 percent of the nation’s seaport tonnage, 92 percent of air passengers, and 93 percent of rail travelers. In short, metro areas are the economic engines of the United States, powered by the clustering of people, the movement of goods, and the agglomeration of economic activity (Berube 2007).

A third major problem in federal transportation policy is that the lack of a 21st-century approach to government means the program is underperforming and failing to maximize efficiencies. Formal cost-benefit analyses are not used, and regular evaluations of outcomes are typically not conducted. The tools that are employed for tracking federal transportation spending and performance data are archaic and out of step with today’s needs. On the whole, transportation, almost uniquely among federal programs, has failed to adopt the past half century’s advances in economic (e.g., cost-benefit) analysis, evidence-based evaluation, and data that support those decision tools. In the absence of a scientific approach, the nation’s largest nonwelfare transfer program from the federal government to state governments has become detached from performance analysis and visionary goals.

Broad issues that transcend state and metropolitan areas go unaddressed; outdated policies pursued under federal law work against many state and metropolitan area efforts to maintain modern and integrated transportation networks; and underperforming grantees means the transportation program has little ability to strongly shape economic competitiveness, environmental quality, and the nation's quality of life.

A New Federal Approach

If our transportation policy is going to achieve critical national objectives in an era of fiscal constraints, it is going to need to focus and prioritize. The nation should settle for nothing less than evidence-based, values-driven decision making. This means the development of a three-pronged strategy for our national transportation program.

1. The federal government must lead in those areas where there are clear demands for national uniformity or else match the scale or geographic reach of certain problems. The United States needs to define, design, and embrace a new, unified, competitive vision for transportation policy—its purpose, its mission, its overarching rationale. The focus should be on investing in infrastructure that supports the competitiveness and environmental sustainability of the nation, rather than on funding individual states or singular needs.

Congress should authorize a permanent independent commission—the Strategic Transportation Investments Commission (STIC)—to prioritize federal investments. The STIC would develop a national priority map that would become the basis of a multiyear federally driven program prioritized on a cost-benefit basis, taking into account multimodal interactions. The identification of these important federal investments should be based on the overarching vision and goals set above. The STIC would focus on only three specific program areas of national importance: the preservation and maintenance of the interstate highway system, the development of a true national intermodal freight agenda, and a comprehensive national plan for intermetropolitan passenger travel.

The STIC should evaluate proposals for expansion of the interstates within a competitive framework, and federal funds should be directed to projects where there is a clear demonstration that they will return value for money, just as it currently is for transit projects. To ensure the efficient intermetropolitan movement of people and goods, the STIC must identify gateways and corridors of national significance. Prime candidates are the congested ports in the largest metropolitan areas and those corridors that connect large places that are fewer than 500 miles apart. These investments would be subject to cost-benefit analysis and outcome measures that go beyond traditional metrics such as number of passengers or cost effectiveness and consider energy and environment, access and social benefits, land use, and other measures.

2. The federal government should empower states and metropolitan areas to grow in competitive, inclusive, and sustainable ways. With the federal government focused on areas of national concern, there are other aspects of transportation policy where metropolitan areas should lead.

The late 20th-century model in transportation retained the standard federalism pyramid structure: with the federal government providing resources that rain down from the state to the metropolitan and ultimately the local level. But while this structure may have been appropriate for 1956, today it is without the meaningful national purpose that the interstates provided. This devolution of responsibility has produced results that are so far uneven and generally disappointing. What we need now

The realignment of responsibilities also means the federal government needs to go beyond funding and give metro areas the critical tools and flexibilities to lead.

is a new 21st-century compact that flips the pyramid and challenges our nation's state and metropolitan leaders to develop deep and innovative visions to solve the most pressing transportation problems.

This means moving to a tripartite division of labor: (a) the STIC recommending major national transportation expansions and investments; (b) the states retaining the primary role on most decision making and in small and medium-sized metropolitan areas; (c) the major metropolitan areas obtaining more direct funding and project selection authority through a new metropolitan empowerment program known as METRO. The availability of these METRO funds not only would provide financing for vital local projects but also would encourage local officials to get involved in the transportation decision making for their region.

The METRO program should be formula-driven based on population and modeled after the Community Development Block Grant program. The program would consolidate several categorical programs that would include not just the Congestion Mitigation and Air Quality (CMAQ) and Transportation Enhancements (TE) funds (which many states already suballocate directly to metropolitan areas) but also the Job Access and Reverse Commute, Transportation and Community, and System Preservation programs, as well as portions of major programs such as bridge repair.



But the realignment of responsibilities also means the federal government needs to go beyond funding and give metro areas the critical tools and flexibilities to lead. For instance, the federal government needs to embrace market mechanisms and establish a national policy for metropolitan road pricing to allow for better management of the metropolitan network.

The federal government should also pursue a strategy of “modality neutrality.” Transportation policy should enable metro areas to meet their goals on economic competitiveness, environmental sustainability, or equity by the best means available, rather than constrain them with rules governing a particular mode (e.g., highway, transit, bike/pedestrian, air).

Last, the federal government needs to assist states and metropolitan areas in developing truly integrated transportation, land use, and economic-development plans to serve projected growth over the next several decades. Sustainability challenge contracts should be awarded

to entice states and metropolitan areas to devise their own visions for coping with congestion and greenhouse-gas emissions. Such a challenge would seek to assist states and metropolitan areas in one of their hardest tasks: transcending the stovepiping of disparate programs that remains a serious cause of undesirable development outcomes. Metropolitan areas need to work over a sustained period with the goal of massively transforming the design and workings of the built environment.

Significant grant money—as much \$25 million or more for each challenge—would be awarded in a competitive process to the partnerships of states, metropolitan areas, localities, or regional business alliances that devised the boldest, most interdisciplinary proposals for improving regional development patterns or reducing carbon emissions. In exchange for some formal endorsement—by a state legislature, governor, or local metropolitan planning organization—of their commitment to adopt innovative solutions, the federal government will fund 80 percent of the development of the strategies.

3. The federal government should optimize Washington’s own performance and that of its partners to maximize metropolitan prosperity. In order to rebuild the public trust, the rationale for the federal program should be abundantly clear to the American people. A tangible set of outcomes must be explicitly tied to the federal program and funds.

While no simple analytical tool can provide all the answers, in this era of fiscal austerity the federal government should take steps to ensure that grantees apply rigorous cost-benefit analyses to any project that uses federal funds (U.S. GAO 2005, 1–49). In this way, there can be some assurances that high returns are being generated and that smaller-scale investments are properly evaluated. Yet for such analyses to be truly useful in making investment decisions, they need to be tightly coordinated with the full range of decisions that local, state, and metropolitan officials make. For one, land-use measures should be improved and incorporated into any economic analysis. They should also examine the distribution of the benefits and costs of investments across social and income groups, as well as geographic areas. Finally, these newfangled analyses need to understand the rapidly changing travel patterns and characteristics of people and goods.

Congress should then allow the U.S. Department of Transportation to maintain an incentive pool to reward states and metropolitan areas that consistently perform at an exceptional level. This includes those places that take full advantage of merit-based decision making, utilizing relevant empirical evidence and resulting in projects that generate very high returns even after accounting for the full range of environmental, social, and geographic impacts. The department should also give high performers relief from regulatory and administrative requirements in order to accelerate project delivery where appropriate.

By the same token, intervention strategies for consistent low performers should be considered. (In designating high and low performers, the DOT should take into account the difficult challenges facing state agencies and metropolitan planning organizations in large and multistate regions.) There is substantial federal precedent for such a national accountability framework in education and welfare, for example. Why recipients of federal transportation dollars should be exempt from such stewardship has yet to be fully explained. The transportation system of governance and finance shares similarities with many other areas of domestic policy and should operate under similar accountability.

Recognizing the political hurdles in linking funding to outcomes, performance, and accountability, states should be allowed to opt out of

the revamped federal transportation program. Those states would be free from most federal regulations but would also forgo their allocation of federal trust-fund revenues. They would still be required, however, to maintain and preserve their portion of the interstate highway system through whatever means they deem appropriate; failure to do so would jeopardize their opt-out status.

Another idea is to revamp existing formulas so federal funds are not distributed based on factors that potentially increase greenhouse-gas emissions, on overly simplistic equity provisions, or on the basis of earmarking. Yet in order to commit to an evidence-based program, a major overhaul is needed in how the federal government collects, assembles, and provides data and information. The federal government requires states to build and maintain the nation's roadway network, but it does not require them to provide the public with accessible, detailed information about state investment decisions about those funds. It is far easier for citizens to discern where private banks and thrifts lend (thanks to the federal Home Mortgage Disclosure Act) than to determine where public transportation agencies spend. We desperately need a sunshine law for transportation data to better inform decision making at the state and metropolitan levels and to regain the credibility of the public.

A frank and vigorous conversation about transportation finance should come only after these accountability and performance measures are put in place. To meet the challenges of the future while also ensuring financial revenues will be available, all options for reinvigorating transportation funding should be on the table:

1. First, to fund the projects of national significance identified by the STIC, the federal government should act as a guarantor of debt and create a National Infrastructure Corporation that would sell bonds to private investors who would take this interest income in the form of credits against federal income-tax liability. (There are many different ideas on how this bank or corporation would specifically function. See, e.g., Ehrlich and Rohatyn 2008.)
2. Second, to empower states and metropolitan areas, the federal fuel tax should be raised—at least to keep pace with inflation. Yet the nation should not be tethered long term to the fuel tax for transportation revenues—so the federal government should also provide stronger incentives for the adoption of market mechanisms like congestion pricing to maximize metropolitan road networks, as well as the expansion of user fees. While a tax that charges motorists for how far they drive (as opposed to how much gasoline they consume) is still years away from practical implementation in the United States, it is an idea whose time has come. (A 2020 goal for a VMT tax was recently set by a national transportation financing commission; see National Surface Transportation Infrastructure Financing Commission 2009.) The federal government can continue to study its feasibility through pilot projects.
3. Third, the federal government should also provide strong incentives for the adoption of market mechanisms like congestion pricing, true guidance on the use of public/private partnerships, and the expansion of a range of user fees.

These ideas about finance and revenue sources should not preclude a comprehensive and inclusive discussion about transportation—a discussion that includes accountability, overall intent, and connection to broader goals of economic growth and personal mobility.

CONCLUSION

If transportation policy is going to achieve critical national objectives around economic competitiveness, environmental sustainability, and social equity in an era of fiscal constraints, it will require a 21st-century transportation vision. By concentrating reforms on three major policy areas—federal leadership, empowerment of metropolitan areas, and optimization of the program—federal transportation policy can move from the anachronistic structure that exists today to something that actually works for the nation and metropolitan America. We must recognize that we are on the cusp of a new wave of transportation policy. The infrastructure challenge of President Eisenhower’s 1950s was to build out our nation and connect within. For Senator Daniel Patrick Moynihan and his colleagues in the 1980s and 1990s, it was to modernize the program and better connect roads, transit, rail, air, and other modes. Today, the challenge is to take transportation out of its box in order to ensure the health, vitality, and sustainability of our metropolitan areas.

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CHAPTER 2.3

The Transportation for America Agenda

Mariia Zimmerman



I have been involved with a group over the last year that has been working with APA and many other partners around the country to think about how we can move forward on a much broader, bolder coalition of voices to try to influence the upcoming transportation bill, climate bill, and energy bill.*

* This is an edited and condensed transcript of a presentation given at the 101st National Planning Conference in Minneapolis, April 26, 2009.

We have a partisan political system, but transportation is usually a bipartisan issue where we can bring people together to think about investments.

This new coalition is called Transportation for America, and it includes many transportation professionals, practitioners, and associations, as well as the American Public Health Association, the National Association of Realtors, and AARP. Also involved are the Transit Workers Union, the Amalgamated Transit Union, the Environmental Defense Fund, and the National Resources Defense Council, all of whom know that transportation is so fundamental to our country and so fundamental to our communities. Currently we have more than 280 partners. We have members in every state in the country, and we're getting out into communities to hold town-hall meetings, to work with partners, to try to help build support and build a vision of how we want to grow as a country and how we can start to do things differently. We need to take charge and help put forward a bold new vision that can realize the potential to meet our changing needs.

The campaign brings together practitioners and people with more of a political bent to think about how we can organize, have influence, and try to get the kind of change that we want. Part of this is recognizing that our country is at a crossroads. We know this. The American public knows this. We have some critical challenges and questions in front of us: What is the legacy we want to leave for our children and our grandchildren? How do we maintain and actually improve our global competitiveness? How can we improve our quality of life for our communities? We have got to be addressing these questions in ways that make our lives better and make our economy and our environment fit together. We have challenges with the economy. We have the volatility of fuel prices. We have energy concerns. Transportation is one of the most energy-intensive industries—the largest and fastest-growing contributor to global climate emissions. We have huge demographic challenges facing the nation, not only as we become older but as we become more diverse.

So we're thinking about how to move forward. But it is really hard to make change happen in Washington, D.C., and that's why it takes all of us working together. We have a partisan political system, but transportation is usually a bipartisan issue where we can bring people together to think about investments. There is a lot of concern about how do we move from the status quo.

In 1991, ISTEA was a transportation bill that restructured our programs and brought planning much more into the forefront, as we began to think about intermodal investments. But in the years since we have created a lot of little programs about things we care about that don't get well-treated in the overall transportation bill. And these are great programs with great purposes, and we put a little bit of money into them. But if we think about doing things in a fundamentally different way, how can we make sure that those programs and their ideals don't get lost or disadvantaged? And at a time when the economy is so tight, changing the status quo is risky. So how do we move forward? We're trying to think how this economic crisis can help us and not hinder us as we move forward.

The goals that Transportation for America is articulating for the next federal transportation bill reflect the input of hundreds and hundreds of Americans over the last several months, as we have been working in communities and doing outreach and working with a lot of good partners. It's exciting to see the nexus between a growing interest in green jobs and the need to green our economy as a whole. How does transportation fit here, not only in terms of green construction but in terms of overall operation as well? We're in a huge transit crisis right now, with more than 80 transit systems across the country potentially raising fares, cutting service, or laying people off. Part of the solution—and part of fostering sustainability in our communities through transportation—is to build a world-class 21st-century

transportation system. President Obama has expressed a commitment to a high-speed rail network, in which somebody could get off an airplane, take high-speed rail to get to their next destination, hop on a transit system and go a couple stops, get off and walk to their final destination, and have it be a seamless and easy experience.

Transportation for America has been working to put forward a broad strategy for transforming the overall federal surface-transportation bill. It focuses on four key areas of reform: improving performance and accountability; restructuring programs; improving our institutions to meet 21st-century challenges; and rethinking transportation finance. It is fundamental for the next transportation bill to clearly articulate what the federal role is in building and maintaining a national comprehensive transportation system. That means preserving the huge asset we have in the interstate system but also building up inner-city rail, freight, and passenger-rail components, building up high-quality mass transit systems in our larger metropolitan areas, improving access in our small towns and local communities, and helping to invest in vibrant bicycling and walking networks. We think it is critical to speak to transportation not as an end but as a means to an end—to see how it relates to the economy, the environment, global climate change, and social-equity issues, as well as public safety, system functionality, and public health. To achieve this, we need more than a long list of planning factors in a federal bill. It's time to move beyond mere investment studies to a truly outcome-based planning system.

We are looking at how to consolidate and elevate a core set of national priority programs (rather than 108 separate programs), how to propose a set of integrated, outcome-based transportation-option programs, and how to incentivize innovation. These core programs would speak to national objectives and outcomes, one of which should be a much more rigorous planning program. We have seen—in places such as Utah; Charlotte, North Carolina; and Sacramento, California—that having a blueprint or scenario plan or comprehensive plan with tangible outcomes can transform the planning process and the project delivery process. This can then lead to much more integrated thinking, with land use, transportation, housing, and economic development concerns all in the mix to create a 20-year vision for transportation investment. We're proposing that the next federal bill contain a metropolitan blueprint program, required for metropolitan regions with over one million people. Communities that develop a certified blueprint plan should receive direct contract authority so they can fund off of the plan and not have to go back and forth with state DOTs and others. This will allow them to tie outcome-based planning to local flexibility and regional comprehensive solutions, while improving project delivery.

We need to move from modal solutions to integrated solutions, where you may have a package of transit, highways, intelligent transportation systems (ITS), and transportation demand management that all can work together. We can't solve the problem with just a highway investment or just an ITS investment. We need to have a level playing field for highway, transit and rail, freight and maritime ferry investments as well, so that we do not have one environmental review process for a highway project but a different intensive review process for a transit project. We should be streamlining the transit investment process, but we also need to consider it fully in terms of costs, benefits, financial risk assessment, and supportive planning mechanisms. Proposals such as the national infrastructure bank offer a great opportunity to move us in that direction. We also want to incentivize innovation, because we know that there are no simple answers or silver bullets. This is a time when there are a lot of questions, and we want to have as much innovation going locally as we can to solve these issues.

We need to move from modal solutions to integrated solutions, where you may have a package of transit, highways, intelligent transportation systems (ITS), and transportation demand management that all can work together.

To think we can solve the problems of today and the future with institutions that have changed very little over the last 50 years is wrong.

To think we can solve the problems of today and the future with institutions that have changed very little over the last 50 years is wrong. As we look at opportunities to move forward, we need to consider how we can better have state, metropolitan, and local entities working together, what kinds of things we are asking from those institutions, and how we can give more authority—in terms of both project selection and project delivery—to the appropriate level of government.

Finally, when you look at the tremendous needs we have to repair and maintain our system, let alone to build out a world-class 21st-century rail/transit/highway/bicycle/pedestrian network, the reality is we need more money. We do not support just putting more money into the current system because we do not think that is going to get us the outcomes we want. But we do support trying to find innovative ways to fund the system and getting performance and outcome to drive that investment. In our current system, we start, for good political reasons, with the belief that we need a user fee to fund transportation. Currently, most of that user fee is coming from people who buy gasoline. Thus, we created a highway trust fund. We fight and fight and fight to get a little bit of that money to create a transit trust fund. And those folks who rightly say we have a lot of problems because we are not investing in freight suggest we need to create a freight trust fund. So everybody has a silo trust fund. Yet in Europe and other parts of the world, you see more unified funds for surface transportation, aviation, rail, and other modes. The question then becomes, “What is the best transportation solution?” As we look at diversifying the portfolio for how we fund transportation, we’ve got to figure out the finance solution. Reforming how we finance transportation is vital in the near term, as the federal highway trust-fund balance has fallen below obligations, and in the long term, as part of more substantive reform. Now is the time to propose a unified trust fund that could invest in the best and most integrated transportation solutions.

So how do we move forward? The federal stimulus funding provides some opportunities. We can make the case for the value created by those investments, but there are also some problems with what the stimulus funds can and cannot be spent on and how it is moving through the system. For instance, the majority of transit money provided cannot be spent to provide short-term operating assistance that could preserve transit jobs and service. In many states, metropolitan areas failed to receive a proportional amount of the funding.

We need to work together to help get our message to Congress, and so we need to be part of the conversation and push to have a new vision and a new program structure. Please work with us. Become engaged with us. Share your good ideas. You’re in communities doing this, so you know what works and what doesn’t work, and we want to have something that can be as practical, but transformative, as possible. I think together we need to recognize that the sum really does need to be greater than the parts. So while we all have our own little programs or pieces we care about, the real issue is how we can advocate for that piece within the larger whole.

CHAPTER 3.1

**Highway-Induced Development:
What Research in
Metropolitan Areas Tells Us***Reid Ewing*

In contrast to the literature on induced traffic, the literature on induced development has not been reviewed comprehensively.* This review finds that in the postinterstate era, major highway investments have small net effects on the growth and development of metropolitan areas, instead mostly moving development around the region to take advantage of improved accessibility. Induced development is very close to a zero-sum game. Highway investment patterns tend to favor suburbs over central cities and thereby contribute to decentralization and low-density development. Corridors receiving major highway investments experience land appreciation and therefore are likely to be developed at higher densities than developable lands outside the corridor. Highways may be necessary, but they are not sufficient, to induce development. To the extent that current planning and zoning caps hold, impacts within the corridor will be moderated.

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Introduction

Interstate 270 angling to the northwest from the Washington, D.C., Beltway was widened in the late 1980s and early 1990s. In 1999, the *Washington Post* ran a story comparing actual traffic volumes on I-270 to projections before construction.¹ The article declared the widening a failure based on the amount of induced travel, which effectively used up the added capacity. By 2000, traffic volume for certain sections of I-270 already exceeded forecasts for 2010.

This was a time of growing interest in the phenomena of induced travel and induced development. The Maryland-National Capital Park and Planning Commission and Metropolitan Washington Council of Governments (MWCOCG) responded with a 2001 study that suggested that highway-induced development was mainly responsible for the high and premature levels of congestion on I-270. Also blamed was the failure to build all transportation facilities in the adopted regional transportation plan of the time. Some projects had been delayed and others dropped.

On the subject of induced development, MWCOCG concluded that “higher observed traffic volumes relative to the 1984 forecast appear to be due in large part to shifts in population, employment, and travel to the I-270 corridor from other areas in the region, rather than to entirely new travel.” Population and employment in the I-270 corridor were, respectively, 23 and 45 percent higher than forecasted in 1984. For all of Montgomery County, they were 7 and 21 percent higher than forecasted. Meanwhile, population and employment were 9 and 23 percent lower than forecasted in Prince George’s County, and 29 and 3 percent lower than forecasted in the District of Columbia. These shifts in development are illustrated in Figures 3.1.1 and 3.1.2.

For the region as a whole, population growth was 5 percent lower than had been forecasted in 1984, while employment growth was 9 percent higher.

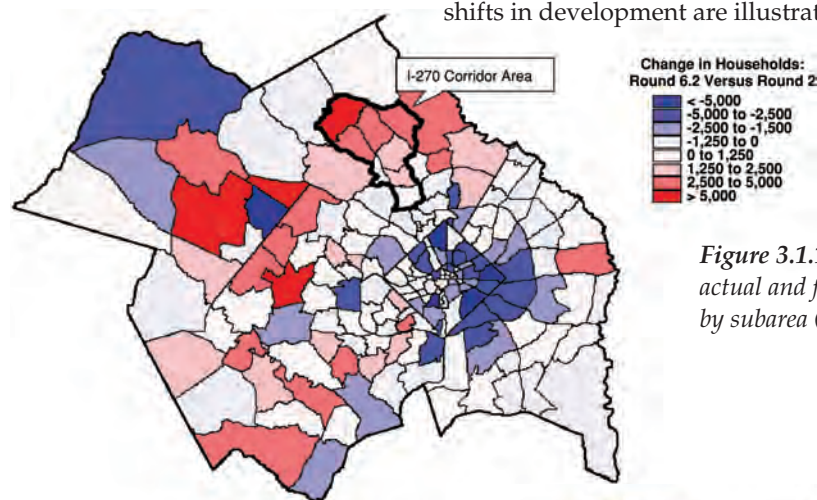


Figure 3.1.1. Difference between actual and forecasted households by subarea (2000).

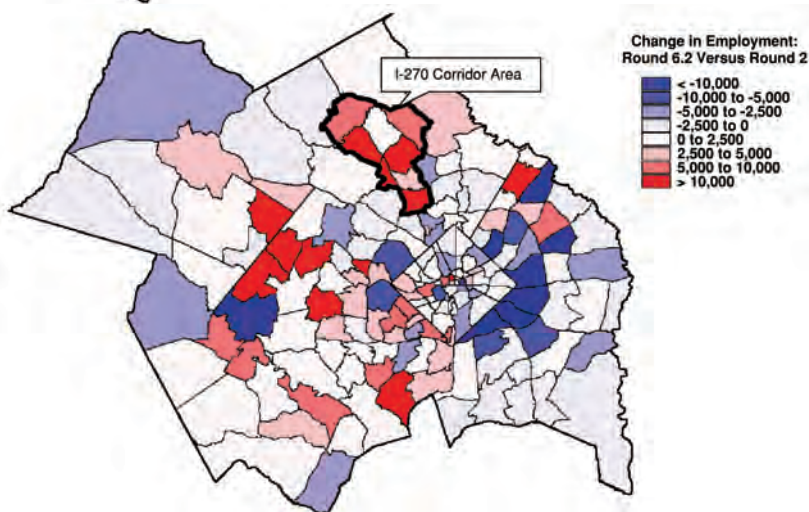


Figure 3.1.2. Difference between actual and forecasted employment by subarea (2000).

For the region as a whole, population growth was 5 percent lower than had been forecasted in 1984, while employment growth was 9 percent higher. To MWCOC, the two facts together suggested small (if any) net impacts of I-270 on regional growth.

It turns out the experience with the I-270 widening mirrors the literature on highway-induced development. There may be major shifts in the location of development within a region but probably not much impact on regional totals. In contrast to the literature on induced traffic, which has been reviewed extensively, the literature on induced development has not been reviewed comprehensively.^{2,3,4,5} This review seeks to fill the gap in the literature.

Induced Development in Relation to Induced Traffic

Induced traffic and induced development are related. One can think of induced development as a cause of induced traffic, not immediate but longer term. To better understand induced traffic and its connection to induced development, it is necessary to explore the behavioral consequences of additions to roadway infrastructure capacity.

In the short run, a variety of behavioral changes can contribute to increased traffic without any induced development. These include route switches, mode switches, and changes in destination. In addition, there is the possibility of new trips that would not have occurred without the addition in infrastructure capacity.

In the longer run, increases in highway capacity may lower travel times so that residences and businesses are drawn to locate in the area near the expanded highway capacity. The question is always whether the new development that occurs in proximity to a highway was induced to locate there as a consequence of the expansion or whether it would have occurred anyway, regardless of the highway. Indeed, the highway investment may be a response to new or anticipated development, rather than vice versa. If the development itself would not have occurred otherwise, the traffic it generates and the development can be considered induced.

Definitionally, a gray area exists if the development that occurs near a highway would have occurred somewhere else in the region in the absence of the investment. Some would call this induced development, others simply redistributed development. We use the term “induced development” liberally to mean any development that would not have occurred at a given location without a highway investment.

Complexity of Induced Development

Induced development is a complex phenomenon, which helps account for the divergent viewpoints regarding its magnitude and even its existence. One reason for the complexity is that highway infrastructure has both spatial and economic properties. On the one hand, highways allow movement, communication, and market exchange. Highway investments may affect household location decisions by lowering travel costs. Household location decisions may, in turn, influence the location decisions of firms and industries, and vice versa.

On the other hand, highways are an input into the production of private goods and services. Lower transportation costs may boost private sector productivity and output generally across the region. If so, highways will affect the economic landscape in ways that no simple location model can capture.

Because of these two distinct properties, the interjurisdictional effects of highway investments may be either positive or negative. Development and other economic activity may flow from jurisdictions without highway investments to jurisdictions with highway investments. Alternatively, posi-

tive spillovers from jurisdictions with highway investments may cause economic growth in neighboring jurisdictions without highway investments. Metaphorically, highway investments may create a rising tide that lifts all boats, or it may lift some as it capsizes others. Which effect dominates is an empirical question—hence the need to review the empirical literature.

Historical Changes in Induced Development

Clearly, impacts of highway investments are less today than they once were. Construction of the Interstate Highway System, in particular, has tied virtually every place in the country to everywhere else. Most studies finding sizeable highway impacts date back to the first round of interstate highway construction, which created huge positive externalities for areas gaining access to the network.⁶⁷⁸ By the early 1970s, the Interstate Highway System was largely complete. Incremental additions or improvements to the network have since produced comparatively small improvements in interregional accessibility.

So, in the postinterstate era, how great are highway impacts on economic and land development? This is a subject of great debate. In a well-known point-counterpoint, Giuliano minimized the importance of highway investments for three reasons: “The transportation system in most U.S. metropolitan areas is highly developed, and therefore the relative impact of even major investments will be minor. The built environment has a very long life. . . . Even in rapidly growing metropolitan areas, the vast proportion of buildings that will exist 10 to 20 years from now are already built. . . . Transport costs make up a relatively small proportion of household expenditures.”⁹

Cervero and Landis countered that “although new transportation investments no longer shape urban form by themselves, they still play an important role in channeling growth and determining the spatial extent of metropolitan regions by acting in combination with policies such as supportive zoning and government-assisted land assembly.”¹⁰ They then challenged Giuliano’s empirical evidence and presented evidence of their own.

Who is right? Giuliano probably is right about aggregate impacts, while Cervero and Landis probably are right about localized impacts. The following sections provide a sampling of research on highway impacts. The sample is by no means exhaustive but is representative of the literature in the two subject areas: aggregate impacts of highways and localized impacts of highways. Other relevant studies (not reviewed herein) are included in the references.

Aggregate (Net) Impacts of Highways

Numerous studies have analyzed the net impact of transportation infrastructure investment and economic growth.^{11 12} Such studies typically estimate production functions, with economic output explained in terms of labor, capital, and other factors of production. Capital inputs include public capital, public capital includes transportation infrastructure, and transportation infrastructure includes highways. Aggregate impacts of highways on economic growth have been estimated for the nation, states, metropolitan areas, and even counties.

Estimated effects of infrastructure investment on economic growth have varied widely among studies. Early studies at the national level showed large effects. Early studies at the state level showed smaller but still significant effects. Effects at the metropolitan or county level were smaller still, presumably because some of the economic advantage of infrastructure investment is lost to smaller areas through positive spillovers to their neighbors.

Later, more rigorous studies (with fully specified production functions and lagged input variables) suggested that public infrastructure plays a more limited role in economic growth, or perhaps no role at all. Here we quote Boarnet: “Once the appropriate econometric techniques are used, even the time series evidence usually shows no statistically significant productivity effect from public capital. Later cross-state studies verify this result. When controlling for unique state and year effects, which in general is the preferred approach, recent studies find no statistically significant effect from public infrastructure ... results are the same when only highway capital is used as an independent variable.”¹¹

As an example, Garcia-Mila et al. modeled gross state product in terms of total employment, total private capital stock, and total public capital stock broken down into three categories—highways, water and sewer systems, and other.¹³ Without controlling for unique state effects, highway capital stock had a large, positive, and significant effect on gross state product. After accounting for state effects, the effect of highway capital stock was small (but still significant).

To date, no compelling studies have been conducted at the metropolitan level. Most importantly, no studies have controlled for unique metropolitan-area effects. However, since effect sizes vary directly with the size of area, it is safe to assume that highway investments in metropolitan areas have only marginal effects on gross regional output. This is not to say that highway investments have no role in the private economy, but rather that in the postinterstate era, there is ample infrastructure already in place and marginal effects on output are likely to be small.

Localized Impacts of Highways

Even if the net economic impacts of highway investments are small, highways may have important impacts on the geographic pattern of development within a region: “it is quite possible that infrastructure investments made by states are indeed productive assets, but that their value is reflected in the fact that economic actors will change locations within, but not across, states to utilize new public works.”¹⁴

In this section, we review evidence of highway impacts at the relatively fine geographic scale of the county or smaller units. Studies reviewed either looked at direct effects on development, growth, or output, or indirect effects through property values.

Lichter and Fuguitt. Lichter and Fuguitt analyzed patterns of growth for nonmetropolitan counties between 1960 and 1970.¹⁵ Some counties were exurban just outside metropolitan boundaries, others more remote. Being on an interstate had both direct and indirect effects on population growth, the indirect effects occurring through growth of employment in export services and tourism industries. From 50 to 80 percent of the total effect was indirect, through employment. An interstate highway’s effect was greatest for nonmetropolitan counties near metropolitan cities. Overall, though an interstate’s effect was small, the “consistency of evidence provided on the relationship between date of completion of the interstate highway and various demographic measures was striking.”

Payne-Maxie Consultants. Payne-Maxie Consultants examined the influence of suburban beltways on the growth of suburbs and central cities in 54 U.S. metropolitan areas.¹⁶ They concluded that beltways have little impact on overall growth of metropolitan areas, but they also concluded that beltways may shift growth from one place to another within a metropolitan area. The magnitude of land-use impacts were deemed dependent on (1) overall local economic conditions; (2) proximity to medium- or high-income residential areas; (3) availability of land to develop; and (4) favorable local zoning policies.

Stephanedes and Eagle. Stephanedes and Eagle studied the interaction between highways and employment for 30 counties in Minnesota over a 25-year period.¹⁷ They found positive impacts of highway expenditures on local employment (beyond the regional trend) only in the Minneapolis–St. Paul metropolitan area. This was attributed to the greater capacity of the metropolitan area to absorb growth, compared to other areas within the state. Positive impacts of highway expenditures in certain counties were offset by negative impacts in adjacent counties.

Carlino and Mills. Carlino and Mills analyzed development densities circa 1980 for 3,000 U.S. counties in terms of economic, demographic, and policy variables and regional dummy variables to capture climatic effects.¹⁸ Population and employment densities were modeled jointly using structural equations to reflect the interaction between them. The density of interstate-highway mileage within the county was among the significant variables in both models. Elasticities suggested that a doubling of interstate-highway mileage would result in approximately 2.8 and 6.1 percent increases in population and employment densities, respectively. Interstate-highway access was not a dominant influence, nor was it inconsequential.

Rephann and Isserman. Rephann and Isserman isolated impacts of interstate highways by comparing highway-impacted counties to control-group counties.¹⁹ They compared output and income series before and after highway construction. Their data covered 334 U.S. counties from 1959 through 1984. Counties were classified as competitive (home to small cities that got interstate mileage during the period), urban spillover (near large cities with interstates but without their own interstate mileage), uncompetitive (predominantly rural with interstate mileage), and adjacent (close to highway-impacted counties but off the interstate). Results showed the strongest impacts in urban spillover counties, followed by competitive counties. Significant negative impacts were found in adjacent counties, possibly reflecting a flow of resources out of the adjacent county toward the new highway investment.

Boarnet. Boarnet estimated a production function using panel data for California counties from 1969 through 1988.²⁰ County gross product was modeled in terms of labor inputs, private capital inputs, and highway capital stock of the county itself and its neighbors. The study used lagged variables to test the premise that causality runs from changes in the highway stock to changes in output, not vice versa. Results showed that output of a county was larger when a county's own highway stock was larger than its neighbors, but county output was smaller when the highway stock in neighboring counties was larger. This suggested not only that highway stock increased output at the county level but also that the output gains were at the expense of neighboring counties.

Hansen et al. Hansen and colleagues employed econometric techniques to study land-use impacts of highway expansion in several corridors in California's four largest urban areas.²¹ Land development was measured in terms of construction permits. Highway expansion had the effect of increasing the number of single-family housing permits in the corridor relative to the level in the region. The results for multifamily housing permits were similar. Highway expansion was found to have an immediate positive impact on commercial but not on industrial development. The effect on commercial development diminished over time. The authors concluded that "highway capacity expansion stimulates development activity, both residential and non-residential, in the corridors served by the expanded facilities."

Voith. Voith examined the geographic distribution of transportation investments in the Philadelphia metropolitan area, focusing on differences between the city and surrounding suburbs.²² He presented county-level estimates of total, per capita, and per user highway investments over the 10 years from 1986 to 1995, as well as fees generated by highway users. The central finding: highway capital expenditures were about 2.5 times higher on a per capita basis in the suburbs than in the city. Even when transit investments were included in the analysis, per capita transportation investments benefiting suburban residents exceeded those benefiting city residents by 47 percent. Voith's conclusion: highway investments have provided an economically significant, although not overwhelming, incentive for suburban population and employment growth at the expense of the city. His "reasonable guess" as to the impact of differential highway investments was a net loss of 40,000 jobs for the city.

Haughwout. Haughwout related county employment growth to state highway investments, other state infrastructure investments, county employment density, and other economic variables for the period 1974–1992. (See note 14.) Results suggested that state infrastructure investments redistributed growth from dense to less dense metropolitan counties. His explanation: by reducing the cost of travel, state highway investments undermined the spatial advantage of dense urban counties. The redistributive effect was limited to metropolitan areas; state highway investments did not appear to spur employment growth in nonmetropolitan counties. Because dense areas tend to be more productive, the net effect of state infrastructure investments could be to reduce overall employment growth.

Boarnet and Chalermpong. Boarnet and Chalermpong conducted a before-and-after study of the impact of new toll roads on single-family detached house prices in Orange County, California.²³ The study used hedonic price analysis and multiple sales price analysis to examine how the opening of the toll roads altered house prices in nearby areas between 1988 and 2000. Highway access was measured by distance to the nearest toll road on-ramp. The evidence suggested rather strongly that changes in accessibility were reflected in home sales prices. The evidence was particularly strong for the Foothill Transportation Corridor (see Figure 3.1.3). Thus it was "reasonable to conclude that new highways will also create changes in development patterns," as higher prices lead to higher densities of development.

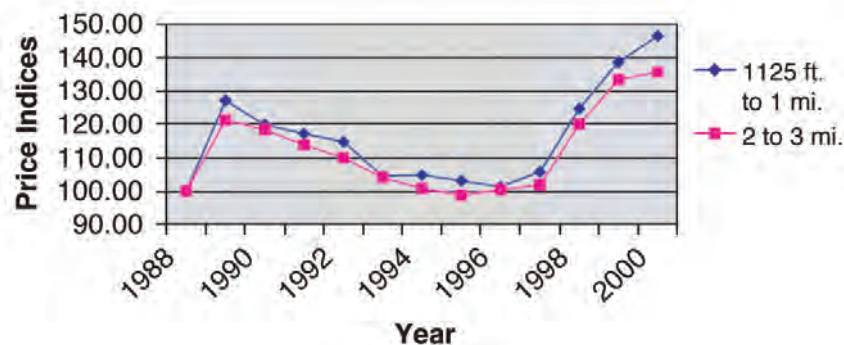


Figure 3.1.3. House price indices for the Foothill Transportation Corridor (as function of distance from highway beyond 1,125 feet).

Kockelman et al. First, Kockelman and associates analyzed nine years of building permit data for planning areas in Austin, Texas.²⁴ The results suggested that highway expansions had no impact on development activity. However, the authors acknowledged that more spatially disaggregate or larger data sets might have led to different results. Then, they studied 17 years of tax assessment records for parcels along an improved highway. They found significant inflation in land values in response to right-of-way

acquisition, subsequent declines from their speculative levels, and then gradual rises as development occurred (see Figures 3.1.4 and 3.1.5). The subsequent statistical analysis confirmed that the year of land acquisition was significant in land-value adjustments and that the value of land on corners and land with frontage on major facilities were much higher than the value elsewhere.

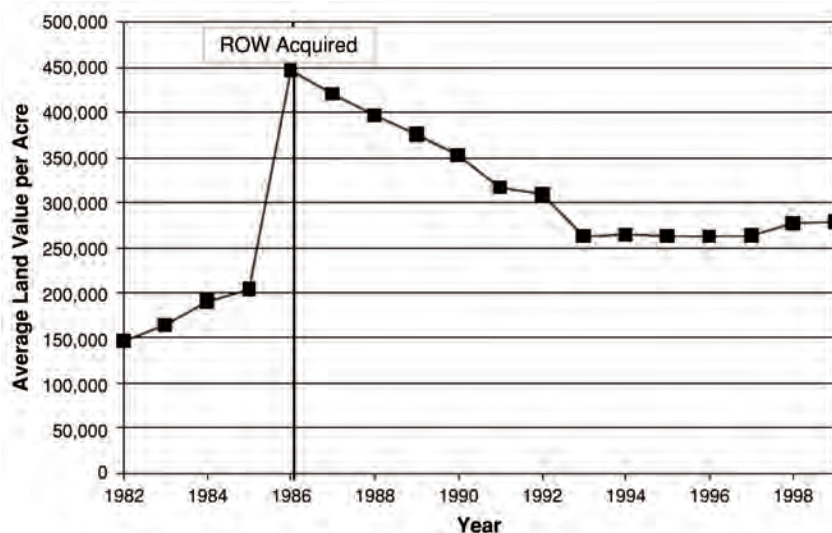
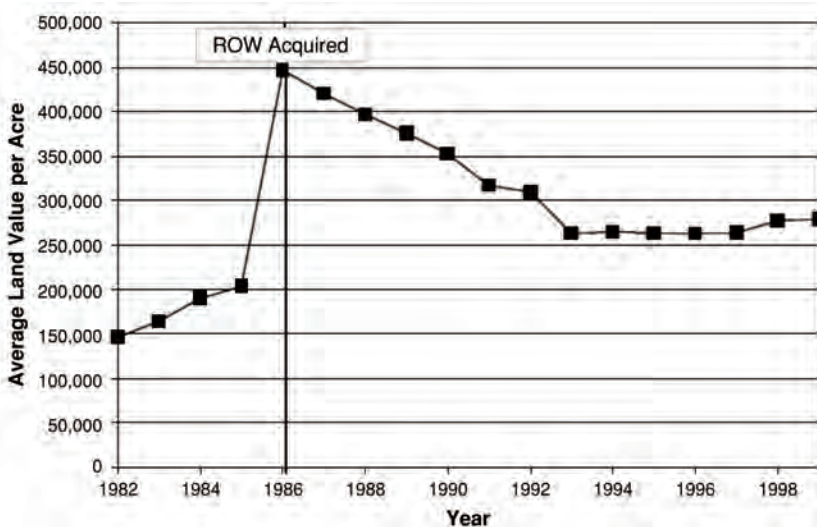


Figure 3.1.4. Average assessed values before and after right-of-way (ROW) acquisition, by average land value per acre.

Figure 3.1.5. Average assessed values before and after right-of-way acquisition, by average improvement value per square foot (SF).



Bollinger and Ihlanfeldt. Bollinger and Ihlanfeldt used data from the Atlanta region between 1985 and 1997 to estimate the change in a census tract's share of regional employment as a function of tax-incentive programs, highway and transit infrastructure investments, and crime.²⁵ The results showed that tracts containing rail-station areas lost employment share, but this effect was never statistically significant. In contrast, highway improvements within the tract caused employment shares to grow by about 0.00006 in the year following the completion of the improvements, which translated into a job gain of about 70 jobs. These effects were stronger than those commonly reported in the literature, perhaps because this was the first study to use panel data at the neighborhood level. "As the geographical area over which the firm chooses its location gets smaller, alternative locations become closer substitutes. Hence, policies [or highway investments] may have stronger effect within a metropolitan area than they have within a state or broader region."

Cervero. Using data for 24 California freeway projects across 15 years, Cervero explained traffic increases in terms of both faster travel speeds and land-use shifts that occurred in response to adding freeway lanes.²⁶ Figure 3.1.6 shows elasticity values from his longer-term path analysis. The share of countywide building square footage and valuations along a corridor increased with the share of countywide freeway lane mileage added three years earlier. Building activities were also highly responsive to average operating speeds two years before. Evidently, lane-mile additions in previous years, confirmed by increased travel speeds, spurred developers to build housing, offices, and retail stores near improved freeways. Consistent with theories of “highest and best use,” offices and public buildings appeared to value accessibility benefits conferred by freeway expansions more than industrial uses. Prior-year gains in travel speeds, but not lane-mile additions, spurred retail development.

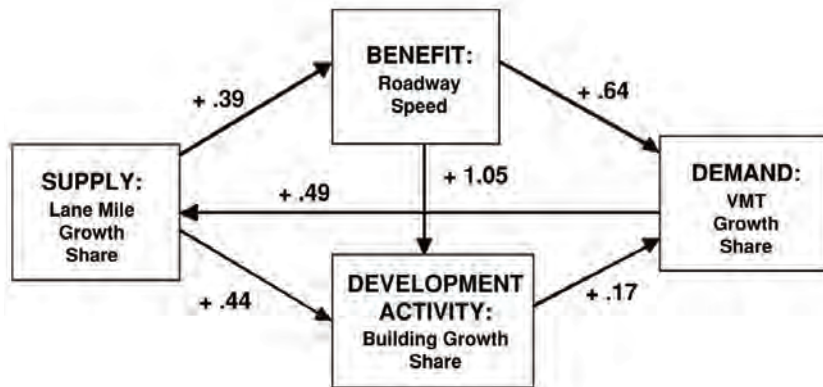


Figure 3.1.6. Longer-term path model summary (VMT: vehicle miles traveled).

Hartgen. Hartgen studied the growth of North Carolina’s 1,551 census tracts during the 1990s in relation to the location of 312 major road projects.²⁷ “Major road projects” were defined broadly, even including widening of two-lane arterials and creation of one-way pairs. Growth went where there was room for it, filling in urban tracts and lower-density edge tracts. Road improvements increased tract decade-long growth by 50 to 550 persons per decade per mile of investment, about 2 to 14 percentage points above the baseline decade growth rate. The effect of road improvements on growth was greatest in Charlotte, North Carolina’s largest city (see Figure 3.1.7). However, relationships were generally weak, explaining only about 15 to 25 percent of the variation in growth and suggesting that other, uncontrolled factors were at work (e.g., schools, taxes, sewer and water, community receptiveness).

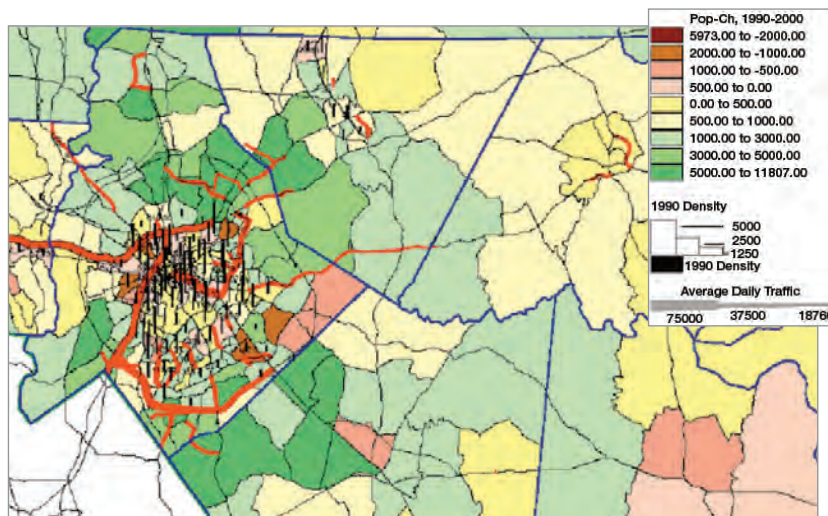
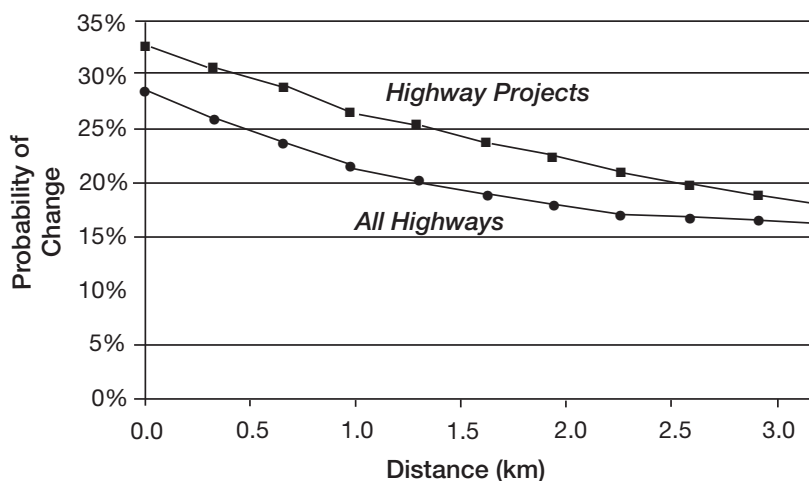


Figure 3.1.7. Population change and major road projects 1990 to 2000, Charlotte, North Carolina, region.

Sanchez. Sanchez studied whether highway projects in 15 Oregon cities induced conversion of land to urban uses between 1970 and 1990.²⁸ Aerial photography was used to delineate the extent of urban development during the 20-year period. A logit regression model related the likelihood of land conversion to distance to the nearest highway, distance to the nearest state highway project, years since the highway project was completed, distance to the city center, zoning classification, city size, and population growth rate. The results suggest that for the 15 cities, the likelihood of urban development declined with increasing distance to the nearest highway, declined with distance to a state highway project, and increased with years since the state highway project was completed (see Figure 3.1.8).

Ryan. Ryan examined the effect of access to highways and light-rail lines on office and industrial property rents in the San Diego metropolitan area between 1986 and 1995.²⁹ Three variables measured access: straight-line distance of each property to the closest freeway on/off ramp; straight-line distance of each property to the closest light-rail transit station; and straight-line distance of each property to the central business district. Results indicated that access to highways had a significant effect on office rents, while access to light rail did not. Neither type of access produced a premium in industrial rents.

Figure 3.1.8. Likelihood of development, 1970 to 1990.



Preliminary conclusion: while freeways provided clear access benefits for office firms, access to light rail transit and to the CBD was valued only when it coincided with significant, preexisting centers of like activity.

Berechman et al. Berechman et al. estimated econometric models for counties and municipalities in the New York–New Jersey metropolitan area using longitudinal data for 1990 to 2000.³⁰ State-level models were also estimated but are not relevant here. The models were production functions with output as the dependent variable and labor, private capital, highway capital, and unemployment as independent variables. Highway investments were found to have positive and statistically significant impacts on output at county level but not the municipal level. As the geographical scale gets smaller, spillover effects of highway investments into neighboring areas get larger. When time lags were applied to highway capital, the coefficients associated with that variable became smaller, suggesting that output responds immediately to highway investments (or may even anticipate highway investments). Output elasticities with respect to highway investments were 0.34 and –0.01 for counties and municipalities, respectively.

Jiwattanakulpaisarn et al. Jiwattanakulpaisarn et al. analyzed the extent to which investments in highways contributed to employment using panel data for all 100 counties in North Carolina, in each year between 1985 and

1997.³¹ Focusing on the actual impacts after the construction phase, they estimated models with the density of highway lane-miles as the independent variable of interest and county private employment as the dependent variable. In a standard model with control variables and dummy variables for fixed effects, the relationship between highway density and employment was statistically significant. However, when lagged employment was added as an independent variable, highway density ceased to be significant. They concluded that omitting the dynamics of employment adjustment could lead to an overestimate of the effect of highways.

SUMMARY OF WHAT WE KNOW

In summary, what does research (reviewed here and elsewhere) tell us about the metropolitan development impacts of major highway projects in the postinterstate era?

- Major highway investments have small net effects on the growth and development of metropolitan areas, instead mostly moving development around the region to take advantage of improved accessibility. Induced development is very close to a zero-sum game.
- Highway investment patterns tend to favor suburbs over central cities and thereby contribute to decentralization and low-density development.
- Major highway investments may actually hurt regional productivity, if they induce inefficient development patterns (read “low-density” development patterns).
- Corridors receiving major highway investments experience land appreciation and therefore are likely to be developed at higher densities than developable lands outside the corridor.
- Highways may be necessary, but they are not sufficient, to induce development. To the extent that current planning and zoning caps hold, impacts within the corridor will be moderated.
- Counties receiving major highway investments attract population and employment growth to a greater degree than they would otherwise.
- Nearby counties may experience more or less growth than they would otherwise, depending on the strength of spillover effects.
- Nonresidential development is more strongly attracted to major highways than is residential development, particularly in the immediate vicinity of facilities.
- The induced-development impacts are wider and deeper for interstate-quality highways than lesser highways and streets.
- It takes many years after construction for development to adjust to a new land use–transportation equilibrium.
- The induced development impacts of major highways extend out at least one mile and probably farther.

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CHAPTER 3.2

Infrastructure and Development: Planning Matters

Robert Cervero



Transportation infrastructure is vital to economic growth, mobility, and quality of life.* The question is not whether to invest, but how. Transportation investments that are coordinated with careful land-use planning are much more effective than an investment-only strategy, and that is where I want to focus my remarks.

* This is an edited and condensed transcript of a presentation given at the 101st National Planning Conference in Minneapolis, April 27, 2009.

An efficient regional network of transportation infrastructure enlarges labor markets and tradesheds, which is a positive thing. Companies in search of specialized skills can match their needs to the labor force more easily when the laborshed is enlarged. Both consumers and firms benefit from larger tradesheds. But does enlargement of tradesheds, laborsheds, and commutersheds equate to sprawl? Unfortunately, in the United States it has. The lack of good regional planning, particularly how we organize densities, has had, I would submit, a major role in contributing to excessive urban decentralization—and as a result the productivity benefits of enlarging laborsheds and tradesheds have been offset by the negative externalities of inducing sprawl.

How densities are organized matters. Curitiba, Brazil, is much smaller than São Paulo, South America's largest megametropolis, 400 kilometers to the north. The blended population densities in Curitiba are no higher than in São Paulo, but Curitiba has linear densities that can better support transit (see Figure 3.2.1). Curitiba has planned densities versus the chaotic, unplanned densities of São Paulo. And Curitiba has a marginally higher transit-ridership rate per capita than does São Paulo, despite being one-fifth



Robert Cervero

Figures 3.2.1 and 3.2.2. The planned linear densities of Curitiba (above) can better support transit than the chaotic, unplanned densities of São Paulo (right).



Robert Cervero

the size of São Paulo. Over the last three years, I have collaborated on a National Science Foundation project focused on urbanization and resource consumption across 370 U.S. metropolitan areas. The research confirms that road density and population density are jointly determined—each in part causes the other. I found evidence of induced growth from highway investments. Highway investments lead to lower density, or sprawl, which indirectly increases vehicle miles traveled (VMT) per capita (Cervero and

Murakami, forthcoming). There was an induced investment effect as well. Areas with high population density were associated with high road density, which was in turn associated with high VMT. Certainly, higher-density places need infrastructure investments, and so some of that investment was no doubt justified to relieve congestion, but a crucial message of a broad view of planning scholarship is that transportation investment must be coupled with land-use planning to meet our economic and sustainability goals.

One of my concerns is that if we blindly invest in new transportation infrastructure, without effective land-use planning, it will further flatten the density gradients and spin off growth to the exurban fringes. For example, if we proceed with President Obama's vision of aggressively investing in high-speed rail, such as in the Central Valley of California, and we site stations on the periphery of cities surrounded by parking lots, we know what will unfold. High-speed rail will induce further sprawl. If land use is simply an afterthought to any kind of regional infrastructure investment, history shows that the added infrastructure will be a force toward more decentralized growth, which in America equates to sprawl. Such infrastructure simply adds a new layer of accessibility to far-flung places, which, absent regulatory controls, unleashes new layers of sprawl. More important, transportation investment without coordinated land-use planning cannot meet the accessibility, mobility, and quality-of-life goals that many persons seek in our urban areas. Consider the example of Los Angeles.

In Los Angeles, population growth tends to be uniformly distributed. Los Angeles is not a low-density city—far from it—but the densities are not organized in a way that facilitates travel other than the private automobile. Los Angeles's densities do not even work well for the automobile. Much of Los Angeles is dotted by three- to four-story garden apartments, horizontally distributed, which in a car-dependent society creates very high trip-generation rates and high traffic densities. The result is very high congestion—according to the Texas Transportation Institute, the worst in America for the past 20 consecutive years. Los Angeles's densities thus do not work well from a highway standpoint. You get overuse of cars and high congestion. Nor are Los Angeles's densities well suited for point-to-point rail systems—unlike, for instance, Stockholm, where rail stops are arranged linearly, akin to a “necklace of pearls,” a configuration that is quite conducive to transit riding. The blended densities of the suburbs of Stockholm are no higher than those in Los Angeles. On a purchasing-power parity basis, the per capita incomes are roughly the same in Los Angeles and Stockholm. But the two cities are organized differently. People in Stockholm own cars; they just don't use them anywhere nearly as much for their day-to-day driving. Stockholmers are much more judicious and discriminate when they drive and when they use public transit. I think the design of the metropolis has had a big part to do with that. Whereas Stockholm's spatial organization is well-suited to rail transit, Los Angeles suffers from “dysfunctional densities”—densities that are too high for a car-based society and (with the exception of the Wilshire corridor) are not configured along lineal axes to support high-quality transit. As a result, the VMT per capita is significantly lower in Stockholm than in Los Angeles or any other U.S. city (Kenworthy and Laube 1999; Cervero 1998; Gorham 1996). Indeed, Stockholm is one of the few global cities where VMT per capita is trending downward, in part due to its well-planned densities.

Perhaps the best example in our country of lineal, rail-supportive densities—the Stockholm “necklace of pearls” model—is the Rosslyn-Ballston corridor in Arlington County, Virginia, part of the Washington, D.C., metropolitan area. The Rosslyn-Ballston corridor mimics the Stockholm model in many respects. One of the nice things you find in this planning approach, where land uses are intermixed along a natural travelshed of roughly four

to six miles, is efficient bidirectional flows. That is, trains and buses are full in both directions during peak periods. Most of Arlington County's station areas, however, are not truly mixed use nodes. Some have more dominant land uses than the others. Rosslyn station, for instance, is dominated by office towers, whereas the Clarendon station is primarily retail-commercial. But collectively across all five stations, one finds an intermixing of most suburban activities, arranged efficiently along the five-mile linear corridor. So what you get by sprinkling the land uses along this travelshed is efficient bidirectional flows. The trains are full in both directions. Mixed land uses produce mixed travel flows.

From the analysis we did a few years back in our "TOD in America" study (Cervero et al. 2004), we found that a significant share of Arlington County's growth clustered around the Rosslyn-Ballston corridor. Stations became a magnet for development and redevelopment. But why? First, there were favorable market conditions. From 1970 to 2000, the Washington metropolitan area experienced as much employment growth as any metropolitan area in the country, as well as a good share of population growth. Also important were factors like building-height limits in the District of Columbia and the federal requirement that all new federal office expansions outside of the District of Columbia but within the regional transit system's service jurisdiction had to be within a walkable distance of a Metrorail station. Growth is regionally determined, so like a tube of toothpaste, if you limit growth in Washington, D.C., it will end up somewhere else. In Arlington County's case, you had a world-class rail system in place to absorb where that decentralized growth went. There were a lot of favorable factors, but a big reason why Stockholm-style growth occurred around stations was proactive planning. Arlington County planners were ahead of the curve. They did not treat transit-oriented development (TOD) as something that the market would automatically create. They leveraged the development with an assortment of tools: density bonuses, as-of-right zoning overlays, and supportive investments like street lighting improvements, undergrounding utilities, and public squares sited near stations.

Lastly, shifting to the micro scale, I'm going to focus on some of the evidence on the effects of rail investments on land prices. We know that rail system benefits get capitalized into land prices. We might expect a fair amount of development activity to be associated with new rail systems as well, but some of that gets suppressed by building limits and not-in-my-backyard (NIMBY) reactions against development. In certain ways, land prices absorb some of the displaced growth effects that might have otherwise occurred if you didn't have constraints on growth. So the land-price premium near transit is likely a combination of accessibility benefits and limits on development.

The lesson I'm going to introduce here is that context matters an awful lot. In a study we did in Santa Clara County, we used real-estate sales transactions to quantify the price premium for properties near rail transit (Cervero and Duncan 2002). Santa Clara County has both the San Jose light-rail corridor and a Caltrain commuter-rail corridor. We used a hedonic-price model to look at the effects on property prices of not only proximity to the rail stations but also accessibility over both the highway and the regional transit network. In addition to the accessibility variables, we also looked at whether the parcel was within roughly a quarter-mile walkable distance to rail stations or the freeway. We found that if you're within the noiseshed of the freeway interchange, there is a negative price effect, as you would expect, on residential property values.

We also looked at a set of zoning/land-use mix balance indicators and a set of neighborhood attributes that we included as control variables, factors

such as income, density, school scores, and crime rates. We found in this particular analysis that the land market benefit (the price premium) from rail transit access was quite substantial, particularly for some commercial properties.

As in the case of Arlington County, conditions matter. My former colleague John Landis did this very same study of Santa Clara County in 1991, when the light-rail system was only a couple of years old (Landis et al. 1994). Landis's study was done during the trough of a recession. Local jurisdictions at that time had not aggressively pursued their transit-oriented development policies. And Landis found no land-price effect—in fact, he actually found a disbenefit (a negative price premium) from being near rail transit.

When we did the study eight years later, it was at the height of the economic expansion during the go-go dot-com boom period. By then, traffic congestion in Santa Clara County—home to the Silicon Valley—was at an all-time high, the worst in the nine-county Bay Area. The rail transit system had matured; it had gone from an early single line to a more complete network. By that time, the Santa Clara County Transit Authority—operator of the light-rail system—had set up a TOD division within the agency, working with the city of San Jose to build affordable housing and improved walkways near rail stops. These were totally different conditions from the early 1990s. Under unfavorable conditions in the early 1990s, there was no measurable impact of the rail system on land prices; in more favorable conditions, we found a quite substantial land-price premium associated with rail transit.

When we look at the effects of transportation investment on land use and market prices, it is premature to expect large impacts in most U.S. metropolitan areas. Transit has to compete with highways and freeways. Highway networks are largely complete, covering the entire metropolis. Most motorists can thus go from anywhere to everywhere at fairly high speeds. In contrast, most U.S. cities with recently built rail systems have only one or two lines in operation. Even for cities with well-developed transit systems, rail often serves less than 20 percent of origin/destination combinations. In most U.S. metropolitan areas, rail systems are incomplete networks. We know that when transit mimics the network connectivity of Paris's Metro or London's Underground, transit grabs high market shares of all trips. When transit is limited to one or two branch lines, it cannot begin to compete with the car for the majority of origin/destination combinations. It is premature to be passing much judgment on our transit investments, particularly rail investments, until they can start achieving some of the spillover network benefits of the chief competitor to transit: the highway/freeway system. I believe that with every additional link we add to our transit networks, we will see synergistic spillover network effects that will begin to produce proportionally bigger mobility benefits.

We should not miss the opportunities to leverage a new generation of green infrastructure. We have plenty of opportunities to “kill two birds with one stone.” We can jumpstart America's struggling economy by investing in green infrastructure while also putting the transport sector on sustainable pathways. I often turn to global examples because, quite frankly, when it comes to transportation/land-use relationships, one finds a lot of dysfunctionalities in the United States. America continues to suffer from gross mispricing of its transportation system—in the form of free parking, underpriced traffic and environmental externalities, impervious road surfaces that contribute to heat-island effects, and so on. To get a sense of what is possible, you can look at the suburbs of Copenhagen where, for example, they have invested massively in green infrastructure—perpendicular feeder bikeways and pedestrian pathways that tie into stations. Not only are Copenhagen's rail-served suburbs economically prosperous, they are sustainable. There,

two-thirds of all access trips to suburban rail stations are by bicycle or foot. In America, the majority of access trips to suburban rail stations, even if under a mile in distance, are by car. Unfortunately, using a car as part of a rail journey largely negates the environmental benefits of transit—in large part due to the disproportionately high tailpipe emissions in the first few miles of a car trip. In Copenhagen and other cities outside the United States, the vast majority of rail users walk-and-ride, bike-and-ride, or bus-and-ride to reach stations. Build it and they will come. You build a parking lot or a freeway ramp, people will drive. If you build better infrastructure to accommodate alternatives to driving, more persons will walk or bike. Investments in bikeways and walkways are not an amenity. They are a basic service.

That said, America can no longer afford a unisectoral, monolithic response to today's transportation problems. Single-sector or single-mode strategies will lead us along a business-as-usual path. Transportation is connected in a multimodal and a systems sense. The key is to plan effectively, to organize land uses in ways that support a broad range of transportation modes, and to invest in transportation in ways that are consistent with long-range regional land-use plans. We have unprecedented opportunities not only to build infrastructure but to coordinate that infrastructure with land-use planning and sustainability goals, and in so doing to improve quality of life and economic vitality in our metropolitan areas.

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CHAPTER 4.1**Regional Transportation Planning
for Southern California Ports***Michael Armstrong*

I work for the Southern California Association of Governments, better known as SCAG.* The SCAG is a metropolitan planning organization comprising six counties and 187 cities. The SCAG region includes all of Southern California, except for San Diego County. If the region was a country, its economy would be the 15th largest in the world, with a gross development product of \$186 billion.

* This is an edited and condensed transcript of a presentation given at the 101st National Planning Conference in Minneapolis, April 26, 2009.

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The region has many planning challenges that transcend local boundaries, such as air quality and transportation, particularly traffic. Some of these challenges even transcend the boundaries of the region. For instance, in aviation, LAX, our international airport, is the only international airport of consequence from San Luis Obispo to the Mexican border. It basically serves all of the southern part of Southern California, even beyond our region. These challenges require a regional approach to be effective.

Most of our future planning challenges are inextricably tied to future population growth anticipated in the region over the next 25 to 30 years. The 5.5 million new people we are anticipating in the region is equal to 14 cities the size of Minneapolis; we are anticipated to grow every two years equal to the size of the current city of Minneapolis, which is about 386,000 people.

Increasing highway congestion and declining average travel speeds is a particular problem that is very much tied to increased population growth. We have been the worst in the nation in terms of average highway speeds since 1982. It's not something we are particularly proud of. We would be very much happy to hand off that trophy to somebody else.

There are major fiscal challenges to mitigating future congestion because of some of the following factors, such as increased construction costs; the declining buying power of the state gas tax, which has remained fixed for many years now and is not likely to change in the near future; the questionable health of the Federal Highway Trust Fund; and the viability of the State Highway Account, given the very serious fiscal problems the State of California is having right now.

Economically, the regional economy is in transition, having lost a significant number of relatively well-paying manufacturing jobs—blue-collar manufacturing jobs—especially in aerospace in the 1980s and the 1990s. This has resulted in a per capita income that has declined significantly, dropping the region from fourth to 17th in the ranking of metro areas around the country by per capita income from 1986 to 2003.

Fortunately, the region has been able to diversify its economy since the very severe recession of the early 1990s, with robust growth in high technology, business services, entertainment, and international trade. The region is particularly well positioned to serve growing trade volumes from Asia with three major seaports, six major airports, and 9,000 lane miles of freeways.

The San Pedro Bay Port Complex, which is composed of the ports of Los Angeles and Long Beach, is the largest port in the United States and the fifth-largest port complex in the world in terms of total cargo tonnage handled. The deep harbors and land site facilities make the San Pedro ports the preferred destination for Asian cargo, particularly imports to the rest of the country. Other competitive ports such as Vancouver and Seattle-Tacoma have limited landside facilities, and other ports such as Oakland are also limited by the harbor depth and their ability to accommodate large post-Panamax ships, which cannot go through the Panama Canal.

Up until recently, there has been a dramatic increase in international trade volumes at the San Pedro Bay ports, and there are very robust unconstrained growth forecasts there, especially compared to ports around the rest of the country. The unconstrained forecast is basically a 450% increase over current volumes—a very, very robust forecast.

The forecast that the ports currently use—and which SCAG employs in its planning, primarily its regional transportation plan every four years—does consider landside constraints for future terminal development, which reduces the forecast to a more modest tripling of cargo volumes by the year 2030. This forecast currently is being revised for reasons I will discuss.

For the Port of Los Angeles, all of its top trading partners are Asian countries—China, Japan, Taiwan, South Korea, and Thailand. For the Port of Long Beach, its trading partners are the same, with a slightly different mix of imports.

Together, the two ports have a regional economic impact of about \$23 billion, with relatively few jobs actually at the ports themselves. There are ripple impacts throughout the region. The San Pedro Bay ports have significant national economic importance as well, since more than 70 percent of the imports entering the ports are transshipped through the region for delivery to the rest of the country.

An extensive logistics industry has developed around the goods movement activity and that has benefited the region enormously. One out of 10 jobs in the regions is now in the logistics sector. The well-paying jobs in this sector have helped the region buffer the decline of average per capita income and the loss of manufacturing jobs. We do have some major costs and impacts associated with goods movement. We have very severe congestion on our freeways resulting from goods movement, particularly the Long Beach Freeway, which directly connects to the ports, and also our east-west porters that deliver our goods from the ports to the rest of the country.

We also have a significant amount of rail congestion. That's considered in our regional transportation plan (RTP). And the rail congestion has some pretty severe impacts on some of the local traffic. At the traffic crossings, many do not have grade separations. We have a number of improvements planned in our regional transportation plan to deal with not only rail capacity issues but also grade separation problems. Our 2008 Regional Transportation Plan (RTP) also recommends a number of projects to mitigate the movement of goods through the region, including a number of dedicated truck lanes on selected freeways, which would be reserved for trucks employing clean technology. It would be a major endeavor to be able to implement those particular projects, because they cross county boundaries. We also have some major rail expansions included in the RTP, which are very important not only in dealing with the rail-congestion issue and also the grade-crossing issue, but in dealing with the issues caused by the fact that our Metrolink commuter service shares the track capacity with the railroads.

It's not going to be cheap to implement these projects. There are big questions about how we are going to fund these infrastructure investments. Clearly we need new sources of revenue, such as user fees and container fees, and also an increase in public/private partnerships to be able to fund these fairly ambitious projects that cross county boundaries in the region. A recent port elasticity study SCAG recently completed demonstrated that even with the imposition of a \$200 fee per container at the ports, there would

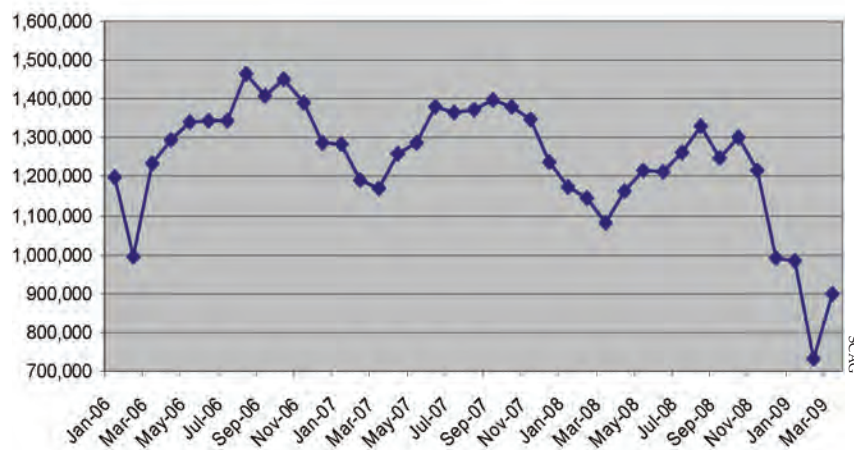


Figure 4.1.1. San Pedro Bay Ports Total Monthly Container Volume (in TEUs); total monthly container volume includes loaded inbound, loaded outbound, and empties.

The ports that are able to best make cost-effective improvements will probably have a competitive edge in the global economy of the future.

be minimal diversion of cargo to competitor West Coast ports, as long as the revenues from the fees were used for capacity improvements.

Unfortunately, all of our previous assumptions in forecasts about regional goods movement are in disarray because of the meltdown of the national economy, which has had brutal consequences for international trade and regional employment in the goods movement and logistics industries. Between August 2006 and February 2009, cargo volumes at the ports dropped by more than 50 percent; comparing February 2007 to February 2009 (to control for seasonal variation), there was a 39 percent drop. This has had severe economic consequences, particularly for the Inland Empire, which are the two counties east of the highly urbanized counties of Los Angeles and Orange. Those two Inland Empire counties (Riverside and San Bernardino) have a very high concentration of warehouse activity. The Inland Empire already has been severely affected by the mortgage meltdown and has some of the highest foreclosure and unemployment rates in the region. According to Dr. John Husing, who is an expert on the Inland Empire economy, from 2000 to 2007 there were about 41,000 warehouse jobs that were created in the Inland Empire. However, they lost about 1,000 of those jobs last year and are expected to lose another 6,000 or so this year because of the ongoing downturn in the goods movement industry.

Besides the potentially protracted economic recession, there are a number of threats to the regional goods movement industry on the horizon, including the planned improvements to the Panama Canal, which are supposedly going to increase the throughput by 100 percent and make it possible for the canal to handle much larger ships; the major expansion of the Port of Prince Rupert in Canada, which has the deepest harbor on the West Coast in North America and also is closer to Asian markets than Southern California; a new port planned in Mexico called Punta Colonet; mainline rail bottlenecks getting the cargo out of the region; and improvements and increased efficiencies and cost cutting at competitor ports in this cutthroat, brutal economy, where shippers are very sensitive to costs. There could be diversion if another port becomes less costly and more efficient. Academics and professionals are debating how much these factors may threaten the San Pedro Bay ports' volume, and there is really no consensus about the magnitude of the impacts on shipments through Southern California. Another future uncertainty is whether and how strong the economic recovery will be and whether there will be fundamental restructuring of our demand-driven, consumer-driven economy. That could have lasting implications for the goods movement industry.

Compared to Asian ports, U.S. ports are rather inefficient when it comes to handling cargo. Even those ports that have land constraints can increase their throughput quite substantially through increased efficiencies. The ports that are able to best make those cost-effective improvements will probably have a competitive edge in the global economy of the future.

The bottom line is that there are a lot of uncertainties out there, a lot of wild cards that make it very difficult to plan right now. Fortunately, SCAG is embarking on a major comprehensive regional goods movement study and implementation plan, and we are going to be reevaluating many of our core projects. Ideally, it would be nice to have a fixed guidepost on which to base planning, but that doesn't look like it's going to be the case. My own opinion is that we will see a return to trend-line growth in a few years, but that is my guess and we will have to see what the future holds.

I would like to close with a message that enjoys widespread consensus among our elected officials in Southern California, and that is that the goods movement infrastructure of Southern California is of vital national importance. The Southern California goods movement complex is a major

gateway for international trade and contributes important to the national economy. This infrastructure needs to be improved and expanded to mitigate ever-increasing congestion and local impacts on communities in our region. Southern California cannot do it alone. We are very much looking forward to a stronger partnership with and support from the federal government to make the cost-effective improvements needed to compete in the global economy of the future.

CHAPTER 4.2

Policy Innovations in Southern California Ports

Thomas O'Brien



Michael Armstrong did a nice job of giving you an overview, sort of the macro level of the issues confronting Southern California as a global trade gateway.* I want to talk about the same area at the micro level, looking at the planning and policy context for goods movement in Southern California—in particular, the planning and policy context as it's responding to the region's environmental challenges. I'll talk about two particular responses: the PierPASS program and the two ports' Clean Air Action Plan and Clean Truck Program. I will conclude by asking whether the lessons that we take away from Southern California are unique, and by discussing what lessons can be transferred to other parts of the country.

* This is an edited and condensed transcript of a presentation given at the 101st National Planning Conference in Minneapolis, April 26, 2009.

When we talk about international trade and the role of urban centers, we're not just talking about the physical movement of goods. We're talking about the use of information tools and ancillary services that support international trade.

It is worthwhile to start with some lessons that we have from Lewis Mumford, because Mumford looked at the interdependent nature of trade, urban development, and the use of technology. From the 18th century up until the early 20th century, we saw the development of the early stages of capitalist industry, the development of industrial centers, and the switch from falling water to steam power. This is where we see the growth of port cities as coal depots to fuel the engines of trade along the East Coast of the United States and Europe and also northern Africa as well, with iron and steel replacing wood as the primary structural material.

This is sort of a rapid historical overview, but it reminds us of the fact that when we talk about international trade and the role of urban centers, we're not just talking about the physical movement of goods. We're talking about the use of information tools and ancillary services that support international trade, which is why you have not only concentrations of warehousing and distribution centers around global gateways but also concentrations of legal and financial services that support trade-related activities. That brings us to in the case of Southern California today. Our proximity to Asia has made the region an important national and global trade gateway.

There are a couple of things that we need to keep in mind, though, when it comes to competitiveness. We can talk about planning at the local level, at the state level, even at the national level, but we have to keep in mind that trade is a function of global market dynamics. For example, most of the ocean shipping companies that ply international waters and bring the goods to major ports are no longer U.S. flag carriers. They are international carriers that are principally subject to international treaty, often outside the jurisdictional arm of local planning and policy organizations, particularly when it comes to questions of air quality. The federal government has been somewhat hands-off. Unlike our counterparts in places like Canada and Europe, we really don't have a national freight policy.

The state and local role then has been to fill in the gap where the federal government has not acted. In California, it's particularly important that the state is involved because the state owns the coastal waters and the ports. The ports operate under a 1911 law called the California Tidelands Trust Act. The state cedes authority to local cities to operate the port on behalf of the state. There are a couple of important implications of this for planners. First, the revenues that are generated from port activities have to go back into port operations. Local city councils cannot take revenues generated by the port and put them into schools or health care. There has to be a direct nexus between port revenues and expenditures, which raises interesting issues for local jurisdictional authorities. Also, the stakeholders involved in trade-related issues may be exempt from local planning regulations because of their involvement in interstate commerce. The railroads are the principal examples of that. The latest trend is that the ports themselves have become regulating agencies, sometimes in response to the threat of state legislation and sometimes in a more voluntary "self-regulatory" fashion.

We've seen rapid growth in U.S. trade up until the recent economic downturn, and the result has been that the benefits have been widespread across the nation. But the impacts in terms of traffic congestion and air pollution are very much local in nature.

Los Angeles/Long Beach is the largest port complex in the United States, fifth largest in the world. It's also the source of 64 percent of all U.S. Pacific containerized trade, including Hawaii; 54 percent of total Pacific containerized trade, including Mexican and Canadian ports; and 35 percent of all U.S. containerized trade. Until this recent downturn Los Angeles and Long Beach, which are neighbors and competing ports, were not only the largest ports on the West Coast but also the fastest growing—they are big in terms of absolute numbers and also rate of increase. The Port of Prince Rupert means that the Canadian ports are the biggest threat, and in 2007 Vancouver supplanted Oakland as the third-largest port

on the West Coast. The impact of trade through our region remains significant for the entire country. This is an argument that Southern California elected officials try to make, sometimes to no avail, to congresspeople and senators from the rest of the nation—the point being that our competitiveness as a region means competitiveness for the nation as a whole as a world leader in trade.

But all of this growth has resulted in a backlash from local communities and elected officials, which has changed the planning landscape. There are a couple of significant events that I want to highlight here. The first was in 2000, when the South Coast Air Quality Management District's MATES II (Multiple Air Toxics Exposure Study) showed the relationship between diesel emissions and exposure to cancer-causing agents such as particulate matter (PM). This was the beginning of our ability to understand and quantify the relationship between trade activities and health. Following that, there were a series of lawsuits that used the environmental impact report and environmental review process to bring about change on the terminals by forcing ports and the terminal operators to implement new cleaner technology on the docks.

I like to show this graphic that came out of that air quality management district study. The large black area stretching out into the ocean and up essentially to downtown Los Angeles is the I-710 corridor. This shows exposure to PM 10 (particulate matter 10 micrometers or less in diameter.) The large black area on the map became known in community hearings and in planning hearings as the Diesel Death Zone. And this changed the landscape of the discussion. No longer were the ports seen only as economic generators; they were also seen as the source of negative environmental and health impacts.

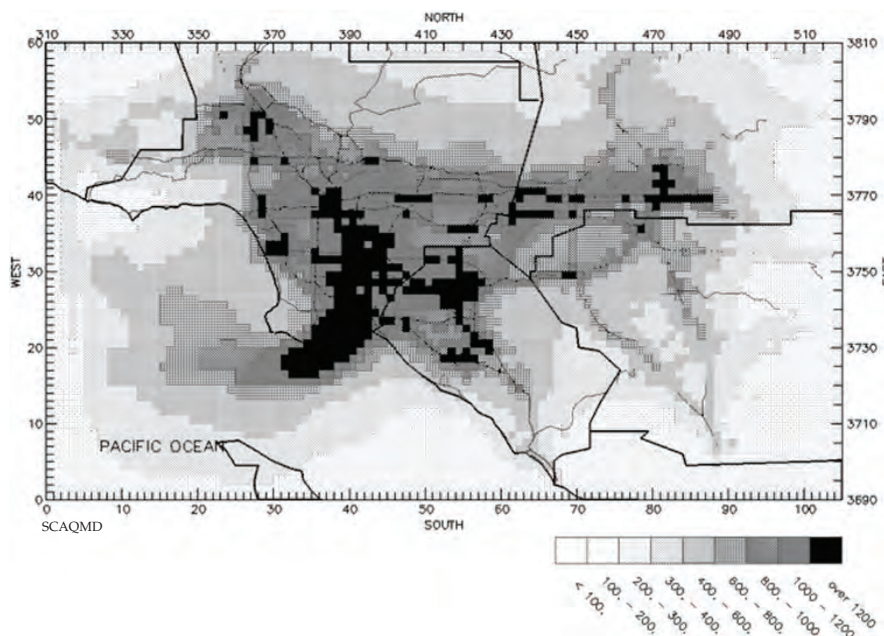


Figure 4.2.1. Mates II PM <10 Exposure Map

Because of the reasons I mentioned—the fact that there is some inability on the part of local jurisdictions to influence trade because of their lack of jurisdictional authority—after 2000 we started to see a rash of bills at the state level seeking to influence the behavior of the ports and industry. One of these bills—Assembly Bill 2041, in 2004—attempted to do a couple of things: to set up a port management congestion district, and also to mandate that ports operate after 5:00 p.m. as a way of spreading out truck traffic. Most ports around the country operate between 8:00 a.m. and 5:00 p.m. for the pickup and drop-off of containers, because the cost of labor in the off-peak is so much higher because of longshore regulations. This bill would have mandated that the ports operate in “off-peak” hours as a way of spreading out truck traffic. Because of the legislative activity,

the terminal operators decided to say, "Thank you very much for your efforts, State Assembly. We get the message. We'll actually implement our own program if you withdraw your legislation." And that's what happened. The terminal operators implemented a program called PierPASS, which assessed a surcharge on containers that were picked up or dropped off in peak hours between 8:00 a.m. and 5:00 p.m. The fee was assessed per TEU, which stands for "twenty-foot equivalent unit." After the onset of the program, we saw a significant shift of containers to off-peak periods. The following results are from a study that we did at the METRANS Center at Cal State Long Beach and USC. The first week was July 23, 2005, and during the first week there was movement of 35 percent of the eligible containers to the off-peak period. Consistently throughout the life of the program, about 35 to 40 percent of shipments have been shifted to off-peak time periods.

We looked at truck traffic in and around the ports. On the I-710 and the I-110 freeways near the ports we did see a considerable impact: a drop in truck traffic during the day and an increase in truck traffic during the off-peak period after 6:00 p.m. We also saw a similar impact on Saturdays, because there were PierPASS gates on Saturdays, so containers could be moved on the weekend as well without paying the traffic mitigation fee. I should mention that that fee that was charged went back to the terminal operators to pay for the cost of additional labor in the off-peak, which was their incentive to implement the program on their own.

So what are the impacts on competitiveness for the region? Well, there was an immediate and continued shift of eligible cargo to the off-peak, which meant there was an improvement in terms of local congestion, which improves the region's competitiveness. From our simulation and models, our best guess is that we offset about two years of port growth by shifting some goods movement to the off-peak. The PierPASS program also demonstrated that industry could support container fees, with the understanding that the fee revenues would be used to improve the system as a whole. A new nonprofit was set up by the terminal operators to collect the container fees, which are now being used for other mitigation measures. It's all controlled by local industry, by the terminals themselves.

The next policy measure is the Clean Air Action Plan and the Clean Truck Program, which is the first component of the Clean Air Action Plan. This is a plan that was jointly adopted as a policy measure by the two ports in 2006, again trying to get ahead of a number of bills that were being considered in Sacramento. The Clean Truck program looks to establish concessions for trucks based on environmental standards. So as of last October 1, if you had a truck or a truck engine with a model year of 1989 or older, you were no longer able to access the port. You had to show that you had a newer vehicle. Over the next three to five years, newer model engines will be phased in.

This has been an interesting case to observe. It has tested the ability of local ports to set policy, and the ports did jointly adopt the program. The problem is, in a couple of key measures, they adopted different standards. The Port of Los Angeles went on the record saying that port concessions would go only to truck drivers who were employees of trucking companies, not independent owner-operators. That is not a provision that the Port of Long Beach required. The argument is that truck drivers will be more likely to invest if they have the financial support of a trucking company behind them. On the flip side, there were arguments made that the only reason this was being done was that it is easier to organize and unionize truck drivers who are employees. After a series of court cases, it looks like the Long Beach proposal is the one that will likely survive and that independent owner-operators will be allowed to get concessions. In late April, a U.S. district judge issued a preliminary injunction that halted some of the Clean

Truck Program rules, including one that prohibits drivers at the Port of Los Angeles from being independent contractors.

In terms of port competitiveness, the fact that the ports are pursuing environmental goals through their own initiatives is a good thing. The hope is that Los Angeles and Long Beach, by getting ahead of the curve and anticipating issues that other ports around the country are going to have to deal with, are going to be well-positioned to set the standard and have programs implemented while other ports are still struggling with these issues. But there's also a general concern that if that's not the case—if the other ports don't follow the example of Los Angeles and Long Beach—that the cost of moving goods through the San Pedro Bay ports, because of additional surcharges and fees, will become too much and cargo will be diverted to other locations.

So finally, the question for those of you from other areas who are looking at the planning and policy arena is, "Is Southern California unique?" Well, its size and its dominance in some ways make it unique. We have a certain level of inelastic demand because of the size of our local market. The scale of congestion and pollution problems are also unique, and the frequency and scope of state regulatory efforts so far are unique. We've seen similar efforts in Northern California, New York, and New Jersey modeled on the Los Angeles experience, but nothing yet in those regions has had the extent that we have seen in Southern California.

The effectiveness of Southern California environmental advocates is also unique. We have environmental advocates across the country, but the Southern California groups have had particular ability to influence and bring about change on the part of the terminal operators and the ports. In the research we've done, we found you need the combination of congestion and political pressure to make some of these policy measures happen, and we haven't seen that combination yet elsewhere.

The ports are links in a system. So you'll see the domino effect of these policy and regulatory measures, first starting in Oakland, then moving to Seattle-Tacoma, and then even places like Vancouver and Prince Rupert. Everybody has a need to fund infrastructure, so container fees will likely be considered elsewhere.

Changing trade dynamics impose local costs and dispersed benefits. Because of the global trading system, that is only going to continue. Ports are less able to influence trade flows than you might think. The ocean carriers determine which ports are used. They have a fair amount of leverage, as do the major shippers. Ports will compete for business, and the shippers will leverage one port against another.

But to a certain extent, the ports, at least in Southern California context, have come to understand that it is important to internalize the externalities because failing to get ahead of environmental issues is bad for business. The challenge is getting the right balance of the fees and making sure that fee revenue goes toward improving the infrastructure in ways that benefit industry, which is important to garner private sector support.

The hope is that Los Angeles and Long Beach, by getting ahead of the curve and anticipating issues that other ports around the country are going to have to deal with, are going to be well-positioned to set the standard.

CHAPTER 4.3

Sustainable Goods Movement in Southern California: The Promise of Collaborative Planning

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The Southern California National Freight Gateway is a regional, national, and international trade powerhouse. The gateway extends from the ports of Los Angeles and Long Beach over multiple truck and two major railway corridors through the Los Angeles metropolitan region and beyond. When measured by the annual volume of container shipments, the combined ports complex is the fifth-largest port in the world and the largest in North America (Wilbur Smith 2008). With the advent of containerized shipping, landside unloading, repackaging, and rerouting has created a robust logistics industry that relies on warehouse complexes and networks of rail and road infrastructure to move the goods from the port to initial processing facilities within the region, and from there by either truck or rail to destinations throughout the United States. That logistics industry is a source of many jobs in metropolitan Los Angeles, but the ports and the landside transportation network are also a source of substantial health and welfare impacts.

The ports and associated overground freight shipments are the single-largest source of particulate emissions in the Los Angeles region (Houston et al. 2008). The California Air Resources Board (2008) estimates that diesel particulate emissions contribute to 3,500 premature deaths in the state each year.¹ The ports and freight gateway complex are at once one of Southern California's primary economic drivers and one of the region's most important environmental health threats.

The challenge is not simply the traditional task of balancing economic development and environmental sustainability. Our silo-like governance and planning institutions were designed in simpler times and cannot well address the web of interlocking concerns, issues, and opportunities that characterize modern planning challenges. The National Environmental Quality Act (NEPA) of 1969 presciently anticipated the growing complexity of our economic and environmental concerns and the need for a multipronged, collaborative approach.² To date, however, the original NEPA call for a holistic and systemic approach toward environmental governance has gone largely unanswered. As the practice of environmental impact statements (EIS) has evolved, the focus has been on post-hoc explanations of decisions and distinct projects rather than on early-stage policy development that spans multiple systems and comprehensive planning.

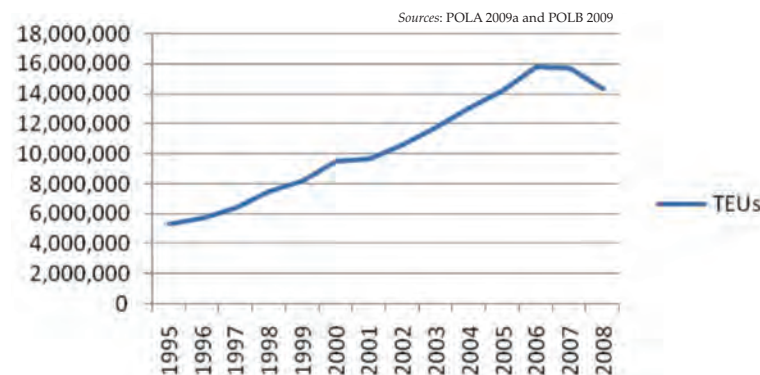
The goods-movement challenge in Southern California is an exemplar of the need for multidimensional and collaborative planning approaches that can create flexible institutional arrangements which span existing regulatory and agency boundaries. This chapter proceeds in three parts. First, we describe the goods-movement challenge in the greater Los Angeles region. Second, we discuss the evolution of environmental conflicts from the project-specific to the systemic and the resulting need for governance structures that can adapt to the resulting complexity. Third, we elucidate the opportunities for and advantages of multiagency collaborative efforts related to the region's goods-movement challenges.

The Ports of Los Angeles and Long Beach

The Ports of Los Angeles and Long Beach are adjacent to each other in San Pedro Bay, approximately 25 miles to the south of downtown Los Angeles. The two ports are governed by distinct authorities, each of which is an arm of the city governments of Los Angeles and Long Beach. While the entities are distinct, the complex can be viewed as one entity in terms of proximity, landside shipment routes, and impacts, both positive and negative.

Shipments into the ports have exploded in recent years, fueled in large part by the increase in trade with Pacific Rim countries. Figure 4.3.1 shows shipments

Figure 4.3.1. Shipments (in TEUs) in and out of the Ports of Los Angeles and Long Beach, 1995–2008.



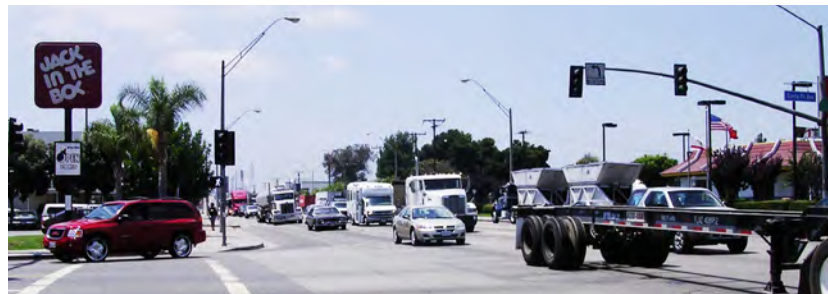
through the two ports measured in 20-foot equivalent units (TEUs) from 1995 through 2008.³ During that time, goods-movement volume has almost tripled. In some years, the year-to-year incremental increase in shipments through the combined San Pedro Bay ports rivaled the total shipments through Oakland (the largest port in northern California).⁴

More than 40 percent of all international cargo entering or leaving the United States now moves through the ports of Los Angeles or Long Beach (Wilbur Smith 2008). Seventy-seven percent of all shipments through the San Pedro Bay ports are destined for locations outside of the Los Angeles metropolitan area (Wilbur Smith 2008). While shipments through the ports have declined during the current economic crisis, the availability of a large local market, deepwater access, accessibility to the Pacific Rim, and strong landside logistics and warehousing capabilities all suggest that the reduction is likely to be temporary and that the pattern of dramatic increases in TEUs will resume absent either a prolonged and deep global recession or substantial increases in barriers to international trade.⁵ Projections from regional agencies suggest a possible doubling or tripling of shipments through the two ports between now and 2030 (SCAG 2005).

Houston et al. (2008) note that 70 percent of the cancer risk from air toxins in California is associated with particulate emissions, that greater than 70 percent of diesel particulate emissions in the state are from the goods-movement sector, and that heavy-duty diesel trucks are responsible for 70 percent of the goods-



*Figures 4.3.2 and 4.3.3.
Trucks on surface streets near
the San Pedro Bay Ports.*



Douglas Houston

movement sector's particulate emissions. Particulate emissions are associated with adverse health outcomes that include increased cancer risk, respiratory illness, and heart disease.

Particulate emissions concentrate near the source. Research has shown that concentrations are highest within 100 meters downwind of the source, dropping to background levels at a distance of 300 meters (Zhu et al. 2002). The pattern of emissions, exposure, and health risk therefore concentrates near the ports and along major transport routes through the urban area. Most of the Los Angeles region's development occurred before the risk of particulate concentrations was well understood, so there are many sensitive land uses near freeways. As an example, Houston et al. (2006) found that approximately 7 percent of California's licensed child-care slots (a total of 57,000) were located within 200 meters of freeways. Given historical development patterns, with lower-income or working-class neighborhoods constituting much of the residential development near ports and highways, exposure to particulate health risks is an environmental justice issue. See Figures 4.3.2 and 4.3.3 (above) for photographs of the proximity of truck traffic

to neighborhoods in the port-adjacent communities. Land-use conflicts from the proximity of communities to highways and rails occur throughout the region.

In addition to air-quality impacts, communities along the goods-movement corridor experience other negative effects. More than 112 freight trains per day move through the greater Los Angeles region, with the highest volumes in Riverside, San Bernardino, and eastern Los Angeles counties (Wilbur Smith 2008). Portions of the primary east-west rail route through the region are still single-tracked, leading to congestion delays. As freight volumes have increased, rail-crossing locations have become an increasingly important land-use and traffic issue in affected communities. The Southern California Association of Governments' (SCAG) 2008 Regional Transportation Plan identifies 132 grade-separation projects in the region (SCAG 2008b).

In addition to grade separations, the warehousing, transloading, and logistics industry creates land-use patterns and transportation challenges that bring both positive economic benefits and the need for planning. A report for SCAG estimated that in 2003 the logistics industry employed 548,278 workers in the region, at an average annual wage of \$45,314 (Husing 2004).⁶ Looking forward, the bulk of that industry will locate in the inland counties of the region, driven by land needs that, in the case of large warehouses, can require sites as large as 60 acres per facility (Husing 2004). While the economic impact on the inland counties of San Bernardino and Riverside can include well-paying blue-collar jobs, the land-use impacts themselves will require careful planning. There are currently 1.5 billion square feet of warehouses and transloading facilities in the Los Angeles region (Wilbur Smith 2008), and increased trade and associated goods movement will necessitate additional land needs in a metropolitan area with limited options for large greenfield development. An aerial photograph of the industrial and warehousing complex near the Ontario airport conveys the scale of the land-use planning challenge (Figure 4.3.4).

Figure 4.3.4. Aerial photograph of a warehousing complex near Ontario airport. (The airport runways appear toward top center, for scale.)



Environmental Conflicts: From Project-Specific to Systemic

Over the past seven years, environmental organizations have engaged the two ports on issues that initially focused on air-quality impacts and specific projects. Organizations such as the Natural Resources Defense Council, the Coalition for Clean Air, and the Coalition for a Safe Environment have raised awareness through litigation and lobbying efforts on issues related to the development of a new port terminal (Coalition for Clean Air 2004), the re-

placement of a bridge (LACAC 2009), and proposed highway improvements. The ports have responded in several ways. The Alternative Maritime Power program (POLA 2009b) provides onshore power that allows ships to turn off their diesel engines while in port. The San Pedro Bay Ports (2006) Clean Air Action Plan created the Clean Truck Program, a combined initiative for both jurisdictions that requires lower diesel emissions from trucks that access the ports. There have been international efforts from the International Maritime Organization (2005) to have ships use low-sulphur fuel as they approach the ports. Also, regional, state, and national air-quality agencies (South Coast Air Quality Management District, the California Air Resources Board, and the U.S. Environmental Protection Agency) have worked to address specific sources of air-quality problems from ships, trains, and trucks.

These efforts have largely been focused on the ports themselves and are more in keeping with the historic battles between industrial and environmental interests. In large part, the environmental conflicts were project-specific and were arguably within the competence of the traditional forums—the ports and the courts—to address. Within the last five years, this has changed. Concerns have expanded to embrace more systemic impacts from the movement of freight through the region, involving several communities, some of which are scores of miles from the ports themselves but near freight routes or the landside logistics industry. The more systemic issues in the mix include:

- Increased congestion of rail lines, accompanied by the increased closure of arterial streets at rail crossings, resulting in local demands for grade separations, particularly in the inland counties in the region;
- Significantly increased truck traffic on freeways within the freight corridors, raising concerns regarding congestion, particulate emissions, and the allocation of the cost of constructing and maintaining highway infrastructure;⁷
- Issues of greenhouse-gas emissions and climate change; and
- The need for increased infrastructure to support the significant anticipated increase in truck traffic and automobiles (resulting from an increase in the regional population as well as increases in goods movement) and to mitigate impacts to local communities.

These concerns have been heightened by studies regarding increased health risks from diesel emissions and by California's recently legislated climate-change regulations. The net effect has been that the conflicts and concerns have shifted from the project-specific to the systemic, and both the issues and possible solutions range well beyond the traditional geographic, disciplinary and institutional domains of the ports themselves or existing regulatory agencies.

The policy response initially focused on several broad areas: regional and state goods-movement plans, funding and institutions (e.g., container fees), and, significantly, collaboration—that is, how the various affected agencies and interests cooperate in scoping, deciding upon, and implementing an effective plan. With respect to planning, following a series of project-oriented efforts, a series of more comprehensive goods-movement-related plans were developed, including a plan by SCAG (2005). In 2007, the California Business, Transportation, and Housing Agency also put forth its own Goods Movement Action Plan (CBTHA and CEPA 2007). In May 2006, the U.S. Department of Transportation adopted a "Congestion Management Strategy," which it updated in 2008, designating a "hot spot" team to address congestion relating to freight movement through the Southern California

The complexity of the goods-movement challenge results in part from the sheer number of interested parties.

region (USDOT 2008). The five county transportation commissions within the SCAG region completed a Multi-County Goods Movement Action Plan in 2008 (SCAG 2008a).

Roughly in parallel, policy actors at the local, state, and federal levels have proposed container fees as a means of financing needed environmental mitigations and infrastructure. Through their Clean Truck Initiative, the ports charge container fees to help fund the conversion of trucks that enter the ports to lower emission diesel engines (San Pedro Bay Ports 2006). State Senator Alan Lowenthal (Long Beach) in 2008 introduced S.B. 974, a container-fee legislation bill, intended to finance regional goods-movement infrastructure. It was passed by the California legislature but vetoed by Governor Arnold Schwarzenegger. At the federal level, Representative Ken Calvert (CA-44) in early 2009 introduced H.R. 947, a national container-fee legislation bill, which is currently still pending in the House Subcommittee on Railroads, Pipelines, and Hazardous Materials (Calvert 2009). While the ports of Los Angeles and Long Beach have the legal authority to charge fees associated with infrastructure and activities within the port areas, efforts to finance mitigation and infrastructure over the much larger gateway area would require state and possibly federal authorizing legislation.

As a third prong, a group of 20 key local, state, and federal agencies entered into a cooperative agreement in 2007, establishing the Southern California National Freight Gateway Collaboration to address the development of a strategy for the movement of freight through the region. This effort is a response to the inherently multijurisdictional and complex context for the goods-movement problem. That collaboration represents a flexible structure operating outside of the traditional boundaries of jurisdictions and disciplines, and it highlights what we believe is the most promising approach to resolving the systemic planning issues that will become increasingly common in the future.

Collaboration as a Framework

The complexity of the goods-movement challenge results in part from the sheer number of interested parties. Government agencies have incentives to focus on their respective regulatory missions rather than a comprehensive solution. The strong and distinct interests of private enterprise (including shippers, railroads, and developers) and public-interest groups further complicate the problem. The current governance structure, with its strong focus on project-by-project funding, permitting, and environmental review, often fails to adequately respond to broader systemic concerns. The result is often conflict, litigation, suboptimal solutions, and significant loss of time and opportunities.

Within the gateway area, the vast majority of diesel emissions come from trucks, trains, and ships. Land-use regulations to require, for example, air-quality setbacks from freeways would require the political, if not the legal, action of the many local agencies and constituencies along these corridors. A user fee (e.g., a container fee to construct and operate necessary infrastructure) is likely to require federal action, for reasons related to interstate commerce and international treaties.⁸ There would be similar questions as well with regard to a state-levied user fee, in addition to questions regarding the “fair share” allocation or “nexus” of such fees under state and federal constitutions and international trade agreements (e.g., regarding “anticompetitiveness” provisions).

In summary, the current fragmentation in governance makes it extremely difficult to address and implement a strategy for the anticipated increase in freight movement through the gateway area. Such a strategy will require that we address concerns ranging from mobility to air quality (including

particulates and greenhouse-gas emissions), energy, land use, community quality of life, jobs, financing, and administration. Yet it is highly unlikely that we can reform the fragmentation in authority among affected local, state, and federal agencies. Rather, we argue what is possible—what is required—is that we focus on the design of a collaborative process that transcends governance boundaries and effectively addresses the development and implementation of a freight strategy for the area.

The Southern California National Freight Gateway Collaboration is a promising pilot effort with ramifications for environmental and planning processes across the nation. The collaboration comprises the 20 local, state, and federal agencies that are concerned with various aspects of a strategy for moving increased amounts of freight through the region.⁹ Its function is to cooperate (as envisioned by Section 101(a) of NEPA) and assist in the “scoping” of that strategy. For this reason, the signatories are referred to as “conveners” and are not a group of super-decision makers. The development and implementation of a strategy comprises three elements: scoping, decision making, and implementation. The collaboration is focused on the first of those three elements.

Scoping is the process of thinking about various topics, selected by the conveners, that relate to the proposed strategy. In public processes, such as those contemplated by NEPA, scoping includes:

- the identification of the affected agencies and interests to be involved;
- research (including studies, literature reviews, and interviews);
- dialogue with the affected agencies and interests (including the solicitation of comments and group dialogues); and
- the preparation of a report or reports that describe the results.

The scoping process is a deliberative one. One goal is to develop a shared understanding of concerns and opportunities, as well as relevant considerations, alternatives, impacts, and possible mitigation measures. This does not imply consensus. Rather, the objective is to accurately and fairly represent the points-of-view presented, and to illuminate shared understandings and divergent views. Similarly, the scoping process is designed to move deliberatively forward, respectful of relevant planning and regulatory time constraints while appreciating the need to effectively inform the relevant decision-making processes. The process is intended not to block or defer decision making but rather to enable better and more expeditious policy action.

To the extent that the process addresses proposed actions or approaches, it normally includes a discussion of significant considerations and a range of alternatives, impacts, and mitigation measures. Because scoping is preliminary to but not part of decision making, the scoping report does not include recommendations. Decision making is left to the formal decision makers.

The Freight Gateway Collaboration preliminarily identified four topics to be scoped, utilizing a neutral facilitator:

- the environmental-review processes that might be employed regarding projects and programs related to the freight gateway strategy, with a view toward making these processes more thoughtful, efficient, and expeditious;
- funding and related institutions (addressing the source of funding; underlying considerations; and the institutions for collecting, disbursing, and monitoring the funding and undertaking the work);
- air-quality attainment (to develop a vision as to how air-quality standards can be met while providing enhanced freight movement); and
- the form and function of related federal programs.

As environmental challenges have grown more complex there is a need for new approaches. Governance structures need to better give voice to a broad range of actors, while building networks of individuals who can work across agencies, jurisdictions, and the public, private, and nonprofit sectors.

The power of the Freight Gateway Collaboration goes far beyond the production of a scoping report. Working to collaboratively focus on a strategy that transcends the historic silo-like and fragmented governance structure creates a kind of common ground among the agencies and interests involved. It promotes networks of key individuals who are familiar with the details of the freight-movement challenge facing the region.

While the Freight Gateway Collaboration provides an effective process for bridging governance boundaries in the scoping of the appropriate strategy, it does not provide for decision making or implementation. There is a significant need to develop similar flexible, ad hoc institutional arrangements that transcend boundaries in the implementation of the strategies scoped and decided upon. Such arrangements are currently being explored in this country and in Europe. An example is the Habitat Conservation Plan concept, under the federal and state endangered species acts (7 U.S.C. §136, 16 U.S.C. §1531 et seq., California Fish and Game Code secs. 2050–68). Generally, this concept envisions a plan, covering a specified area, developed by various local, state, and federal agencies, and focused on the conservation of biodiversity in the midst of urbanization, which is then implemented through an agreement to which these agencies are signatories.

By analogy, a “special area” could be established comprising the Southern California National Freight Gateway Area. Within this area, a pilot effort could be pursued that would tailor national, state, and local policies to the challenges within this region. For example, it could provide an approach that would assure that regional measures, including container and user fees, are consistent with the exercise by the federal government of the interstate commerce power and the provisions of international agreements. Taking into consideration that this gateway services more than 40 percent of the nation’s imports, it would be an appropriate pilot effort and potential model for other areas and even for other problem domains.

CONCLUSION

The idea of collaboration can revive the concept of holistic environmental management as envisioned in NEPA. Four decades ago, the federal involvement in environmental regulation was prefaced with an understanding of the need for boundary-crossing approaches that could embrace broad systems. Yet a more project-focused and often silo-bound regulatory approach evolved. That approach has yielded many benefits, but as environmental challenges have grown more complex there is a need to answer the call for new approaches. Governance structures need to better give voice to a broad range of actors, while building networks of individuals who can work across agencies, jurisdictions, and the public, private, and nonprofit sectors. The Southern California National Freight Gateway Collaboration is an example of a promising approach to such boundary-crossing planning frameworks.¹⁰

ENDNOTES

1. The evidence on the health impacts of particulate emissions is evolving. The general trend has been that more recent studies suggest more severe health impacts. The Air Resources Board’s estimate of premature deaths has an uncertainty interval of 1,000 to 6,400. For more on these and associated methodological issues, see California Air Resources Board 2008.
2. Section 101a of NEPA (42 USC §4331) comments on the “the profound impact of man’s activity on the interrelations of all components of the natural environment, particularly the profound influences of population growth, high-density urbanization, industrial expansion, resource exploitation, and new and expanding technological advances.” That section continues: “It is the continuing policy of the Federal Government, in cooperation with State and local governments, and other concerned public and private organizations,

to use all practicable means and measures . . . to create and maintain conditions under which man and nature can exist in productive harmony.”

3. A TEU is a 20-foot cargo container. Many rail and truck freight shipments now use 40-foot cargo containers (two TEUs), which are typically moved from ship to dock and then to truck or rail as one unit, for offloading and (potentially) transloading and rerouting elsewhere.

4. For example, from 2005 to 2006 the increase in containers through the San Pedro Bay ports was about two-thirds the level of shipments (in TEUs) at Oakland in 2005 (POLA 2009a, Port of Oakland 2009, and POLB 2009).

5. Due to the current economic recession, the Port of Long Beach saw a 20 percent drop for the month of March 2009 compared to imports in March 2008, while the Port of Los Angeles saw a 10 percent drop from the previous year (POLA 2009a; POLB 2009).

6. Husing’s (2004) estimate of logistics jobs is derived by summing year 2003 employment in the following sectors: wholesale trade (NAICS 42), truck transportation (NAICS 484), support services for transportation (NAICS 488), nonlocal couriers (NAICS 492110), general warehousing and storage (NAICS 493), air transportation (NAICS 481), rail transportation (NAICS 482), and water transportation (NAICS 483). That definition includes logistics functions that go beyond those related to international trade. The largest portion of Husing’s estimate of logistics employment, slightly over 400,000 jobs, is in the wholesale trade and truck transportation categories (NAICS 42 and 484). Husing states that many of the region’s wholesaling operations, in particular, are related to international trade.

7. On cost allocation, see, e.g., the U.S. Department of Transportation studies at www.fhwa.dot.gov/policy/otps/costallocation.htm.

8. The ports can argue that they have the authority to levy a charge on those utilizing them (e.g., lessees and related transportation operators), as a matter of contract, to fund port operations or lessen the impacts from those operations. However, that argument is more difficult to make as the distance of such infrastructure projects and impacts from the ports increases.

9. Signatories include:

- the five Southern California county transportation commissions, including the Los Angeles County Metropolitan Transportation Authority (MTA), Orange County Transportation Authority (OCTA), Riverside County Transportation Commission (RCTC), San Bernardino Associated Governments (SANBAG), and Ventura County Transportation Commission (VCTC);
- the Southern California Association of Governments (SCAG, the metropolitan planning organization);
- the Southern California Air Quality Management District (AQMD);
- the Ports of Los Angeles (POLA), Long Beach (POLB), and Port Hueneme, Oxnard Harbor District (PH);
- key state agencies: Business, Transportation and Housing (CBTH), the Department of Transportation (Caltrans), the Environmental Protection Agency (CALEPA), the Air Resources Board (CARB), and the Resources Agency; and
- key federal agencies, including: the Department of Transportation (USDOT), the Environmental Protection Agency (USEPA), the Army Corps of Engineers (ACOE), the Fish and Wildlife Service (USFWS), the National Oceans and Atmospheric Administration (NOAA), and the National Marine Fisheries Service (NMFS).

10. Ty Schuiling and Jean-Daniel Saphores read and commented on an earlier draft of this chapter, and those comments were very helpful. The authors alone are responsible for the content of this chapter.

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CHAPTER 5

America's Commercial Waterways: An Underutilized Resource

Noël P. Comeaux, AICP



Today, there are more than 25,000 miles of coastal, intracoastal, and inland waterways in the United States that have been providing transportation since before our highways extended from one coastline to another, before our cars and trucks began populating cities and interstates, and before our bridges spanned rivers and lakes.* Our nation's commercial waterways are currently operating at less than capacity but can help get commodities to their destinations faster, easier and with less indirect effects than current means. Trucks can roll directly onto scheduled services (thereby not adding to the average annual daily traffic), and trains can move containers to marine terminals where river- and oceangoing barges are waiting to be loaded and unloaded and where shippers can specify short sea services to move freight inland.

* This article is based on the professional expertise and opinions of Noël P. Comeaux from his 15 years of maritime planning and security experience. It does not represent the views of the Maritime Administration or any other government agency or organization.

There are currently more than 25 intermodal (container-base) maritime services operating in the United States, not including clean-diesel ferry services in more than 20 states.¹ The question is how to better integrate America's commercial waterways into the surface transportation system. Prior to the expanded development of bridges, which made car and truck travel easier, America's waterways served an important purpose: they connected "disconnected" land masses. In the early 20th century, boxcars were transported by barge on the East River in New York. In addition, the site of the current Maritime Museum in San Francisco was home to a thriving vehicular terminal that could get one across the bay to either Oakland or Sausalito. Yet in the 1950s, America began developing more highway infrastructure to improve the economy, provide for urban evacuation, improve the transport of freight between urban metropolitan areas, and improve local economies. This development was beneficial, yet the role of maritime transportation in an integrated intermodal system has been overshadowed by higher-profile investments in road and rail. Ironically, the neglect of maritime transportation has contributed to congestion on highways, urban streets, and even rail corridors.

Urban Corridors

Prior to 1998, trucks used Tchoupitoulas Street or Louisiana Avenue, a historic corridor in the Upper Garden District of New Orleans, to access Port of New Orleans' facilities along the riverfront. In 1998, the port developed the Clarence B. Henry Truckway, a dedicated truckway on the opposite side of the floodwall from Tchoupitoulas Street, a corridor that stretches almost the entire length of Uptown New Orleans. (See Figure 5.1.) Trucks stopped using historic neighborhood streets to get to Elmwood Industrial Park in Jefferson Parish, near Harahan, but instead began to use the Pontchartrain Expressway and I-10, creating a safer environment for uptown residents and visitors.

Figure 5.1. Trucks on the Clarence B. Henry Truckway, New Orleans.



Chris Bonura, Port of New Orleans

Waterway Transportation between Cities

Trucks can be removed from urban expressways or interstates using scheduled roll-on/roll-off (Ro/Ro) vessels. Opportunities include the Gulf Intracoastal Waterway and Mississippi River from Baton Rouge and Memphis and the service Osprey Lines offers from the Port of Houston to New Orleans. Ro/Ro operations can divert a share of truck traffic from highways to commercial vessels. For example:

- Currently, the Average Annual Daily Traffic (AADT) for I-64 in southeast Virginia is about 40,000 to 80,000 vehicles. A new service, the “I-64 Express,” carries containers on barge on the James River between Hampton Roads to Richmond, removing containers from the highway corridor.
- SeaBridge recently started a container-on-barge service from Brownsville, Texas, to Port Manatee, Florida, replacing trucks on I-10.
- A float operation moves railcars on barge between Greenville, Jersey City New Jersey, and Bay Ridge, Brooklyn, New York.

International goods can be moved by a commercially navigable waterway, providing additional available capacity to the complete surface transportation system. In 2007, the California intrastate trucking market generated 1.27 million trips—or 41 percent of the truck trips in the I-5 corridor—between the Los Angeles and Oakland areas, signifying a need for a commercial waterway ser-



Figure 5.2. Cars loading onto a ferry.

vice.² A Ro/Ro operation could minimize loading and unloading time, reduce congestion on I-5, reduce accidents in the corridor, and improve the air quality for Oakland and Long Beach (both of which are classified as “nonattainment” areas, meaning they do not meet federal air-quality standards).

Most United States waterways that can be transited by commercial vessels are currently underutilized. As denoted by the Federal Highway Administrations’ Freight Analysis Framework and the report “Estimated Cost of Freight Involved in Highway Bottlenecks 2006,” many urban interstate corridor intersections are at or above capacity, needing funds to accommodate trucks.³ Further, key East Coast and Gulf Coast rail corridors are operating above capacity, and some railroad right-of-ways are becoming bike trails or becoming obsolete, further reducing the capacity of our total surface transportation system. For example, the Missouri River between St. Louis and Kansas City offers great potential for commercial navigation. I-70 incurs greater than 160,000 AADT near St. Louis, contributing to congestion, delays, and local air-quality issues. St. Louis is an EPA-designated nonattainment area, per Particulate Matter 2.5 standards. The Missouri River can decrease I-70 AADT and eventually help make St. Louis an attainment area.⁴

The basic economic factors of transportation—time and money—are ones that shippers consider in moving products throughout the United States. The system supports both international and domestic cargo movements (including via ferry services such as a new potential service between Portland, Maine, and Nova Scotia, Canada), but while trucks offer more flexibility, rail and water are options to move international goods inland or domestic goods interregionally. Currently, most ports and some distribution centers

have direct rail access, due to the historic focus on rail (for long hauls). Yet it is becoming evident that shippers understand the benefits of commercial waterways by considering them for distribution.

Port Infrastructure

While there is no ideal port terminal in design, Ro/Ro operations seem to provide the best solution to extending the surface transportation system to America's commercial waterways. While these facilities cost taxpayer dollars, they are not associated with additional roadway congestion and are ideally located in the same port facility as cargo entering from overseas. Trucks can move from distribution centers to ports for scheduled weekly service, minimizing storage fees, the cost to transfer products to containers, and idle time. Environmental benefits can also be significant, as seen by the use in distribution centers across the United States of marine terminal equipment that is powered by electricity (e.g., container-gantry cranes) or compressed natural gas (forklifts, rubber-tired gantries, etc.). For example, Tote Marine Services currently provides Ro/Ro service from Tacoma to Alaska in just over three days, helping lower shipping costs by not incurring U.S. Customs fees through the Port of Vancouver. And the trailers that would otherwise move via a 2,422-mile (60-hour) trek over Highways 97 and 29 are instead moved in a more direct and efficient water route along the U.S. and Canadian coastlines.

Another option to minimize infrastructure investment is direct ship-to-barge (midstream) transfer, as seen in Asia. Either the barge or ship must be outfitted with a crane. These types of transfers are Lift-on/Lift-off (Lo/Lo), instead of Ro/Ro. However, Lo/Lo is typically associated with higher transport costs from origin to destination due to additional lift fees, which are passed onto consumers.

CONCLUSION

Urban planners should do their part by adding or expanding the role of commercial waterways in their state, regional, and local transportation improvement plans. They can support projects professionally, publicly, and financially, helping improve traffic and thereby air quality in their own urban areas; this also means safer roads, potentially less energy expended, and even reduced infrastructure costs. What results is a net improvement for the entire surface transportation system and not just for bicyclists, pedestrians, and commuters, which are typically the focus of urban planners.

It is time to recognize that local or even regional commercial waterways are a vastly underutilized resource. Incorporating waterways into integrated transportation planning can help reduce freight shipments on other modes, reduce traffic congestion, increase safety, and improve the environment.

ENDNOTES

1. See www.eco-transport.com/uploads/DOT%20Maritime%20Administration.pdf.
2. See the Federal Highway Administration, Freight Management and Operations' Freight Analysis Framework, available at http://ops.fhwa.dot.gov/freight/freight_analysis/faf.
3. See http://ops.fhwa.dot.gov/freight/freight_anaylsis/faf/index.htm and www.fhwa.dot.gov/policy/otps/freight.cfm.
4. It is important to note that truck lanes are currently being considered between St. Louis and Kansas City, Missouri (see www.improvei70.org), yet marine diesel engines can be operated on low sulfur fuel, ultralow sulfur fuel, CNG, and Liquefied Natural Gas (LNG), exemplified by deepwater LNG vessels. EPA's Clean Diesel Program supports these alternatives. Also, on October 13, 2008, the Louisiana Department of Environmental Quality (LDEQ) announced it would receive \$295,320 from the EPA to retrofit two ferries for clean diesel (see www.dotd.louisiana.gov/pressreleases/Release.aspx?key=939).

CHAPTER 6

Sustainable Streets: An Essential Emerging Practice

Ellen Greenberg, AICP



This chapter is about moving toward the next frontier for the public right-of-way.¹ For decades, American practice has been in a state of transition, as professionals and community leaders have struggled to achieve harmony between the goals of community on the one hand and societal demands for ever-increasing mobility on the other. Moving forward means embracing the progress of the past decades in recognizing both the value of urbanism and the necessity of context-sensitive solutions. It also means incorporating into street design, management, and operations meaningful responses to the urgent agendas of reversing global climate change and achieving environmental quality, social equity, and robust economies. This next frontier is the creation of sustainable streets.

Moving Toward Sustainable Streets

Sustainable design principles are in widespread use throughout the design professions. These principles generally focus on long-term energy efficiency and waste reduction (see, for example, Brown et al. 2005). However, sustainable streets must go further than incorporating generalized principles of sustainable design by responding to the unique role of public streets in networks that shape human settlement and movement patterns. A provisional definition for sustainable streets is:

Sustainable Streets are multimodal rights-of-way designed and operated to create benefits relating to mobility, ecology, and community that together support a broad sustainability agenda embracing the three E's of environment, ecology, and economy.

This definition offers a preliminary framework for design professionals, decision makers, and members of the public that can be used as a basis for selecting objectives and features of sustainable streets as well as for policy making, priority setting, and evaluation. This inclusive approach to defining sustainable streets is borne out by the objectives and features identified by the sustainable streets review of practice that is the focus of this chapter (Table 4.1).

TABLE 4.1. SUSTAINABILITY OBJECTIVES AND FEATURES FROM SUSTAINABLE STREETS REVIEW OF PRACTICE

Movement Objectives	Movement Features	Ecology Objectives	Ecology Features	Community Objectives	Community Features
• Increased pedestrian activity • Reduced vehicle miles traveled through increased transit and nonmotorized mode share and shorter average trip length • Increased transit use • High level of overall accessibility • Increased bicycle use	• Highly connected multimodal network • Supporting land-use pattern with diverse activity in close proximity and connected by multimodal system • Vehicle speeds compatible with context, community objectives, and energy-efficient operations • Facilities for transit services, safe and convenient bicycling, and walking • Midblock pedestrian paths	• Improved air quality • Water conservation; improved water quality; compliance with NPDES permitting requirements • Enhanced ecological health and productivity • Mitigated urban heat island effect • Efficient use of energy and resources overall • Reduced greenhouse-gas emissions	• Tree preservation; newly planted trees • Biofiltration (variety of features) • Permeable paving • Use of energy-efficient traffic signals and lighting • Preserved natural features and processes • Use of dark-sky lighting	• Placemaking: identity, distinction, and beauty through natural and built features • Supporting compact and infill development • Local priorities, traditions, and resources reflected in design • Positive public-health outcomes: physical activity, safety, noise exposure • Sociability and community life • Creating value for adjoining properties • Significantly reduced travel-related injuries and fatalities	• Public gathering places, parks, outdoor dining • Narrowed travelway width; “road diets” reducing number of travel lanes • “Shared streets” and “festival streets” designed for community events • Custom lighting, furnishings reflecting local design traditions • Access to adjoining properties • Landscaping with native plants • Improved aesthetic environment

This chapter is based on a one-year project investigating the emerging practice of sustainable street design, sponsored by the University of California–Davis’s Sustainable Transportation Center’s Visiting Practitioner program.² Project research included documentation of 12 exemplary projects, which are discussed here.

The mechanisms by which streets can contribute to sustainability objectives are straightforward in some cases and highly complex in others. For example, a chain of effect associated with installation of stormwater-management

features such as pervious paving offers direct and reliable ecological benefits associated with reduced urban runoff. This predictability contrasts with the more complex and less reliable chain of effect required to increase transit, walk, and bike mode share—intermediate outcomes ultimately associated with reductions in vehicle miles traveled (VMT) and associated benefits relating to carbon emissions, energy savings, and air and water quality.

This greater complexity generally stems from the fact that human factors play a significant role in outcomes and consistently correlate with less reliable results. This is particularly true when benefits are contingent on a specific behavioral response to the built environment, such as mode shift out of single occupant vehicles (SOVs). This complexity is just one reason that sustainable streets should be viewed and implemented as one component of integrated sustainability programs that comprehensively address development, conservation, and movement at multiple geographic scales.

Sustainable Streets: An Emerging Practice

Many localities are in planning and design phases of sustainable streets, a small number of localities have pilot projects under way, and an even smaller number are actively working to change their standards so that sustainable streets will become business as usual in their agencies. Nationally, a small number of early adopters are incorporating very innovative work in numerous projects. Chicago, Seattle, and Portland stand out. (Each of these cities offers in-depth material on their websites, which readers can consult for additional documentation of projects and methods.) In addition to sponsoring infrastructure projects, early-adopting agencies are incorporating sustainable-street elements into design guidance, codes, and regulations. Because projects are new and applications still limited, methods for measuring performance specifically associated with sustainable-streets objectives appear to not yet be well established.

Examples

Examples of sustainable streets can be organized into four groups in recognition of their principle objectives:

- Neighborhood Plus
- Revitalization Plus
- Stormwater Plus
- Movement Plus

Collectively, these groups of projects are notable because they provide a shift away from a design paradigm in which the transportation functions of the street are the overarching determinant of design. The emerging view is of streets as multifunctional spaces that serve community and natural resource needs as well as a mobility infrastructure role. In some cases multiple functions readily coexist because travel volumes are low or rights-of-way are generous. In other cases, creative design strategies are being used so that space can be allocated differently at different times—for example, using a curb lane for parking at some times and for outdoor dining at others, as in the case of downtown Livermore, California, described below. Where space is available, sustainability features can be designed directly into the street, as is shown in several of the examples. The City of Chicago's Green Alleys Program (Chicago Department of Transportation 2007) is a result of a conscious decision to focus ecological improvement on the parts of the circulation network on which there is the least vehicle travel demand and where property owners would be likely advocates for a green-streets approach that minimizes the threat of flooding to their homes.

Neighborhood Plus. Three projects at the neighborhood scale illustrate the use of a variety of features. Each was designed and implemented to emphasize multiple objectives, from a tripling of density on an infill site at Garrison Woods in Calgary, Alberta, to fully embracing low-impact development principles at Pringle Creek in Salem, Oregon, to a comprehensive approach to sustainability at New Columbia in Portland, Oregon. In each of these neighborhood-scale projects, the street plan is one part of the broader approach to achieving the development program.

In Garrison Woods, a 165-acre former army base has been reused for compact residential and mixed use development. The development has an urban character and is within the inner-ring suburbs around Calgary's downtown. Key aspects of the plan are increased residential density and street connectivity, along with customized design of all streets. The project increased the total dwelling unit count onsite from 600 to 1,800 units. Designers customized the street cross sections to make a network of safe pedestrian and biking opportunities. The system was designed without a clear-cut road hierarchy. The street network offers multiple connections to adjacent neighborhoods to allow for easy, short trips in and out of the development. Site planning preserves established trees in private yards. Traffic calming is offered by neighborhood design. With respect to the community dimension, Garrison Woods illustrates how a site's history can be celebrated and used to enhance the community identity of a contemporary infill project—in this case through emphasis on the military history of the property and of Canada as a whole.

New Columbia is a HOPE VI project initiated by the Housing Authority of Portland (HAP). The project transformed an 82-acre public housing site into a mixed use, mixed income community. The street plan was intended to help facilitate the social and economic integration of the site into the greater community, as well as to maximize connectivity and encourage sociability and community by creating a "safe" place for pedestrians and cyclists. The primary objective of the redevelopment was to reduce the concentration of poverty by building a neighborhood of varied types of housing (rented and owned) and to attract residents of different income levels. A secondary objective was to redevelop the site in the most ecologically sustainable way possible, in particular by effectively managing stormwater onsite. Inclusion of space for small-scale retail businesses as well as a school and other public services, a central park with community garden and basketball courts, and TriMet bus routes through the neighborhood all support conditions for short trips and a high level of non-SOV use. This is further supported by the project's location close to downtown Portland's job, shopping, and cultural destinations. New Columbia's street plan features:

- A high degree of connectivity within the site and to the surrounding neighborhood, meeting City of Portland connectivity standards through a combination of streets and midblock pedestrian pathways
- "Skinny" streets and alleys, including alleys with a pervious central paving strip
- Stormwater management features in travel lanes (permeable paving), park strips, and parking lot perimeters (rain gardens)
- Tree preservation

Downtown Revitalization Plus. Revitalization projects are focused on Smart Growth principles—in particular, facilitating successful infill development. Each treats street design as a key element of intensification and creation of walkable urbanism. Movement and community objectives are thus central to these projects, while the inclusion of ecological function in the right-of-way is a secondary consideration or not a consideration at all.

Each example in this category makes street design one part of a multi-pronged effort. The overarching objective of a major civic effort in Chattanooga, Tennessee, was supporting compact development and reuse of land while revitalizing the downtown. In Chattanooga's Riverfront Parkway and Downtown Streets project, a five-lane limited-access highway was narrowed to a two-lane surface street with an adjoining riverfront esplanade and realigned to create sites for new housing. On-street parking and crosswalks were added, travel lanes narrowed, and posted speed limits lowered for a new emphasis on livability rather than the accommodation of through-travel movement. In the center of downtown, two one-way couplets were converted to two-way streets, with parking and landscaping added. These improvements were included in an ambitious strategy to attract investment to downtown Chattanooga.

In downtown San Jose, California, as in many other locations nationally, revitalization has emphasized new transit services along with housing and civic buildings. San Jose's innovative design places transit in the pedestrian zone, with light rail running at sidewalk rather than carriage-way level. The San Jose experience highlights the need to link transit-oriented street designs to land-use and demand-management programs that will spur ridership.

Many revitalization programs, particularly in smaller cities and towns, focus on the redesign of main streets. "Road diets," angled parking, period lighting, custom furnishings, and landscape designs are typical elements. The Livermore, California, example includes a more dramatic "Main Street" design along First Street, the downtown spine (Figure 6.1). The streetscape was a "catalyst project" recommended by the city's 2004 Downtown Specific Plan (Livermore, California 2004), which set out a program of actions for the city to establish a compact, livable town center and a vital downtown. In this



Gregory Tung, Freedman Tung & Sasaki. © Freedman Tung & Sasaki (www.ftscities.com)

Figure 6.1. Livermore's First Street, between J and K streets, after redesign by Freedman Tung & Sasaki (2007). State highway CA-84 was rerouted off of First Street, which was redesigned as a two-lane street with "flexible parking zones" that are convertible from angled parking to outdoor café use.

project, vehicle capacity took a backseat as the main street was redesigned to function as a community centerpiece intended to stimulate investment, business activity, and the community's social life. The First Street Project reduced four vehicle lanes to two, creating curbside "flexible zones" that can accommodate either angled parking or outdoor dining as needed by revitalizing businesses. The perception of territory dedicated to cars is reduced with the use in the curb lane of pedestrian paving, newly planted street trees, and architectural trellises in the "flexible zones" that enhance the street's comfort and appeal.

Stormwater Plus. Best practices for stormwater management are most frequently associated with sustainable-street design in practice. The National Pollution Discharge Elimination System Municipal Separate Storm Sewer System (NPDES MS4) permitting program is a key driver of municipal stormwater-management activity (U.S. Environmental Protection Agency 2008). Often, natural drainage is implemented on facilities where travel volumes are low and allocating space in the public way to ecological functions does not require a change in the transportation design or function of the street. A number of projects are modest in scope but are testing new designs and functions for wider use. On a single block face of 12th Avenue in Portland, Oregon, for example, preexisting landscape space was replanted as rain gardens. Likewise, Gresham, Oregon, is using modest projects as pilots and opportunities for community education. The city's pilot project retrofitted a local street, supplementing a failed drywell infiltration in a community with inadequate stormwater-capacity system with three rain gardens and one parking strip of porous asphalt. In northwest Seattle, the Broadview Green Grid project installed extensive stormwater-management features in a 10-block area, adding sidewalk and retaining preexisting vehicle space. These improvements were possible because the full width of the right-of-way had not previously been improved.

On 12th Street in Kansas City, Missouri, a downtown streetscape project integrates the mobility, ecology, and community dimensions of sustainable streets along seven blocks of the city's main east-west corridor, traversing the downtown financial and government districts. The project includes stormwater management and landscape features as well as a redesigned roadway, reducing the number of travel lanes from four to three and adding on-street parking. An added community benefit is the introduction of trees and other landscape features into what had been a bare downtown environment.

A more dramatic example of an integrated approach to sustainable streets is being undertaken as part of a redevelopment project in the Denny Triangle neighborhood of Seattle. Taylor 28 is an urban infill project in which green building is integrated with a street redesign including stormwater features and public gathering space (Figure 6.2). Stormwater-management features are designed to result in zero discharge during a 25-year storm event within

Figure 6.2. In this rendering, the Taylor 28 landscape plan is shown from a pedestrian point of view.



the plaza street zone. To create this zone, Taylor 28 reallocates street space from the travelway to the pedestrian realm, along with a shift from parallel to angle parking.

Movement Plus: Projects on Arterial Streets. Arterial streets pose a particular challenge to reconciling the themes of mobility, ecology, and community. Essential movement functions are served with long-distance capacity and connectivity. High volumes often are perceived as eliminating any possibility of reallocating space to either community or ecological functions. The adjoining roadside typically is the place most strongly affected by the negative impacts of vehicle noise and emissions, which can discourage pedestrians and make adjoining properties difficult to design and develop. In the worst cases, arterials lack even continuous sidewalks that offer basic provisions for pedestrian safety.



Figure 6.3. This rain garden along NE Sandy Boulevard in Portland, Oregon, is one of several that were designed to manage stormwater, improve pedestrian safety, and help green one of Portland's busiest urban streets.

Sustainable-street projects on arterials can be classified into two types. The first are arterial designs that are largely remedial. These focus on safety and access for pedestrians and bicyclists, access management, and aesthetics. In one example—Aurora Boulevard in Shoreline, Washington—stormwater management was also integrated into the design. These projects result in a facility providing essential multimodal accommodation and safety improvements supporting low-impact, healthy travel along a high-volume route. They are not directly associated with changes to land use or development policy. Their near-term impact on walking and biking is likely to be limited by dominant land-use patterns that are characterized by low densities, long distances between origins and destinations, and auto-scale design. Examples also include Sandy Boulevard in Portland, Oregon, and East Market Street in Greensboro, North Carolina, with the latter including a “road diet” from a six- and seven-lane cross section to a four-lane cross section with landscaped medians. (See Figure 6.3.)

The second category of arterial projects represents more comprehensive efforts that are using street design as one element in improving the corridor right-of-way and supporting revitalization of adjoining properties or districts. Chattanooga's Riverfront Parkway is one such example. In that example, thoroughfare changes are seen as setting the stage for desired land-use changes. Another is the redesign of Pacific Boulevard in the False Creek neighborhood of Vancouver, British Columbia, following the creation

of a very high-density residential district. Pacific Boulevard's reconstruction is intended to activate and green the street while improving facilities and conditions for transit, bicyclists, and walkers; help complete Vancouver's "Living First" strategy by supporting compact residential development; and reduce the urban heat island effect by increasing the city's urban forest.

Supporting an Evolving Practice

Sustainable streets are evolving in both concept and practice. Their development, implementation, and ultimate success in advancing sustainability objectives requires support. Needed are: (1) an expanded knowledge base for sustainable streets; (2) an improved palette of design techniques for each of the three dimensions of movement, community, and ecology; and (3) new planning tools that support sustainable streets in practice. Each of these is discussed briefly below.

Knowledge Base. The most significant need is for data on the effectiveness of specific techniques in all areas of sustainable-street design and operations. A robust knowledge base will be multidisciplinary, addressing topics as diverse as the long-term performance of pervious pavement materials and the impact on real estate values of streetscape improvements. Understandably, there is also particular interest from public agency personnel in having available data on implementation cost, maintenance requirements, and performance over time with respect to the full range of sustainability concerns, as well as information on roadway operations and safety concerns. (These and other implementation issues are investigated in U.S. Environmental Protection Agency, Smart Growth Division 2009.)

A subject of particular importance, as well as complexity, is the role of the street in contributing to transportation performance as measured through the lens of sustainability. Metrics of particular interest are those that are associated with reduced vehicle use and associated pollutant and CO₂ emissions, such as non-SOV modes, VMT, vehicle trips, and vehicle delay. There is increasing acceptance of the fact that, on a per capita basis, compact communities produce lower levels of VMT and vehicle trip making. Higher-density development patterns correlate with reduced vehicle travel per person, despite higher VMT on a per-square-mile basis (Manville and Shoup 2005). Relationships between compact urban form and travel behavior have been studied in great detail in recent decades. Reviews of the literature on these topics (e.g., Saelens and Handy 2005 and Krizek 2005) point to the complexity of the factors involved but nonetheless support practical efforts to use planning and design strategies to influence transportation outcomes.

Agencies and individuals advancing the practice of sustainable streets need to know specifically about the contributions of street and network design to supporting different types of places that, in turn, offer sustainability benefits stemming from their travel performance characteristics. Among the factors of interest, a number of roadway network characteristics have been widely studied. These include sidewalk completeness and street connectivity (Saelens and Handy 2005).

Synthesizing the results of academic and agency research into a practical application for use in scenario planning, the "Index 4D Method" developed for the U.S. EPA (Fehr & Peers and Criterion Planners, 2006) has quantified the relative impact on travel performance of four factors labeled design, density, diversity (land-use mix), and destinations (regional accessibility). The design variable exclusively represents street and network characteristics, reflecting the elements of street network density, sidewalk completeness, and route directness. The analysis indicates that route directness is the most powerful of the three design-factor elements, and a location's design index is negatively correlated with trip making and VMT.

While network and sidewalk connectivity have been addressed extensively, the street designer is faced with far more choices that might have a bearing on sustainability outcomes, such as total travelway width, number and width of travel lanes, and inclusion of buffers that separate the pedestrian realm from moving traffic. For example, it is axiomatic among practitioners focused on placemaking that narrow streets with easy crossing distances are superior—or even essential—for walkable downtown shopping and residential districts that encourage transit use and discourage vehicle use while offering high levels of accessibility. A number of investigations have focused on how such features influence travel, respond to preferences, and support local economic activity (Transit Cooperative Research Program 1998). Less well researched is which among these features is successful in supporting urban places that have higher-than-typical nonmotorized mode share and lower-than-typical VMT per capita.

With public agencies sponsoring street design efforts in pursuit of a very wide range of objectives, increasing evidence is needed about the role of the street and the street network in supporting places where land use, urban design, and circulation system characteristics combine to reduce per capita SOV use, measured in miles or vehicle trip making. Cities such as Washington, D.C., are tracking the economic impact of their street design investments, keen to demonstrate positive impacts and cost effectiveness. In the United Kingdom, the Commission for Architecture and the Built Environment has sponsored a study designed to show how to measure the impact of street design improvements on market prices as revealed through retail rents and residential flat prices (Commission for Architecture and the Built Environment 2007). Their methods may have relevance for work in the United States and elsewhere seeking to relate fine-grained street characteristics to both travel and economic performance.

Design Innovation. New objectives for how streets function are stimulating design innovation, as illustrated by the examples described briefly above. These innovations, in turn, will require testing over time for their performance and cost characteristics. Where designs change roadway characteristics, safety and operational impacts will be of particular importance.

Innovation in street design is particularly challenging because of the practice's reliance on design guidance published by a small number of organizations (notably the Federal Highway Administration, the American Association of State Highway and Transportation Officials [AASHTO], and the Institute of Transportation Engineers) with wide influence on standards adopted by state DOTs and local governments. As noted above, design creativity is being prompted by support from all three entities for the use of flexibility in highway design and by the expanding application of context-sensitive solutions processes.

The sustainable streets projects reviewed illustrate a number of innovative design elements in context. Some, such as the use of a single moving lane (10–14 feet wide) on local streets, are relatively well established (National Association of Home Builders et al. 2001). Others, such as the use in Vancouver of the multiway boulevard cross section, have only limited recognition in professional design guidance. Still others, such as the flexible use of the curb lane in Livermore, are not yet documented in national design guidance. Additional techniques illustrated among the examples cited are based on the creativity, judgment, and experiences of the sponsoring agencies and originating designers. The utility of these designs for meeting objectives relating to movement, community, and safety needs to be investigated, and safety and performance characteristics need to be documented.

Further innovations, ideally leading to updated design guidance, will be needed to support changes in vehicle management and technology. These are likely to include implementation of electric vehicle-charging units and

shared bike programs that demand both space and infrastructure in the right-of-way. Early adopters of these systems overseas and in the United States are already pioneering these techniques. Innovations in bicycle facilities may also be informed by international practice.

New Planning Tools

The concept of using street design to advance a wide range of sustainability goals represents a sea change. Mainstreaming this concept into the practice of street design and management will require that projects advance from the ad hoc practices seen in the Sustainable Streets Project research to consistent and systematic efforts. Basic planning tools such as thoroughfare classification need to evolve in order to support this change. Because it serves as a guide to funding, design, operations, and communication of policy intent, functional classification is a particularly rich area for innovation. In fact, innovation is occurring, particularly in connection with the merging of movement and community functions.

The 2004 edition of AASHTO's *Guide to Geometric Design of Streets and Highways* ("The Green Book"; American Association of State Highway and Transportation Officials 2004) established the basic three-tiered thoroughfare hierarchy that is in widespread use, identifying local, collector, and arterial streets. In the context of innovative planning activities, localities such as Denver and Seattle have elaborated on this basic framework with classification systems that reference land use or community-character types. State agencies including Caltrans and New Jersey DOT have recognized that certain thoroughfare types—particularly the downtown main street—need to be designed to support their placemaking qualities, and that standard designs reflecting their place in the hierarchy of functional class are inadequate; special design guidance is needed (see, e.g., Ewing and King 2001).

The Institute of Transportation Engineers' context-sensitive solutions (CSS) guide for walkable urban thoroughfares (Institute of Transportation Engineers 2006) emphasizes a new approach to functional classification as fundamental to supporting walkable urbanism. The guide uses a hybrid system that retains AASHTO's functional classifications as complemented by a thoroughfare-type designation of street, avenue, or boulevard, each defined in the document.

These innovations in the area of functional classification demonstrate the ability to change an established system in order to reflect emerging values. However, the surge of interest in supporting sustainability aims has outpaced the development of the CSS guide, and ecological functions are not included in the functional classification proposed by the document.

Agencies responsible for planning and implementing projects could use an expanded classification system to identify the mix of functions for a range of thoroughfare types. A functional classification system for sustainable streets should:

- Identify the role of the individual facility within the circulation network, in terms of connectivity and relative capacity
- Identify functional emphasis for facilities with particular characteristics or contexts (e.g., busy downtown streets would be likely to place top priority on community features, while high-volume highly connected streets would have a through-movement focus, and local residential streets might provide opportunity for equal attention to (slow) movement, ecological functionality, and the establishment of a positive setting for social activity
- Establish baseline standards for each of the three functional areas:
 - Movement: a high level of connectivity (except where prohibited by natural barriers) and a strong complete streets commitment

- Community: continuous sidewalks, use of native planting, noise and visual buffering between sidewalk and travel lanes
- Ecology: specified levels of onsite filtration, use of recycled materials, energy-efficient lighting, and infrastructure-supporting emerging vehicle technologies

(Design of individual facilities would have to meet baseline standards and could surpass them whenever possible.)

- Anticipate emerging technologies and practices with implications for street design, ranging from short-term storage space within the right-of-way for shared-use vehicles (such as loaner bikes and neighborhood electric vehicles and public recharging stations) to landscape design for food production and habitat value
- Be used as a systematic tool for funding, design, implementation, and management of the street

CONCLUSION

The practice of design of sustainable streets is still in its infancy, though it rests on many established trends emphasizing livable and context-sensitive streets. A few early efforts, such as the City of Chicago's Green Alleys program, are components of ambitious, comprehensive efforts to minimize the impacts on the environment of urban development while improving the community and economic life of the city. With as much as 40 percent of carbon emissions attributable to the transportation sector, and with 20 to 30 percent of total land area in many cities under public ownership and control as street right-of-way (Manville and Shoup 2005), street design is a worthy element of such efforts. Moving from ad hoc efforts to systematic ones depends on continued evolution of design and planning tools, as well as on an expanded knowledge base that provides information about the effectiveness of the strategies available.

Sustainable streets can create harmony among the goals of community, the demand for mobility, and the mandate for environmental stewardship. Pioneering public agencies are already pointing the way to a new practice that embraces these aims. Support from the research community can help build a foundation for long-term success of this essential emerging practice.

ENDNOTES

1. This paper is based on "Sustainable Streets: Foundations for an Emerging Practice," presented at the Transportation Research Board 2009 annual meeting. The original paper and the research on which it was based were completed during the author's appointment as Visiting Practitioner at the University of California at Davis' Sustainable Transportation Center during 2007–2008. The author gratefully acknowledges Dr. Susan Handy, director of the Sustainable Transportation Center at the University of California–Davis, for her support of the Sustainable Streets Project and her comments on this article in its original form. Morgan Kanninen did an extraordinary job as project researcher, and her assistance is deeply appreciated. The UC–Davis team is grateful to all of the individuals who contributed information and images of their organization's work in order to help us understand and document the emerging practice of sustainable streets. Clark Wilson and John Thomas of the U.S. EPA Smart Growth Office have also provided support and feedback in connection with this work.

2. For a more complete description of project methods and results, see "Sustainable Streets: Foundations for an Emerging Practice," in *Proceedings of the 2009 Transportation Research Board Annual Meeting*, available at <http://pubs.its.ucdavis.edu>.

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CHAPTER 7

Reducing Greenhouse-Gas Emissions from Transportation

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Transportation accounts for approximately one-third of greenhouse-gas (GHG) emissions in the U.S., two-thirds of oil consumption, and about half of urban air pollution (U.S. EPA 2008; Davis et al. 2008; NRC 2006). Transport GHG emissions are also growing faster than those from any other sector, making transport a prime target for climate policy.

REDUCING GREENHOUSE-GAS EMISSIONS FROM TRANSPORTATION

Taken together, the following two central facts make it extremely difficult to reduce GHG emissions from transport: (1) The transport sector is linked to almost all other economic activity, and therefore demand for transport overall is both strong and inelastic, and (2) Other large energy-using sectors can choose from a variety of fuels that vary in their GHG emissions, whereas the only widely used fuel for vehicles is oil.

Those interested in reducing GHG emissions from transport face additional challenges as well. First, an increasing share of goods movement is by truck, a more energy-intensive mode than rail. Second, introducing new low-carbon fuels and new fuel-efficient propulsion technologies is difficult due to the necessity of coordinating innovation between the fuel and vehicle industries, large up-front infrastructure investment, and the need to overcome entrenched consumer expectations and habits. Finally, petroleum production is becoming *more* rather than *less* carbon-intense as easily accessed and high-quality reserves are depleted, remote sources of fossil energy are tapped, and additional refining is required to upgrade fuel quality.

For all these reasons, cost-effectiveness studies generally find that transportation GHG reductions are more expensive to achieve than reductions in most other sectors, implying that we should hold off on implementing climate policy for the transport sector. Though these challenges are daunting and many trends undermine efforts to reduce GHGs, we argue that these analyses are incomplete and misleading because they fail to consider the cobenefits of policies to reduce GHG emissions from transport and gloss over the huge energy cost savings from minor changes in behavior. Here, we holistically examine strategies to reduce transport GHG emissions and their cost-effectiveness. Our analysis focuses on the two largest components of the transportation sector—passenger automobiles and commercial freight trucks—which together make up more than 80 percent of transportation GHG emissions in the United States (U.S. EPA 2008).

Greenhouse-Gas Strategies

Despite analyses indicating that transport-sector opportunities to reduce greenhouse gas emissions may be less cost-effective than others, there are many reasons why these analyses are misleading. First, there are large, highly valued cobenefits to reducing transport GHG emissions. Most strategies to reduce transport GHG emissions would also reduce petroleum use, thereby contributing to energy security. Most would also reduce emissions of local conventional pollutants, and those that involve reduced vehicle use would reduce traffic congestion and road infrastructure investments. Second, many incremental low-cost technologies exist to reduce energy use in this sector. Innovations in engines, transmissions, aerodynamics, and lightweight materials have continuously yielded greater efficiency. Many additional fuel-saving innovations are being pursued. Third, the automotive industry has become highly competitive, with companies seeking ways to distinguish themselves. The halo created by the successful Prius hybrid has proved extraordinarily valuable to Toyota, suggesting the great value in pioneering high-profile green technologies in the automotive industry. Fourth, historic inelasticity of demand for energy efficiency in vehicles is likely to soften as consumers come to appreciate the attractions of electric-drive vehicles.

The universe of transport GHG mitigation strategies may be grouped into three categories: vehicle efficiency, GHG intensity of fuels, and reductions in vehicle use. We exclude a fourth category suggested by some, system efficiency and infrastructure, because improved vehicle flow provides small GHG benefits and most other systems proposals can be included within our three categories.

Increasing Vehicle Efficiency

Available and emerging vehicle-efficiency improvements can be categorized into three groups: incremental vehicle technologies, advanced technologies, and “on-road” operational practices.

Incremental improvements include more efficient combustion (variable valve systems, gasoline direct injection, cylinder deactivation), more efficient transmissions (five- and six-speed automatic, automated manual, continuously variable), and overall vehicle advances (aerodynamics, weight reduction). GHG emissions rates can be reduced by 20 to 30 percent with these technologies in new vehicles. Most studies show that the lifetime fuel savings from these improvements can more than outweigh the increased vehicle cost, often by a large amount. Similar technology packages yield substantial GHG reductions and net positive benefits for commercial freight trucks as well (Lutsey and Sperling 2009).

Much greater GHG reductions are possible with electric-drive propulsion technologies. These include the increasingly popular hybrid gasoline-electric vehicles, “plug-in” hybrids that use both electricity and petroleum fuels, battery electric vehicles, and hydrogen-powered fuel-cell vehicles. Such technologies can double vehicle fuel efficiency. If low-carbon electricity and hydrogen are used, it is possible to reduce lifecycle GHG emissions from vehicles by 60 percent or more (see, e.g., Kromer and Heywood 2007; Bandivadekar et al. 2008). However, these advanced technologies involve either larger initial costs—for electricity and hydrogen storage—or have high development costs, uncertain future costs, or both. Because vehicle turnover is slow and it takes a long time to deploy a new energy distribution system, it will take a long time to realize the potential reductions (see, e.g., NRC 2008).



Figure 7.1. Part of the endless stream of traffic on the San Francisco–Oakland Bay Bridge.

The third group, “on-road” efficiency improvements, involves a combination of consumer education, vehicle maintenance practices, and “off-cycle” vehicle technologies. Improvements to on-road vehicle efficiency can reduce GHG emissions, perhaps by as much as 20 percent (ECMT and IEA 2005). Improved vehicle maintenance practices in regard to tires, wheels, oil, and air filters ensure vehicles operate at their design efficiency levels. Technologies in new vehicles that aid driver awareness of fuel use include

dashboard instruments that present instantaneous fuel consumption, efficient engine rpm ranges, shift indicator lights, and tire inflation pressure. Other changes include replacing the conventional air-conditioning refrigerant, hydrofluorocarbon HFC-134a, with lower global-warming-potential gases (MES 2004).

There are many possible policies to bring about these vehicle improvements. Many are already in place, though in a limited way. Education and informational initiatives (such as public education campaigns and product labeling requirements) can help, but the most effective policies are likely to be a mix of performance-based regulations and market instruments. For those vehicles operating on nonpetroleum fuels, there will also have to be policies targeted at overcoming start-up barriers to those fuels (discussed below).

The decades of high fuel taxes in Europe suggest that high fuel prices lead to the manufacture and sale of smaller and less powerful vehicles, but that high prices are not sufficient by themselves to accelerate the use of advanced technologies. Despite European fuel prices that are typically two to three times higher than U.S. prices, European cars are remarkably similar to U.S. cars. The major differences are smaller engines, smaller sizes, and greater use of diesel. But the types of engines, their energy efficiency (controlled for size and power output), the use of lightweight materials, and the fuels they consume are essentially identical. Advanced technologies do not come any more quickly to Europe, even with the much higher fuel prices (and often higher vehicle registration and sales taxes), than to the United States. In fact, gasoline hybrid electric vehicles were embraced more quickly in the United States than in Europe. Neither region has seen electric or fuel-cell vehicles commercialized widely, though many small companies have sold small numbers of battery electric cars in both places.

Recognizing the limited effects of high fuel taxes, European policy makers have recently adopted aggressive GHG performance standards for vehicles. In 1998, they adopted a goal of reducing GHG emissions per vehicle-kilometer by 25 percent by 2008 (from approximately 180 grams of carbon dioxide per vehicle-kilometer to 140 g/km). These standards were imposed as voluntary goals on the auto industry. The industry's principal response was to insert diesel engines in vehicles. Because such engines emit about 20 percent fewer GHG emissions per vehicle-kilometer than gasoline engines, they proved effective in realizing improvements. But despite the high fuel taxes and the GHG-reduction targets, vehicles became bigger and more powerful, following a pattern seen in the United States, though less extreme (Zachariadis 2006). The net result is that GHG reductions stalled at about 165 g/km, equivalent to roughly 40 miles per gallon in a U.S. CAFE system (based on An 2007), well short of the 140 g/km target. The European Commission, frustrated by the shortfall, responded in 2008 with mandatory performance standards on automakers, requiring them to achieve 130 g/km by 2015 (European Commission 2008).

The U.S. policy history is quite distinct. Other than a gas-guzzler tax imposed on a small number of high-consuming vehicles (mostly sports and heavy luxury cars but never on SUVs), no effort was made to significantly raise fuel or vehicle prices. Instead, beginning in 1978, fuel economy standards were imposed on vehicles. Those standards essentially flattened out in the late 1980s, with only minor increases in light-truck standards in recent years. In December 2007, Congress passed a law to increase the standards by 40 percent by 2020 (to 35 miles per gallon). The Obama administration just accepted proposed standards in California and at least 14 other U.S. states to effectively increase the standards another 10 to 20 percent (measured as grams of GHGs per vehicle-mile, rather than as mile per gallon).

Low-Carbon Fuels

Increased use of low-carbon fuels—or, more accurately, fuels with lower life-cycle GHGs—can greatly reduce overall transportation GHG emissions. However, most alternative transportation fuels face a combination of infrastructural and economic barriers. The easiest action is to blend small proportions of biofuels into gasoline and diesel fuel. Biofuels are not necessarily less expensive (depending on world oil prices and biofuel technologies employed), but the processes for converting abundant agricultural feedstocks such as corn and sugarcane into ethanol are well known, and ethanol is easily blended into gasoline for use in conventional vehicles. The life cycle GHG emissions of sugarcane conversion can be substantially lower than gasoline, but corn ethanol achieves only about a 10 to 20 percent improvement (Farrell et al. 2006; Wang 2005; Hill et al. 2006; CARB 2009). Future biofuels made from agricultural residue or cellulosic energy crops could have life cycle GHG benefits of 70 to 95 percent higher (Wang 2005; Unnasch et al. 2007; Farrell and Sperling 2007). A similar array of biofuel feedstocks can be used to produce biodiesel, which can be mixed into conventional diesel fuel. These types of biofuels can be mixed with gasoline or diesel in proportions of up to 10 percent for use in conventional vehicles and up to 85 percent in vehicles with slightly modified engines.

There are also other transport fuel options involving wholly different fuels and fuel distribution systems that can greatly affect GHG emissions. Marginally lower GHG-producing fossil fuels like compressed natural gas and liquefied petroleum gas have continued to make small contributions to transportation, mostly in fleet vehicles. On the other hand, next-generation fossil fuels produced from oil shale, coal, and tar sands would have much higher GHG emissions than conventional petroleum—unless the carbon from production of such fuels was captured and stored underground (e.g., Jaramillo et al. 2008).



Figure 7.2. Electric-powered transportation is only one of the options we need to explore in reducing GHG emissions.

Large potential GHG benefits can be achieved by powering vehicles with hydrogen (and fuel cells) and electricity (with “plug-in” hybrids and battery electrics). Electric-drive vehicles, powered by low-carbon versions of these fuels (made with biomass, wind, and nuclear energy or with fossil energy coupled with carbon capture and storage) could yield much greater GHG reductions than vehicle efficiency improvements alone.

Governments around the world have sought time and again to promote the use of nonpetroleum alternative fuels, usually for energy security or local air-pollution reasons. The United States and the world have caromed from one alternative to another, some gaining more attention than others but each one eventually faltering. These have included ethanol, methanol, compressed and liquefied natural gas, battery electric vehicles, coal liquids, and hydrogen. In the United States, the *fuel du jour* around 2002 was hydrogen; a few years later it was corn ethanol; in 2009 it is electricity for use in plug-in hybrid electric vehicles. Worldwide, the only nonpetroleum fuels that have gained significant market share are corn ethanol in the United States and sugar ethanol in Brazil. Corn ethanol use exceeded nine billion gallons in the United States in 2008, representing about 4 percent of gasoline use on an energy basis (Davis et al. 2008; RFA 2009). Sugar ethanol use in Brazil was just a bit less in quantity, but because the country uses much less gasoline, this represented about 30 percent of total gasoline consumption (Goldemberg 2006). In both countries, ethanol has benefited from substantial subsidies and a variety of other supportive policies. In the United States, the total subsidies amount to over one dollar per gallon of gasoline replaced (Koplow 2006), approaching \$10 billion in 2008.

Various policy instruments are available and have been tested. Carbon and fuel taxes are compelling. But taxes attract political opposition and public ire and by themselves are of limited effectiveness—unless quite sizable—at least with respect to transportation fuels. In electricity generation, carbon taxes are likely to create changes because electricity producers can choose among a wide variety of commercial energy sources—from carbon-intense coal to lower-emitting natural gas to zero-emission nuclear or renewable energy. A tax of \$25 per ton of carbon dioxide would increase the retail price of electricity made from coal by at least 20 percent, widening its cost differential with clean renewables. This would motivate electricity producers to seek out lower carbon alternatives, resulting in innovation, change, and decarbonization of electricity production.

But transportation is a different story. Given today's transport system, a carbon (or fuel) tax would have to be huge to induce change. Producers and consumers would barely respond to even a \$50-per-ton tax, well above what U.S. politicians have been considering. Oil producers would not respond because they have become almost completely dependent on petroleum to supply transportation fuels and cannot easily find or develop low-carbon alternatives within a short time frame; besides, a transition away from oil depends on automakers as well, who must produce the cars that run on the new fuel.

Drivers also would be unmotivated by a carbon tax. A tax of \$50 per metric ton would raise the price of gasoline only about 45 cents per gallon (based on 8.9 kg of direct carbon-dioxide emissions per gallon of motor gasoline and 1000 kg per metric ton). This would not induce drivers to switch to low-carbon alternative fuels because virtually no such fuels are available. In fact, it would barely reduce their consumption; price swings of more than this amount have become a routine occurrence.

At the other end of the policy spectrum from taxes are fuel mandates. They do not work well either because it is impossible for government to know which fuel to back. The potential for new fuels with dramatically lower emissions is real, but there is no clear winner yet.

Congress imposed a volumetric mandate for biofuels in December 2007, requiring a massive 36 billion gallons by 2022. (Total gasoline consumption at the time was about 140 billion gallons.) The act allows up to 15 billion gallons to be corn ethanol, but most of the rest must be advanced biofuels made from cellulose, algae, waste materials, or other nonfood materials.

Congress added a GHG performance metric, defining advanced biofuels as achieving at least a 50 percent reduction in life cycle GHG emissions and cellulosic biofuels as achieving at least a 60 percent reduction.

Another approach, which we favor, would be to broaden the policy instrument to accommodate all fuels—including electricity, hydrogen, and natural gas—and to set performance-based GHG targets. Under this policy, the winners would be determined by industry and consumers, not government. This concept of a low-carbon fuel standard has been adopted in California (CARB 2009), is being slowly adopted in Europe, and has been proposed as part of climate change legislation in Congress (for more details, see Sperling and Yeh 2009). The concept involves setting requirements for fuel providers to reduce the GHG intensity (life cycle emissions per unit of energy) of their fuels, with an annual tightening of the targets. Setting future targets and creating a transparent GHG crediting system for fuels allows oil companies and other fuel providers to partner and compete to deploy lower GHG-intensity fuels. For example, those corn-ethanol producers who use low-GHG energy inputs and efficient production practices would have lower emissions and could sell excess credits to others. Those who provide advanced cellulosic biofuels, electricity, and hydrogen to vehicles would tend to receive even greater GHG-reduction credits (depending on how the fuels are produced), which they can also sell.

Reducing Vehicle Use

Strategies that affect the transportation choices that individuals make can be sorted into three broad categories: information and communication technologies to provide new and more efficient mobility services; incentives and pricing schemes to encourage the use of less GHG-intensive modes of travel and to discourage higher-emitting modes; and denser land use that more efficiently sorts businesses, residences, and services so as to reduce the overall demand for travel. Here, many of the same technologies and practices that are already used by local governments to manage vehicle travel and traffic congestion also reduce GHG emissions.

Information and communication technologies can be used to simultaneously improve mobility and reduce transport GHG emissions. Incremental enhancements include automating urban traffic signals to streamline traffic and reduce stop-and-go conditions; implementing integrated “smart cards” to facilitate multimodal travel and increase transit use; providing real-time traffic data to traffic managers, vehicle drivers, and transit riders to improve efficiency; and optimizing logistics for more efficient goods delivery. More substantial changes are possible by creating entirely new modes of travel, such as smart carsharing that allows convenient short-term rentals, smart paratransit that provides door-to-door service without advanced reservations, and dynamic ridesharing that facilitates carpooling.

Various incentive and pricing schemes can be designed to reduce GHG-intense travel. Road pricing—either area pricing in city centers or tolls on specific stretches of highway—can reduce both congestion delay and GHG-intensive travel by inducing travelers to carpool, switch to alternative modes of travel, or perhaps even decide to make fewer trips. Pricing of parking can have the same effect as pricing of roads. Revenues generated from pricing policies could be used as a source of funding for complementary policies such as enhancing public transit, reducing transit fares in key areas, or subsidizing carsharing. Additional effective parking policies to reduce transport GHG emissions include installing park-and-ride lots near transit facilities and implementing parking “cash-out” programs by employers (Shoup 2005). Incentives by workplaces to promote telecommuting and carpooling can also help mitigate peak-time congestion travel while also reducing GHG emis-

sions from transport. These sorts of policies, in conjunction with coordinated improvements in transit service, attract travelers to higher-occupancy—and thus lower-GHG—modes.

In the long term, densification of land use may be the most effective way to reduce both the overall demand for travel and the use of GHG-intensive modes of travel. Research shows that residents in more densely populated areas and in areas with better mixes of land uses tend to emit far fewer GHG emissions from their travel (Norman, MacLean, and Kennedy 2006; Ewing et al. 2007; Rodier 2009). They tend to walk more, use transit more, and drive less. This is certainly true when comparing the per-capita transport GHG emissions from built-out dense cities with those from built-out auto-oriented suburbs. However, transforming today's auto-oriented suburbs into tomorrow's low-emission towns and cities is an enormous planning challenge—one that can only happen slowly, with large GHG reductions realized over decades rather than in the next year or two.

Zoning-based policies that local governments could implement include restructuring zoning ordinances to stipulate off-street parking maximums rather than minimums, requiring density minimums rather than maximums, reducing building setbacks, and relaxing building envelopes to more efficiently use space. In many areas, the cost savings from reduced parking requirements in housing developments could make downtown projects more profitable for developers. Mixed-use zoning could be introduced or expanded, allowing shops, offices, and homes to be located in close proximity. Transit-oriented development can have an impact as well. These tools can be used to encourage mixed use, dense, and transit-, bike-, and pedestrian-friendly urban environments that reduce GHG emissions by creating environments in which people simply demand less travel.

In addition, there is a small but growing movement toward the use of “soft” transport policy measures to reduce vehicle kilometers traveled and GHG emissions from transport. These include measures such as personal travel plans, which research has shown can reduce vehicle travel by as much as 10 percent, reducing both GHG emissions and the time people spend traveling to accomplish their daily tasks (Moser and Bamberg 2008; Taylor 2007).

Summary of Transportation GHG Strategies

In Table 7.1 (opposite page), we categorize transportation-sector GHG mitigation options into near-term and midterm options. For the near term, or “today’s measures,” we highlight options that are currently available and easily implementable but would require implementation of policies or shifts in practice to achieve more widespread adoption. For “tomorrow’s measures,” we highlight those technologies where deeper emissions cuts seem possible but where there is greater uncertainty in how technology costs will drop over time and how industry will act.

CONCLUSION

Many transportation strategies to reduce GHG emissions are highly cost-effective. Some generate cost savings over the life of an investment (in a particular energy-saving technology or product) when future energy savings are calculated using normal discount factors. Consideration of cobenefits, such as improved energy security or reduced traffic congestion, makes many more transport GHG-mitigation options attractive. These findings are counter to the conventional wisdom that often ignores cobenefits and emphasizes near-term resistance to the broad suite of technology and behavioral options.

Policies to bring low-GHG technologies and practices to widespread deployment have already emerged and proven successful in limited venues. The available policy strategies are diverse, including local policies that

TABLE 7.1. SUMMARY OF TRANSPORTATION GHG MITIGATION OPTIONS AND POLICIES

Category	Today's measures (deployable 2009–2015)	Tomorrow's measures (deployable 2015–2030)	Supporting policies and practices
Vehicle efficiency	• Incremental efficiency improvements in conventional gasoline automobiles and diesel trucks • “On-road” improvements in maintenance practices, technology, driver education and awareness	• Widespread vehicle electrification (hybrid gas-electric, plug-in hybrid, battery electric) • Fuel-cell vehicles	• Vehicle efficiency performance standards (fuel economy, CO₂ emission rate) • Voluntary industry commitments • Vehicle purchasing incentives (rebates, feebates for low-CO₂, high fuel economy) • Government and company fleet efficient vehicle purchasing
Low GHG fuels	• Mixing of biofuels in petroleum fuels • Use of lower GHG-content fossil fuels (e.g., diesel, compressed natural gas)	• Electricity (in plug-in hybrids and battery electrics) • Lower GHG intensity renewable biofuels (e.g., ligno-cellulosic pathways) • Hydrogen from renewable sources • Mobile air-conditioning (MAC) refrigerant replacement	• Biofuel blending mandates • Low GHG fuel standards • Carbon tax on fuels • Government and company fleet incorporation of alternative fuels
Reduced Vehicle Use	• Intelligent transportation system (ITS) technologies to improve system efficiencies • Mobility management technologies • Inclusion of GHG impacts in land-use and transport planning • Incentives and rules to reduce vehicle use	• GHG budgets for households and localities • Modal shifts (road to rail freight, public transit systems) • ITS technologies to create new, more efficient transport modes	• Road, parking, congestion pricing • Investment in public transit and bike/ped infrastructure • Public awareness, outreach, education campaigns • Zoning law changes

integrate transport and land-use decisions, as well as state and national policies such as fuel efficiency and GHG performance standards on vehicles, outreach and incentive campaigns to instill energy-saving attributes into all consumer decisions, and government and corporate research partnerships that promote technological breakthroughs and reduce infrastructural and implementation cost barriers.

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CHAPTER 8

Big City Planning Directors' Approaches to Infrastructure

*William R. Anderson, FAICP; Armando Carbonell, AICP;
Barbara L. Sporlein; and Harriet Tregoning; with William R. Klein, AICP*



Good afternoon.* This session is sponsored by APA, the Lincoln Institute of Land Policy, and the Harvard Graduate School of Design. My name is Bill Klein. I'm director of research and advisory services at the APA Chicago office. APA's research unit, which has been around for 60 years, is responsible for bringing you APA's Planning Advisory Service and the monthly publication *Zoning Practice*. It also works closely with APA's policy and outreach units in Washington to run the National Centers for Planning, which engage in policy-relevant research and education. Under the umbrella of the National Centers for Planning are three research centers: the Green Communities Research Center, the Hazard Planning Research Center, and the Planning and Community Health Research Center.

* This is an edited and condensed transcript of a panel discussion at the 101st National Planning Conference in Minneapolis, April 28, 2009.

Our purpose here today is to share with you a taste of the discussions that unfolded in the 2008 Big City Planning Directors Institute that was held on November 16, 17, and 18. The institute was originally conceived by the Lincoln Institute and the Harvard Graduate School of Design 10 years ago. In 2003, the Lincoln Institute and the GSD invited APA to join them as an equal partner. What we do is invite the planning directors from the 30 largest cities to come to Cambridge, Massachusetts, for a two-and-a-half-day experience. We bring in some experts on particular high-priority topics. But more important, we provide an opportunity for big-city planning directors to share best practices, compare notes, sometimes cry on one another's shoulders, sometimes boast about the hot new things they're doing in their cities, all behind closed doors, without the press around, which is kind of exciting. It really is a very, very exciting once-a-year colloquy, and for many of you big-city planning directors I'm sure it's the pause that refreshes.

A major theme of last fall's institute was the role of big cities in framing and implementing a national infrastructure strategy. We have here today the planning directors of San Diego, Minneapolis, and Washington, D.C., who will share with us what they're up to in infrastructure planning.

Here is what we asked our speakers to address: we suspect that with a new federal administration in Washington and the stimulus bill off and running, your role with respect to infrastructure planning has changed a great deal. What are those changes in scope and mission? What kinds of infrastructure planning are you involved with? What are you doing now that you weren't doing last year? What problems did you face to adequately staff up for these responsibilities? What would you have done differently if you had a chance to do it over again? And what advice can you give other planning directors?

Armando Carbonell will now give some framing remarks.

Carbonell: Thanks very much, Bill. The Big City Planners Group works on a changing curriculum. Every year, the planners themselves propose the most important topics in their own work. Prior to dealing with infrastructure in this past year, we had been working with the planning directors on climate-change mitigation and adaptation. This year's topic provides continuity in the way that we're dealing with underlying issues that are not going to go away very soon but are very exciting issues for planning.

There are things going on in the policy world that we really haven't seen for some time that make it particularly important that we are prepared in our planning work in cities and around the country to strategically use some of the new resources and opportunities that are developing at the national level. Last fall, we imagined we'd be talking about things like reauthorization of the Surface Transportation Bill, which was the next big thing to come along. And indeed, that is still coming along. It is still worth thinking about. But in the midst of that came the stimulus package, and this presented itself as a tremendous amount of money going to infrastructure—it turned out to be slightly less tremendous but still significant—and as something capable of having a transformative effect on the country. So we worked very hard on establishing some ways of thinking about stimulus that were more than shovel-ready, more than business as usual, and thought about some of the fundamental changes needed in terms of the support systems that our country runs on.

The directors made great progress in establishing principles and ideas about how to use stimulus funding. We tried to communicate these to folks in Washington. I think they heard some of it. We are certainly pleased to see things like \$8 billion for high-speed rail, very substantial and transformative investments in energy, and other very significant contributions geared to the short run.

I should mention that we just had a program in Cambridge with a group of leading journalists from around the country and talked about stimulus and infrastructure in a number of sessions, including some briefings from Washington from other folks who are keeping their fingers on the pulse of national policy and execution of programs. On high-speed rail, some people treat the \$8 billion as no more than a down payment and maybe something we shouldn't take too seriously because we know it's going to be much more expensive. Michael Dukakis, former governor of Massachusetts and vice chairman of Amtrak, said that \$3 billion to \$4 billion a year over time would actually support, in terms of the capital cost, the development of a higher-speed rail system for the country. This is not unattainable and not way out of line with the kind of resources we're likely to see, especially with states contributing in the way that California, for example, has stepped up to the plate with its own transportation bond for high-speed rail.

It was also interesting to learn from an official at the Office of Management and Budget, who actually supervises all of these line agencies that we're interested in, including Transportation and HUD and Energy and a dozen others, that the Obama administration appears to be using the budget and OMB itself to coordinate federal policy, to actually get this kind of transportation effect out of the hundreds of specific programs that constitute the federal budget. I feel very encouraged that there is some intelligence here being applied to what does add up to not trivial resources that, if applied consistently, might actually change the world.

I'm going to turn this over to the West Coast and Bill Anderson, and we'll see what's happening in San Diego.

Anderson: Good afternoon. Let me tell you just a little bit about San Diego: we're a city of 1.3 million people. We're the second-largest city in California, in a county of about 3.2 million people in a binational metropolitan area of about 6 million people, because our twin city is northern Baja: Tijuana, Ensenada, Rosarito, and Tecate. While most of our infrastructure is related to our city, we also have state and national infrastructure that enables trade with Mexico.

We've been very much involved in the capital improvement program (CIP). It is policy that every CIP project has to be consistent with the city's general plan. Every year, the planning commission must make a finding that the CIP is consistent with the general plan.

We updated our general plan, which we call the City of Villages Strategy, in March 2008. It's really a sea change from 200 years of growth through land development, since now less than 4 percent of our city is available as greenfield development. Most of the city is already developed, already entitled, or preserved as open space and habitat area. From here on out, the future is urban infill. It's been a big shift for us, and we're still at the early stages of that transformation.

The first guiding principle of the general plan is open space. San Diego is a city built on a canyon system. This open space system helps define our communities and create a strong sense of community identity. Within the communities, we have villages that are mixed use, walkable, serviceable by transit, and everything else that follows from that. One of the principles is to have high-quality, affordable, well-maintained public facilities, which is really the Achilles' heel of the whole strategy, because if the future is urban infill, that means greater demand on urban infrastructure. Communities are resistant to take that increase in density until we address their concerns about urban infrastructure.

We do have a specific element in our general plan that focuses on public facility services and safety, where most of the policy discussion about capital improvements is located. But capital improvements are also associated with

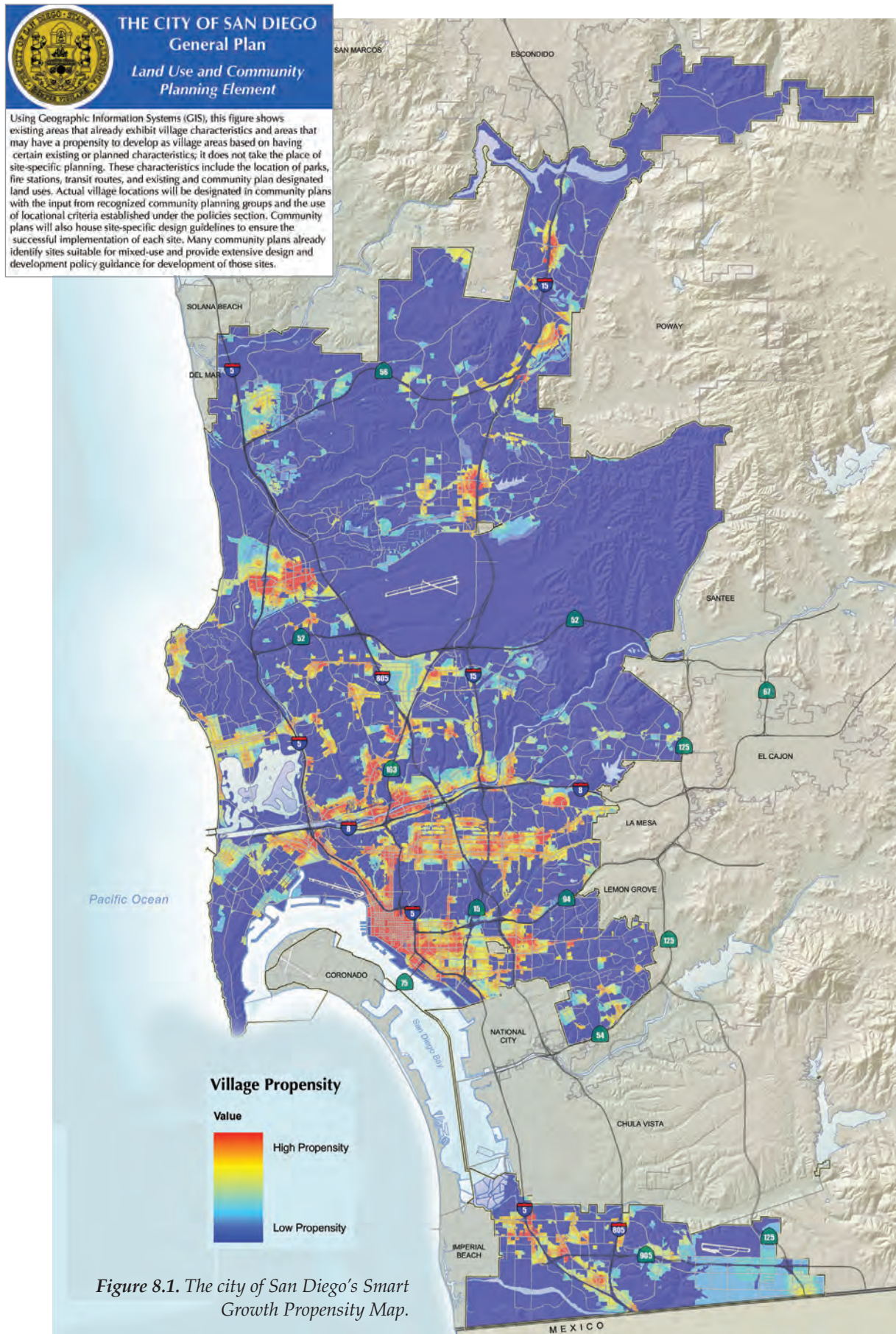
mobility, urban design, economic prosperity, and many other elements. So it permeates the plan. Because the city is large geographically (340 square miles), the planning issues in San Ysidro at the border are very different than in the planned communities in the northern part of the city, in coastal communities, in the postwar suburbs, and in the prewar craftsman communities. Therefore, the general plan does not regulate individual properties, like it might in a suburban city. Instead, it sets the citywide policy framework. We get into specific policies that are tailored for a community, and that affects zoning—what you can do with a particular parcel—in our community plans, which are technically part of our general plan. Each community plan has a public facility element, which identifies public facilities, phasing, and public facility financing.

Our Smart Growth Propensity Map (Figure 8.1) is in the general plan, and it illustrates our land-use policy of trying to cluster in smart growth fashion. It shows the areas where we look to consider villages and higher-density development. It doesn't mean smart growth has to happen in these areas, but that is where we see the conditions that are conducive to support it. It also shows the public that most of the city's land area is not going to be affected by higher density. Most of it is either preserved as open space, or there are stable single-family neighborhoods in which we won't be planning higher density, to allay those fears that we're trying to put Midtown Manhattan in a single-family neighborhood.

That map is tied to maps of smart growth place types created by the San Diego Association of Governments (SANDAG), the regional agency. In fact, their place types for the City of San Diego come from our general plan. Thus, our plan is connected with the Regional Council of Governments' plan. This is very important for our CIP because it means we're integrated for one of our most important forms of capital improvements: transportation, particularly public transportation. Also, we're integrated with SANDAG in terms of regional open space, because they have a role in habitat planning as well.

So we do look at open space as part of our CIP, as well as recreation, conservation, and ways to implement some of the conservation policies related to climate change. We also consider infrastructure for economic development in our base sector industries—which gets to goods movement and how to coordinate trade corridors with our industrial base and land uses. But the real problem we face is that while we have general public support for the concept of smart growth—clustering future development, understanding that it's good for the region's environment, that it's part of the climate-change initiative—most people want to see it happen in someone else's neighborhood, someone else's community. And even the communities who are supportive say that they don't want to take on more density until we have a strategy to deal with the infrastructure issues where we have 40 or 50 years of backlog: parks, street improvements, and water and sewer capacity.

The basis of our facilities financing plan in San Diego is development impact fees and facility benefit assessments (FBA). This is something that we developed in the 1980s as part of the growth management strategy in the last general plan. Basically, the FBA areas were areas that were still primarily greenfield development. They were in what we called our "future urbanizing" areas and "planned urbanizing" areas. For development to occur under the city's growth management plan, future urbanizing areas had to be approved for what we called a phase shift and become planned urbanizing areas. Development in the planned urbanizing areas had to pay for all of its infrastructure needs. It was basically a marginal cost, marginal benefit system. So when these areas were new greenfield developments, the development that was planned and the infrastructure tied to it was calculated, phased, and folded into an FBA that was applied to the land and is updated every year.



However, in the urban areas of San Diego already built, we still had urban infill occurring. In fact, it was occurring faster than anyone anticipated. So development impact fees were applied there as well. But this strategy basically dealt with the marginal impacts on infrastructure, as opposed to the FBAs, which were dealing with the whole package of infrastructure associated with new greenfield development. When dealing with just the marginal costs of urban infill development, we weren't getting enough revenue to pay for the facilities.

New community plans now identify capital improvements so we know what to do, roughly how much it's going to cost for construction and completion of a project. Still, we have a difficult time with coordinating public facilities, particularly with urban infill and trying to implement our park policies, because the cost of buying park land and building parks in urban communities is very, very expensive and results in high impact fees for our marginal infill development. This provides a disincentive for infill development, particularly for affordable housing. We're trying to solve that problem.

There is also a concern with timing, because these mechanisms were a pay-as-you-grow mechanism. Because this process is dependent on anticipated development, which is risky, and on cash coming in as development occurs, we couldn't bond against it. We would collect the money, though, and if we had enough we'd fund a facility. That meant we had a timing issue with concurrency. The demand might be there, but we wouldn't be able to fund the improvements to serve that demand for several years, until we'd collected enough cash. In large land holdings with an FBA, we were able to solve this cash-flow problem, usually with developers who fronted the cash and had reimbursement agreements with us. They would be reimbursed with interest over time as fees were collected. Otherwise they wouldn't have been approved to move forward with their development.

As an example of the fees, the FBA costs per unit range from \$23,000 to almost \$100,000 in Torrey Highlands, which is near Del Mar. It's higher in FBA areas because most infrastructure that converts raw land to improved land and public facilities to serve the total new community population is funded by the FBA. The development impact fee (DIF) for our urban infill development, however, is much lower since it funds only the marginal impact of new development occurring in existing built communities that already have some infrastructure and public facilities in place. DIF revenue in the city ranges from approximately \$1,500 to \$14,000 per unit.

We have other mechanisms that we rely upon to address the capital needs of and the deficiencies in serving the existing population. Some of these needs in the urban neighborhoods developed over 60 years. The neighborhoods that were built before World War II or in the 1950s or the 1960s did not have impact fees to finance development. In addition, standards have risen since those times, so there has been an accumulation of financing responsibilities over time.

We have the Regional TransNet Funds, which is a sales tax overage for public transit and arterial road improvements through SANDAG. We're looking at special taxes more, like community facility districts, and we have business improvement districts (BIDs) and assessment districts. Tax increment financing is a major vehicle, which is handled by our department's redevelopment division. We also enter into development agreements to get private development to build extraordinary benefits beyond what they're required to do as offsets for their impacts. We also use—or would like to use more often—mitigation and exaction fees, community development block grants (CDBGs), enterprise funds, and general obligations, which would

be the lowest-cost mechanism to address existing deficiencies for general-purpose public facilities and infrastructure. We just can't get the voters to vote for it. It requires a two-thirds vote in California. Of these funds, the planning department has influence over the FBAs, the DIFs, the CDBG, and tax increment financing.

In the last year, we have been forming—not in response to the stimulus, though it is timely—a CIP Review and Advisory Committee (CIPRAC). All the directors and the chief operating officers meet monthly to go over CIP priorities and proposals. It's like an investment committee. The departments have to pitch what they want to put into the CIP. We have to make certain findings, including whether a project is consistent with the community plan or the general plan and other council policy criteria. Then we as a group recommend priorities to the mayor and council, who make the final decision.

We have council policy-prioritization factors for nontransportation assets. I won't go into each one, but they include project readiness, health and safety, and the extent to which the facility is already funded and perhaps just needs that marginal additional amount to get it financed. We also have similar priorities for transportation assets. We use a prioritization scorecard, on which we start by identifying the needs in the community plans, first developed as part of the community-plan update process, and by preparing a preliminary prioritization with the "asset-owning departments," such as parks and recreation, the streets division, the fire department, and the library department.

One of the criteria is having enough operating money to support the infrastructure. Because of the fiscal crisis right now—like every other city, we're down in fiscal revenues—we have park projects that are ready to go, but we're not pulling the trigger because we don't feel we have the operating money. We do have a step in the process where we can decide to pause, which is frustrating for some folks, but it helps us with our fiscal management.

Finally, we're working on a facilities financing strategy for the city as a whole, and we have basic financing objectives—to plan efficiently and to reduce cost burdens—that requires planning and coordination with the general plan and community plans. For concurrency we try to get the revenue up front, which means we're looking at bonding structures. We already have mechanisms to ensure that new development pays its way, though we need to maintain our fees current, but we'll probably have to go to the voters to address the facility deficits to serve existing populations. Aligning payment responsibilities with those who benefit is controversial, particularly with regard to facilities that serve the city or the region as a whole. I like to say that people want government to work like a business until it does. So when we actually say, "Okay, well, you have to pay," people say, "Well, no, why?" One thing we're doing with the stimulus money is focusing on facilities that we've already identified, planned, and vetted. We're short on capital, but we have some, and the stimulus money helps us leverage our funds to get projects going. As a result, we can have a greater impact on job generation, which is the intent of the stimulus money. We couldn't do this if we hadn't integrated facilities financing into the planning process.

Klein: Thank you, Bill. It's certainly heartening to see a city so actively engaged in capital improvements programming on such a sophisticated level. And now, Barbara Sporlein is going to talk to us about Minneapolis.

Sporlein: Thank you. There are some similarities and a lot of differences between Minneapolis and San Diego. We're slightly smaller, at 390,000; we have about three million people in the seven-county metro area. We do not

have any development impact fees, and we are fully developed, so we're all about infill redevelopment, either on a very small single-family-by-single-family basis or in larger redevelopment areas of former industrial land that's moving to mixed uses or other new bioscience industrial use.

Something unique about the Twin Cities area is that we do have a regional form of government, the Metropolitan Council. They do have planning authority for their comprehensive plan, which is the development framework for the 189 jurisdictions within the seven-county metropolitan area. All of our local comprehensive plans have to be consistent with the regional 2030 Development framework. In addition, the council is also the operational authority, and it runs the regional transit system. It is responsible for the regional transit projects, all of which start, end, or go through the City of Minneapolis. And the council is also responsible for the sewer and water systems and regional parks and open spaces.

This is kind of preaching to the choir, but infrastructure matters, and that's why we include it in all of our comprehensive planning. We are accommodating and anticipating growth. We've been growing steadily since 1990 and hope to add 50,000 by 2030. Of course, it's a lot smarter to invest in areas where the infrastructure already exists, although you need to reinvest in that infrastructure.

The city defines infrastructure in many of the traditional ways, but also more recently has started to include trees. The Public Works Department was going to sell off some land because it had no reason for it for the traditional categories, and we actually convinced the department that the tree farm on it is part of our essential infrastructure, which we need to keep in our inventory. And so they said, "Well, trees are now part of our infrastructure." So that was kind of groundbreaking.

As in San Diego, we as planners have a lot of authority. Infrastructure and planning naturally go together—or should naturally go together—as a good idea, a smart thing to do. But here it's legally required, which is always helpful.

We have a number of statutes at the state level, among them a state land planning act, and our comprehensive plan and infrastructure plans have to be aligned. We review not only public land acquisition and land sales for consistency with our plans, but also all capital improvements.

The city charter goes further, requiring that the City Planning Commission review and approve all projects. There is quite an elaborate process set up for that, but it wasn't until this past year that we really got thinking more about infrastructure and the linkages among infrastructure, land use, and design. The stimulus certainly triggered additional focus on implementation, but there were two other things that happened here. One was the great success of the Hiawatha light-rail line, which opened in 2006. We surpassed 2020 projections for ridership, so either we are really bad at projecting or it's really been successful; I think it was probably a combination of both. And then the 35W bridge collapse in August of 2007 certainly brought attention to the significant lack of infrastructure investment at all levels of government. And so things since then really started to ramp up, if you will.

We have a Capital Long-Range Improvement Committee (CLIC) that is comprised of 35 citizens appointed by the mayor and city council. All city departments have to submit capital improvement proposals. It's a five-year plan that is updated every year. The committee evaluates these recommendations for city infrastructure priorities. And much as in San Diego, our chief financial officer, our director of public works, and the director of community planning and economic development jointly review and recommend projects. So the mayor has the benefit of this CLIC report, this citizen advisory report, which is a very elaborate review process, and then the co-prioritized

investment decisions. Sometimes they're consistent, and sometimes they're not. We also staff that process, and we are required to do location and design review. It's about 120 capital improvement projects annually.

These are the same categories of consistency that the Metropolitan Council uses when it reviews our local comprehensive plans. The council looks at conformance between the regional planning framework and existing policies, regulations, and projected future plans; consistency with all the legal requirements of the comp plan; and compatibility—is it compatible with what our neighbors are doing? We do review sessions with all the jurisdictions that touch the city.

We augment our comprehensive plan with a series of strategic small-area plans where there is land-use change under way or development pressure. That's much more detailed, and those always include a very detailed infrastructure component. And then each department has to develop work plans and performance measures that include infrastructure.

We have transportation planners in both the Public Works and Planning Departments. The two directors hired staff together; I hired an engineer with planning tendencies, and our public works director hired a planner with engineering tendencies. Those two individuals shared an office for a while, and it worked out well. We also have a transportation engineer on our community planning team. The city is divided into planning sectors, and we have public works liaisons on all those teams. In the last year or the last 18 months we have been trying to integrate transportation infrastructure with planning, land use, and design.

We have 1,000 miles of streets, but often the corridors are owned by somebody else, and that's very challenging. Either the county owns them or the state owns them, and all the freeways, of course, are federal through MNDOT, the Department of Transportation. We don't control when they're going to be reconstructed; we don't control the design. We have some local design review, but it's pretty thin. Our focus is all about technical assistance, having a seat at the table, and using persuasion. We sit on a lot of advisory committees in order to accomplish that.

Figure 8.2 (page 106) shows the city. The Hiawatha LRT connects downtown to the airport, and then there's a little mall south of there. The new Twins stadium will be at the location of the new multimodal station, for the Northstar Commuter Rail, which will start operations at the end of this year. The Hiawatha line will extend, and the Central Corridor will connect the two downtowns. That 10-mile corridor will be up and running in 2014.

We have a regional Transportation Advisory Board with a queue of projects; the stimulus money is not changing which projects we're going to do, it just moved things up in the queue. We have a 20-year state transportation-investment program and a 10-year transportation action plan. The Metropolitan Council has a transportation plan, and Hennepin County has a five-year capital investment program. If your project is not in one of those, it will not get funded. And there is way more to do, especially with the bridges. We have a lot of bridges here, and after the I-35W bridge collapse we closed several others and reallocated funding to bridge maintenance.

We have criteria for joint planning with our economic development and public works counterparts. We examine all those adopted plans for projects and for need. What will be a proposed project's impact? How ready is the project? How will the project contribute to sustainability? Interestingly, public art is also an infrastructure element. Two percent of our net bond debt goes to commission original public artwork on the city's rights-of-way and public land. And by direction of the council a year ago, every single infrastructure project or any new city facilities have to include art at the design stage, instead of including art afterward as a component if money falls out

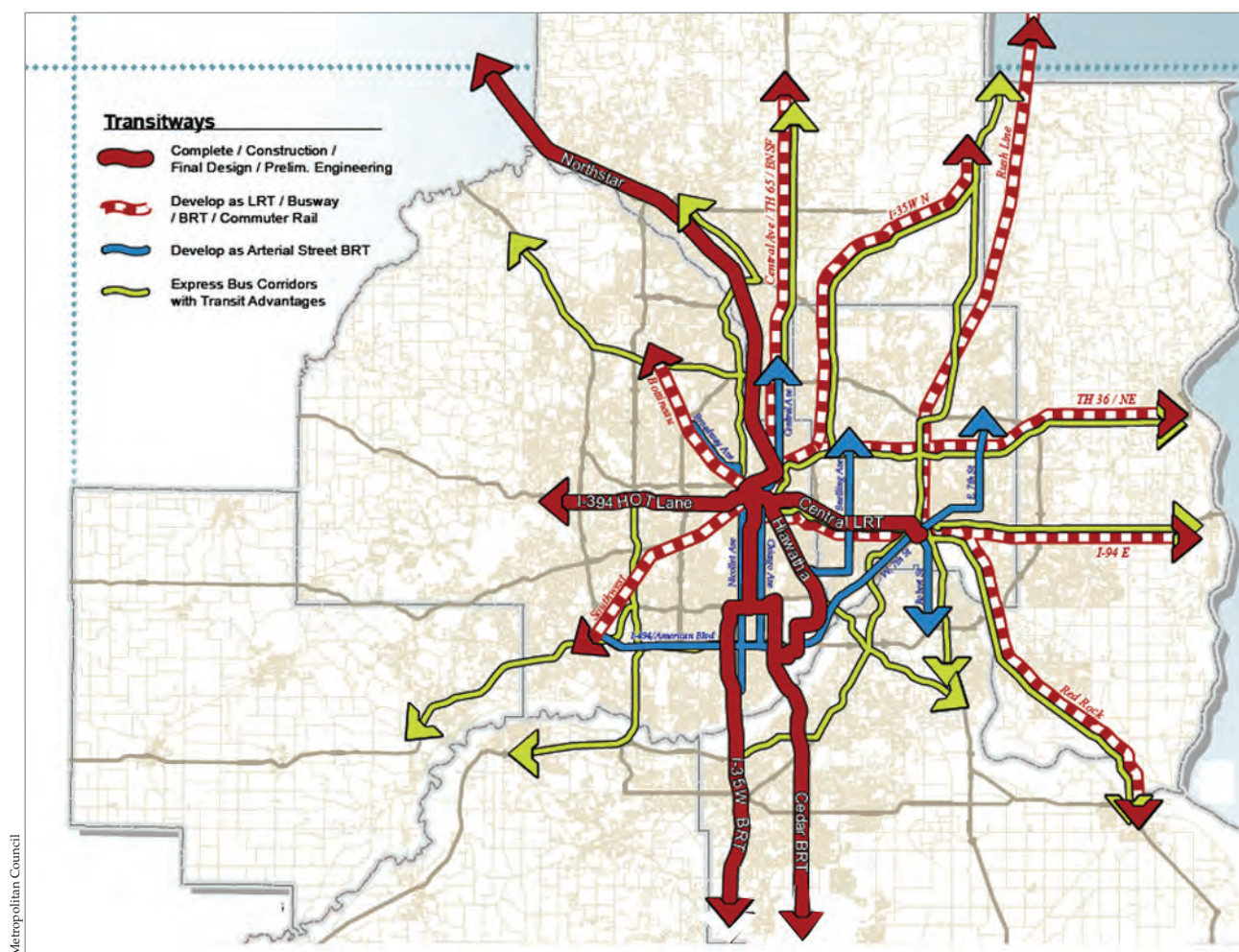


Figure 8.2. The big picture of transportation-infrastructure development in the Twin Cities and surrounding counties.

of the air or if a foundation funds it. The public art doesn't always have to get built, but it has to be included in the process. That was a real mind shift for the public works folks.

Klein: Great, thank you, Barb. I think so far the first two of you are beginning to disprove the conventional wisdom that planners and planning departments have ceded the active role in capital improvements programming to others. This is a great inspiration to many of us. Now, Harriet Tregoning of Washington, D.C. will speak.

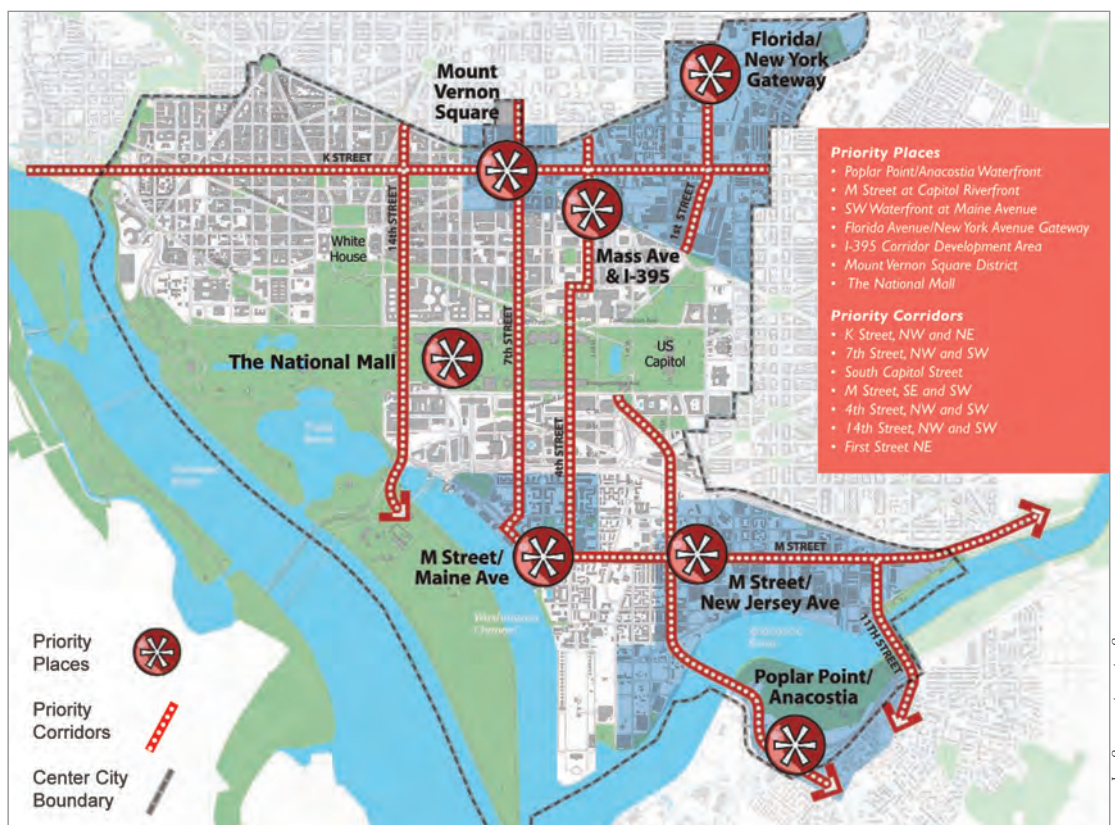
Tregoning: Thank you. I want to frame my remarks in the context of the American Recovery and Reinvestment Act (ARRA). We have a lot of challenges, like every city. Washington, D.C., is a city of nearly 600,000. Our daytime population is one million. That's one of the biggest jump-ups of any city in the country. We also have 20 to 25 million visitors who come every year to visit 68 square miles of land, only 43 percent of which is taxable for us. We have some definite challenges.

Those challenges have a lot to do with what we aspire to be when we grow up: a place that has a lot more mobility and a lot more choice on transportation, as well as an energy infrastructure that meets our clean energy goals for our city and for our region. A lot of the energy that's produced in the Washington area is coal-based, so we do very poorly in terms of greenhouse gas emissions and carbon footprint. We have issues related to aging infrastructure. It is difficult to figure out how to refinance that infrastructure. We want improvement in environmental performance. We want to invest in a

way that reduces the disparities that exist across a lot of areas in our city. So I want to offer four principles for ARRA investment.

- *Principle Number One: First, Do No Harm.* Like most jurisdictions, the District has received a lot of stimulus money. The big debate in transportation is whether that money is being used to expand capacity or to repair bridges and roads. We are expanding capacity, but the capacity we're expanding is bike capacity, through our Smart Bike program. We are not expanding any of our road capacity. We're building bridges, and we're rebuilding bridges. We're rebuilding roads. We're implementing streetscapes. But we're trying to use ARRA funds to do a lot of other things that help implement our plans. Like most places, we are largely doing things that were already in the pipeline because there just isn't enough time to plan some great new project and get it entitled and built. This does suggest that we should have some grand plans in our hip pockets, right?
- *Principle Number Two: Fortune Favors the Prepared Mind.* Infrastructure has always been a key part of our comprehensive plan and our small area plans that help us implement the broader comprehensive-plan vision, particularly in our case to improve walkability and urban mobility. We have a lot of land, surprisingly enough for our small city, that was heretofore undeveloped—former federal reservations, land that had another use but for all intents and purposes is going to be entirely reused. There are hundreds of acres of land in the city that will be reused, which is pretty unusual. We've identified and are redeveloping an area that is essentially five times the size of our current downtown. (See Figure 8.3, page 108.) We've developed the Center City agenda in the blue areas of the map. Much of the area is fortunately already served by transit, though some of it needs some additional transit investment, and some has no modern infrastructure at all. Washington, D.C.'s downtown is projected to be entirely built out in the next four years. We have a height

Figure 8.3. Priority places and corridors in Washington, D.C.



limit, most of the footprints of our sites are nearly fully built out, and the few redevelopment sites there are already under construction. There is continuing demand for commercial space in the city. So we have the highest rents, we have the lowest vacancy rates, and we're gaining in regional office-market share. Do we just call a halt to all of that? It is better for the region if jobs can be located near housing and near amenities and served by transit. So that's been the rationale behind this expansion of what was downtown and is now becoming Center City.

We're also looking to provide new mobility of all kinds. Barbara very helpfully mentioned a bike route intersecting a transit station or a transit hub to create a new multi-modal center. That's exactly how we're thinking. It's no longer just where transit might meet commuter rail. It's where bikes come together with car sharing and several different types of transit, all of which are legible so you can tell what's there. Even at our most multimodal point in the city, Union Station, I pity the poor person who is trying to figure out how to avail themselves of these wonderful transportation choices because the information isn't easily available. You know, this is not rocket science. We are working to improve the legibility of the transportation options.

Another big development area for us is St. Elizabeth's, a former mental hospital—historic, beautiful, and one of the most wonderful places in all of Washington, not visited by many people. (See Figure 8.4.) The West Campus is about to become the new location of the consolidated Department of Homeland Security headquarters. It's a big development.

Figure 8.4. The city is encouraging development alongside the West Campus of St. Elizabeth's, which is about to become the new location of the consolidated Department of Homeland Security headquarters.



It will bring 14,000 employees just to this campus. We own the campus across the street, and the two properties combined can be a dynamic, lively, mixed use community. We've tried to help make that happen by getting Homeland Security to put some part of their program on leased land that we own.

But we've also identified a lot of other opportunities. This all needs infrastructure, but St. Elizabeth's is located in the poorest ward in our city. One of the things that we're doing is devoting our weatherization resources here and using job training funds to promote gateway jobs. We're doing intense job training for environmental auditing, for energy-efficiency auditing, and for weatherization in order to get people into pre-apprentice programs that would enable them to qualify for construction jobs. This is a construction project that will probably go on well through 2018. There can be many years' worth of employment here if we play our cards right. And we're spending a lot of money to make sure that that happens through the stimulus.

- *Principle Number Three: Move the Needle.* Why do we do planning? We want to make a difference. So for ARRA and for the other planning we do, we should be looking at things that move the needle, and we should be figuring out up front what are those things, what are the outcomes we seek, and how to measure them. If we're wildly successful, how will we know it—besides that the money is gone? So what kind of things do we really care about?

For us, we picked a few things that we're really going to focus on in terms of spurring revitalization, keeping in mind that the constraints that are put on the ARRA funds are real constraints and put real limits on what you can do. We do have some transit projects that are ready to go. We're not only taking the formula money, but we're frankly—probably as a lot of jurisdictions are—competing for a lot of the competitive grant funds that are out there for transit projects and for other things. This is something that's unique to Washington: \$2 billion in federal funds are going to be spent in the Washington area, most of it in the District of Columbia. The federal government is going to be doing things like rehabbing their own existing buildings for energy efficiency to make them high performance. That job market that we're trying to train people to get into is a pretty big market, even if you don't count all of the things that the city is doing. That makes it even more imperative that we link our job training funds and our workforce development to enable people to compete for those jobs.

We're looking to make it possible for there to be much more local energy production. We're installing photovoltaics on D.C. public schools. Through our building code and our zoning we are making it possible to incorporate a lot more types of sustainability features into buildings in the District. Energy is a big part of what it costs to be in a home. So things we do in low-income households to improve energy performance, that makes housing more affordable, which is a big issue in most of the small area plans that we do. We're also using our ARRA funds to reduce environmental hazards. Lead paint continues to be an issue in the city, affecting the development of children in every ward.

Transportation choices are a huge thing for us. We're using \$3 million to expand our first-in-the-nation shared bike system. We have a paltry 10 bike stations, but we want to expand to 60 this year and 100 next year. We think that there are going to be so many more of these cycling stations in neighborhoods that it will support efficient door-to-door trips. A lot of people can get to the transit station, and these shared bike stations should enable more transit-bicycle trips.

- Principle Number Four: Encourage Unnatural Acts.** My final principle is to encourage unnatural acts. In this case, I'm talking about regional planning and cooperation. That federal ARRA money we are talking about today does not fall from the sky very often, but that is the kind of thing it takes to get people to put aside their differences and to put aside the kind of log-rolling that typically takes place in a metropolitan planning organization—or, I'd even venture to say, among agencies that are looking at capital projects. They approve other people's projects so that their projects will get approved. No one gets to ask the question, "What would we do if we weren't fiscally constrained? What is it that we would invest in that would be a game changer in our region?"

In our case, it's a rapid-transit expansion project that we had to cajole our MPO to let us begin, once it was clear that this money was going to be coming. It got off to a slow start, but the working map that we're using now is even much vaster than this. (See Figure 8.5.)

We're going to have a network of proposed rail on the books. We previously never had a rail investment plan, not since Metro began. Everyone

Figure 8.5. Proposed rail expansion in and around Washington, D.C.





Figure 8.6. A visualization of the K Street transitway, with bus rapid transit (BRT).

Washington, D.C., Planning Department

has a road plan. We know where all the future roads are going to go. Now we have a rail investment plan. Strangely enough, we're getting as much uptake in the outlying areas as we are in the center, which means that the political constituency for transit is also beginning to change.

The hallmark of this for us, and the most expensive piece, is the K Street transitway, which is going to be an official transitway, with bus rapid transit (BRT). (See Figure 8.6.) This is some of the densest, most intense development in the city. The K Street corridor contains 27 percent of the city's land, 36 percent of the jobs, and 61 percent of the commercial tax base. It's one of the few continuous east-west corridors that we have. It is residential in some places, but there is almost no residential in the center. We're doing a design/build for this project, but we're having to address some issues in very short order to compete for the \$1.5 billion in funds. We want it to be a model complete street. It's going to continue to carry vehicles as well as transit. We're adding bike lanes. We're expanding the sidewalks.

So I'll conclude with this—my four principles of infrastructure in an ARRA era:

- First, Do No Harm
- Fortune Favors the Prepared Mind
- Move the Needle
- Encourage Unnatural Acts

Klein: Thank you, Harriet. Harriet has always been good on principles. You've got some dynamite principles that have come in over the years.

Speaker: Let me ask a question. Are there any interest groups in your cities that have fought against the planning department taking a leading role in capital programming?

Anderson: No.

Klein: So you got pretty widespread support. That's good.

Anderson: They don't want us to take the lead role, but they want us to be part of the lead team.

Speaker: What about the log-rolling? My experience with capital is that one of the challenges is by the time you get to look at a project, they've already

sort of decided this is the project to do, so the ability to influence whether it gets funded or not—I mean, you can veto. But other than that you’re kind of marginally saying, can we make this project better? Can we change some of its performance characteristics, as opposed to saying, “Not that project. I want a project that you haven’t proposed. That’s the project that should really be built.” How are you dealing with those issues?

Anderson: Well, for us, because we have a CIP system where things are tied together, on transportation, it pretty much has to be in the system already. But the energy fund is actually where we’re looking for support for new projects that haven’t really been on the books. They’re improvements to existing building stock that would be perfectly consistent with all of our policies. But they’re not as tied to a whole system like transportation is.

Sporlein: I agree with Bill in that with all of our transportation projects, especially regional transit projects, we have many more projects ready to go and agreed upon than can be funded. So it’s the road projects where there is some arm-wrestling going on right now, not the transit projects. For transit, there is sort of an order: it’s going to be Northstar first, and then Central Corridor LRT, and then Southwest Corridor LRT, and then Bottineau Boulevard LRT, and the three bus rapid transits are all going to be done. So there’s a rhyme and reason to all of that. The additional money is going to speed up when they can be done. With road projects is where you see more questions, especially between urban and suburban areas, like, “Are you going to use that money to expand one freeway system in the northwest suburbs, and it will take all the money? Or are you going to use that money to complete three central-city projects that are really needed?” That’s a little bit where that log-rolling is going on.

With weatherization, nothing was really set until this money became available, and we’re doing the same things: training people to do the energy audits and to do weatherization, especially for low-income households, and really use it as a job creation tool. It’s just a win/win. It’s like a threefer: you make housing more affordable, you train workers, and you get more sustainability.

Speaker: One of the things I’m interested in is investment planning. You talk about single investments that have multiple outcomes. That’s always been one of the challenges in terms of undertaking equitable investments and moving the needle in terms of infrastructure planning. I wonder if you could all reflect on your assessment methods: have those methods have changed in this new round of investment planning so that the balance between those different benefits are being evaluated in different ways?

Anderson: We’re actually trying to shift that in the opposite direction by identifying multiple investments to achieve a single purpose, such as leveraging our access to stimulus money. We have state bond money now for infrastructure with Proposition 1C and other sources. We have tax increment funds. We have Community Development Block Grant money. And rather than just spreading it around opportunistically, one project at a time, we’re thinking maybe we should focus on certain geographic areas, because our ultimate objective is the transformation of those areas, and use these funds for the different aspects of that transformation. We’re finding that having that kind of strategy and implementing it makes us more competitive for other grants, too, because funders know that their dollar is going farther because it’s leveraging other investments.

Tregoning: Part of it is developing the metrics—understanding what it is that’s different if you make a series of investments. And some of these metrics are only just becoming things that we routinely evaluate. One of those things is the combination of housing and transportation costs. Washington

is considered a very high-cost area for residential, but it turns out when you put housing and transportation together that we're a pretty moderate cost area because our transportation costs are very low. In our city over the last three years, motor vehicle registrations have gone down by 15,000. That's 6 percent of our motorized vehicles, gone from the city. This is when population went up by about that same amount, when income went up, and when household size even crept up—all things that you would normally think would bump up motor vehicle registrations. According to the American Automobile Association (AAA), you're spending between \$8,000 and \$10,000 to own and operate a car. So that's putting between \$120 million and \$150 million back into the pockets of our residents. We think part of the reason our sales tax revenue has been steady and our property values have been steady might be due to that extra spending money.

So by investing in different transportation choices, we wouldn't have measured sales tax revenues as a consequence, but yet that's where we're finding the impact. We're still learning what metrics are useful, but I think there's some good ones out there that cities need to be using.

Mitchell Silver: I'm on the board of directors but play director in the City of Raleigh, North Carolina. I would strongly encourage Lincoln Institute and Harvard to start looking at the other top 50 cities. Numbers 30 through 50 are growing rapidly and will soon replace some of the top 30. Raleigh jumped from 54th-ranked city in 2005 to 46th, right on the heels of Minneapolis at 385,000. We're growing very rapidly, the fastest-growing region in the country, and we could benefit from all the wisdom of our colleagues across the country.

My question is that you talk a lot about the challenges of infrastructure, and I want to focus specifically on public works and transportation. While we in planning know the important linkage between land use and transportation, it seems that public works and transportation departments don't always get that linkage. I know in our jurisdiction, transportation means moving traffic and not people. I want to know if you have any lessons learned, or if you're not experiencing that tension and you're both on the same page, because we're struggling still with complete streets. How does the relationship between planning and public works function in your city? Because I'm sure that there is a pot of money that public works departments might rather spend on expanding a roadway instead of dealing with a complete street or doing other things that maybe are foreign to a traffic engineer's concepts but that are commonly discussed in planning. So if you could share your experiences, that would be helpful.

Sporlein: I'd say we have come a long way, but it's a little "two steps forward, one step back." Back in 2003, there was a consolidation of the city's disparate planning and development functions into our current department. The consultant also recommended that transportation planning be moved from public works to our department, which we call CPED: Community Planning and Economic Development. It really was just too much for the mayor and city council to take on, given all the other consolidations.

But short of that, there has been a real concerted effort to really have the two departments work more closely. I described the capital improvement process. But also, they were the lead on the 10-year transportation plan. But we really cowrote it, and it's completely consistent with the comp plan. And when you hear the engineers and the division directors in public works cite all the policies in the comprehensive plan, it warms the cockles of your heart. And this 10-year transportation plan is mode neutral. It really is multimodal. It's not primarily the car. It's all modes, and we're measuring the different modal splits. And some of the products include a pedestrian plan and design guidelines for streets and sidewalks. It's just fundamentally

different approach. But when push comes to shove and they say, “You go find the money. That’s a development-related infrastructure project”—you know, there is still a little bit of that. But we’ve come a long way with a lot of City Council support.

Anderson: We still have some of that, but it’s gotten a lot better. What helps us is that SANDAG, which controls the purse strings for much of the region’s transportation system, is also with us in terms of public transit, smart growth, and alternative forms of transportation. So really on the directors level, and certainly with the council who adopted the General Plan, we work together. We’re working much better.

Where we have conflicts is really at the staff level because we have a mobility group and they have traffic engineers who are doing forecasting and corridor planning and things like that. We have the street design manual. They want the street design manual. And it’s at that staff level sometimes where we have conflicts. It’s more turf. It’s not necessarily disagreements on what should be done, but it is turf, and they would have a tendency toward keeping the roads moving freely, while we’re thinking of ways of slowing down traffic.

Tregoning: I think there are two sorts of strategies. One is a personnel strategy. I would really encourage you to take your very best people and send them over there into enemy territory when there is a vacancy. Offer to lend them somebody and offer to take their people so you have some cross-fertilization.

I am a very unconventional hire for a planning director. We’ve just hired as our transportation planner the former regional president of Zip Car. So we have a lot of stuff going on in our city, but it does signal that we’re headed in a different direction.

The other thing I would say that’s true for all of us is that we’re in a massive natural experiment right now with the recession. Things have played out very differently economically for different communities. It’s not true everywhere, but in many communities there is a real gradient in terms of foreclosures per 10,000 and changes in property value from the center to the edge. Now why is that? What are those properties of center places that allow them to hold their value longer? Is it that they have more transportation services and more transportation choice, more convenience, higher walk scores, better land use? Now is the time for us to be saying, If you care about economic resiliency, then let’s try to understand what’s happened here and what makes for resilient communities, and let’s try to give more of our communities those characteristics and invest more in those things that seem to have long-term value.

Klein: Well, you’ve gotten a little bit of a taste today of only three of the 23 big city planning directors that were in Cambridge in November. And just another example of why it’s so exciting to be in planning right now.

CHAPTER 9

A New Paradigm for Transportation Planning

Robert E. Passwell



“Fragile Foundations,” published in 1988, alerted Congress and the public to the terrible state of American infrastructure (National Council on Public Works Improvement 1988). This report found that much of our infrastructure was in a state of disrepair and that our cities and states were unable to meet existing needs or future demand. More than 20 years later, growing congestion, rapid increases in goods movement, sustained problems with air quality, and rapidly rising energy costs have made concerns over our infrastructure even more pressing. Our transportation institutions, grounded in the 20th century, are inadequate for the challenges of a 21st-century economy. This chapter begins with a review of current practice and then proposes needed reforms for a new paradigm in transportation planning.

Transportation Planning: Origins and Mandates

The Federal Aid Highway Acts of 1916 and 1921 had a profound impact on the national transportation landscape for the remainder of the 20th century. Defining elements of Title 23 of the U.S. Code, the body of law and regulations describing federal aid to highways and bridges, still determine how our surface transport systems are planned, designed, and funded. No matter how influenced by current standards of environment, energy use, and economic considerations, Title 23 has kept the following requirements: In order to be eligible for federal aid to highways, each state must have a department of transportation (originally, in 1921, a highway department); and federal aid for highways will be given directly to these state departments of transportation (DOTs).

From the second decade of the 20th century until the 1980s, the thrust of the U.S. highway program was to add supply, to increase the paved miles of roads throughout the United States, and to continually improve the design and capacity of those roads. By 1960, there were 1.2 million paved miles of roads in the United States; this increased to more than 2.2 million paved miles by 1990 (U.S. Department of Transportation 1996). The incentive for states to increase supply at such a rapid rate was the low amount of state funds needed to match federal dollars (10 to 50 percent from the state, depending on the level of the road).

Transportation planning came of age during the post–World War II period. While planning makes good sense for states and communities, the mandate to have planning as a requisite for funding was both the carrot and the stick. By 1962, the Bureau of Public Roads (BPR) required states to conduct long-range transportation planning in a comprehensive, coordinated, and continuing manner (the 3-C process). In 1964, the Urban Mass Transportation (UMT) Act added support for then hungry mass-transit systems. Finally, in 1966, the U.S. Department of Transportation was formed, bringing together the new Urban Mass Transportation Administration and the Federal Highway Administration (FHWA, formerly BPR) under one roof. But the funding still was specified for highways under Title 23 and for transit under the UMT Act. Because of the almost geometric growth of our transportation networks, the demands and competition for funds from the states was enormous. DOT continued—under the direction of Congress through succeeding highway and transit legislation—to place regulations, guidelines, and restrictions on how states obtain funds. (A good summary of the history of transportation planning can be found in Weiner 1997.)

The 1991 Intermodal Surface Transportation Efficiency Act (ISTEA) attempted to change the supply-driven context of transportation planning. It allowed transfers between highway and transit programs, deemphasized new highway additions, stressed the importance of dealing with environmental issues—particularly congestion and air quality—and even linked funding to meeting EPA air-quality standards. Subsequent renewals of the federal aid highway act, TEA-21 (1997) and SAFETEA-LU (2003), continued the thrust of ISTEA with sustained attention to environmental issues, but in these successive bills new attention was paid to the growing crisis in the ability to fund increasing needs for infrastructure improvement and growth. However, throughout all of this legislation, no matter how tied to growing concerns over environment in the broadest sense, funding mechanisms and regulations were carried out through Title 23 and the UMT Act. Funds for roads continue to be directed to state DOTs, and funds for transit continued to go to the transit operators.

An integral part of the planning process as defined by federal regulations is estimating travel demand for new projects and for transportation improvement program (TIP) certification. Travel demand models now used

and implicitly approved by the FHWA and the Federal Transit Administration (FTA) had their origins in simple computer models designed for regional transportation studies in the 1950s. Developed primarily by civil engineers and economists, these models' objective was to forecast changes in travel flows due to changes in system demand and supply. Over the last several decades, these models have become increasingly sophisticated, using applications from travel behavior theory and from economic utility maximization to help planners determine how and why—and at what cost—individuals travel. The aggregate numbers from these trip-purpose and mode-choice models are then put into networks to help establish supply-demand relationships and finally to answer a basic question, “Do we need new capacity?” and its corollary “If so, where and in what form?”

Overall, the period from 1950 to 1980 could be characterized as one dedicated to the growth of transportation supply in an era of cheap energy and a robust national economy. This supply, as has been so well documented, had profound impact on the structure of our urban and suburban areas, the economics and politics of those areas, and the varied lifestyles they supported. The period from 1980 to 2000 marked a slowdown in the growth of supply, as community and environmental concerns became predominant. Managing infrastructure became the initiative of transportation agencies and institutions, while energy still remained cheap.

The current period represents a radical shift in planning and managing our transportation infrastructure. While post-World War II planning started with regional goals and objectives, 21st-century planning adds the pressures of globalization and the elevated importance of economic growth. Energy has become expensive and will become scarcer. The debate over policy approaches and responses to global warming has opened the question of whether any auto-oriented urban form, even “green autos,” should still be the preferred form for most Americans. (For an excellent discussion of the changing role of the U.S. auto industry and the U.S. auto, see Rothschild 2009.) This debate encompasses the role of the government in policy approaches that will be sufficient and appropriate to address this complex and major 21st-century problem. To cope with this shift, changes in the transportation planning process and the regional planning process must be made.

There are three factors that must be incorporated into a newly invigorated transportation planning process: (1) new approaches to finance infrastructure; (2) institutional change; and (3) the implications of the adoption of new technologies—in particular computers and communications technology—into aspects of transportation operations and management, as well as into everyone's daily lives.

Current Limits to Infrastructure Finance

Old funding mechanisms, arising out of decades of program specifications, are inadequate for our current challenges.¹ The source of federal funds—user fees paid into the Highway Trust Funds (HTF)—has reached its limit. Many predict the HTF will be out of cash as soon as 2012.

Because of how program funds, by law, are allocated, there is a direct linking of project funding to a specific transportation agency (DOT, transit agency, planning board, etc.). Each agency has a well-defined mission that narrowly defines projects to maximize their chance of receiving funds. Title 23 and the UMT Act both contain carefully defined programs and criteria that highway agencies or transit properties must follow in order to be funded. Planning in each agency means optimizing funding opportunities.

In the last several decades, megaprojects have taken center stage; smaller projects and asset rehabilitation—while critical—often get less attention. Megaprojects are high-dollar, long-timescale projects; examples would in-

The current period represents a radical shift in planning and managing our transportation infrastructure.



The Leonard P. Zakim Bridge, Boston, part of the “Big Dig” Central Artery/tunnel project.

clude the Second Avenue subway in New York (current estimate: 15 years to build, \$17 billion cost) and the Boston Central Artery/tunnel project (16 years to complete, \$15 billion cost). By law, the Metropolitan Planning Organization (MPO) is, on paper, the equalizer among agencies. In large urban areas, strong agencies (state DOTs, large transit operators) carry disproportional weight, influencing decision making in the attempt to maximize funding, especially for megaprojects. Revamping of Title 23 and the UMT Act to give stronger power to a balanced MPO would be one step in bringing regional planning and funding into some harmony. But this cannot be accomplished without rethinking the mission of 21st-century transportation agencies. Agencies must be more integrative with respect to regional goals, while funding should be tested against goal optimization. Surface transportation planning must be done in the context of three major considerations:

1. The context and environment for which transportation changes and additions are needed: land use, environment, equity, energy use, and community/regional priorities. Such planning must have a sign-off by those engaged with the responsibility for those issues.
2. Investment and risk: social-benefit cost analysis must be applied
3. Good engineering practice: lifecycle analysis applied to modern practice must remain at the heart of infrastructure planning.

A complementary issue is the need for obtaining operating funds for transit as well as operating and maintenance funds for roads and bridges. Regional and local governments have been unable to sustain the constant need of these systems for maintenance, upkeep, and modernization, as evidenced by the American Society of Civil Engineers (ASCE) grade of for our roads (C- or worse), and the current condition of most U.S. transit systems: less than a state of good repair.² Fares, tolls, and local taxes are the norm for funding. The current planning process is driven by the need for capital funds for system modernization and expansion; maintenance and operations remain underfunded local issues.

Available finance drives the capital process and hence the current planning process. A shift of funding as new transportation systems are put in place would consider full project lifecycle costs, guarantees of operating support, and improved project estimation—including better evaluation of risk and new institutional models for funding and oversight. At present, there is concern among transportation agencies that the current user-based system—gas taxes—are inadequate to finance future needs. Proposals for funding future systems include a broad variety of taxes: on users and businesses, tolls, private buy-ins, and other means. They include new approaches to combine funding and environmental response that include carbon trading, carbon credits, and implementation of “pay as you drive” systems. Each represents a departure of funding tools from the last 60 years and would force a rethinking of transportation organizational structure, mission, and responsibility.

Shifting Institutional Arrangements

Every public transportation agency or organization has a clearly defined mission and mandated rules and responsibilities. Rather than clarifying the steps from plans to implementation, these organizations make regional transportation planning quite difficult. Holguin-Veras and his coworkers showed the complexity and costs of governing freight movements in the New York City region by describing the number of regional agencies and bureaucracies and their constrained missions that influence or impact such movements (see Holguin-Veras and Paaswell 2000). At a time when modern real-time logistics drive 21st-century goods movement, organizational structures that govern movements, intermodal activity, street ordinances, and the like are locked in 20th-century structures. Planning cannot overcome this handicap. Twenty-first-century planning efforts must be directed to system management and integration, tasks not developed through 20th-century programs.

In a rational transportation planning process, a public planning group, such as an MPO, would have responsibility for undertaking the steps beginning with the formulation of regional goals and objectives and culminating with a recommended set of projects. These projects would be based on discussions with stakeholder groups, with assessments of regional economic and demographic changes, with regional strategies for growth or other specific desired objectives, and with knowledge of regional agency ability to deliver specific types of projects. In a large metropolitan area such as New York, this type of procedure has become unworkable. First, the MPO does not possess the real power to carry out such a process. It does work on the mandated long-range plans and the transportation improvement program, but these are outcomes of a negotiated process among powerful stakeholders. Second, individual public agencies have extraordinary powers and specific legislated areas of interest and responsibility, rendering them difficult negotiating partners. Third, there are a number of persons or assemblies who are the decision makers; often, they are unable to come to consensus on specific regional projects that have emerged from a rational planning process.

A recent example is the conflict between the mayor of New York City and the state legislature that led to the defeat of congestion charging for lower Manhattan—a plan endorsed by all of the professional regional planners. This failure is not unique to New York. Why do rational but controversial plans and projects have difficulty being implemented? Why do new projects get to the table and receive funding, though they have not emerged from long-range planning or regional planning efforts? One response is that capital projects have become increasingly politicized. Another is that the agencies that arose from the requirements of Title 23 and the UMT Act are no longer able to meet the needs of today’s stakeholders. Agencies, once designed and

dedicated to their initial mission of building supply (or rebuilding transit), have become overly bureaucratic, locked in process and past practices. While these agencies attempt to modernize, they still must maintain and support existing assets and find it difficult to transform into what is needed for the 21st century.

Agencies must be restructured to accomplish the following needs and to be responsive to planning:

1. The 21st-century agenda for infrastructure must start with the environment, quality of life, and sustainability. We have made enormous investments in infrastructure in the past 100 years. Those investments must be managed in the context of changing household structures and behavior, appropriate use and conservation of land, and the changing role of the car.
2. Agencies must be transparent in their planning to be effective and responsive to stakeholders.
3. Stakeholders will be both regional and global; they will have professional capabilities and will be able to communicate instantly. Agencies must address both regional economic growth and local environmental needs. They must have the capability and leadership to do so.
4. New models of funding, not so dependent on fuel taxes, must be put in place. More attention now is being placed on user fees and charges and beneficiary fees and charges, as well as on innovations such as carbon credits. Proper involvement of the private sector—whether as partners or contractors—will focus the structure of agencies as oversight, planning, operating, or funding entities. Great Britain and the European Union have, over the last decade, made significant changes in how transportation systems are planned, funded, and operated. These changes include rethinking organizational responsibility and structure, separating planning and oversight from operations, involving more private-sector participation, and having planning agendas emerge from nontransportation agencies. Some Asian models are significantly more entrepreneurial.
5. Agencies not only must be truly multimodal in structure but must incorporate considerations of land use and the environment. Regional planning, zoning, development plans, environment, and quality of life must be seen not as external “to be addressed” issues but as integral parts of the outcomes of transportation planning and investment.
6. Agencies must embrace and make use of the availability of real-time information and modern communications technology. This is greatly changing both the supply of and demand for traditional forms of transportation.

The stovepipe model of transportation is still practical for the implementation and operating of transportation systems and system components. But planning, funding, regulating, and having oversight of the complex, information-driven contemporary transportation systems must be part of new governance models. In sum:

1. Federal transportation legislation must be totally rewritten, integrating programs, funding, and regulations governing all surface transportation. Agencies such as FTA and FHWA should still oversee the engineering practice requirements of specific infrastructure, but broad-based planning, especially in the multidisciplinary context of environmental issues, must be focused in a multicabinet government policy unit.

2. Local and state agencies should mirror new federal approaches to the planning, oversight, and engineering of infrastructure. The regulations regarding MPOs should be written to ensure that new regional and local goal setting is done in the context of national priorities.

The Role of Technology

The advances in transportation planning after 1950 were due, in large part, to the rapid growth of computing power. Computers used to estimate simple planning models in the 1950s evolved to powerful and small personal computers that could calculate complex supply-and-demand models, operate traffic signals, and control remote variable-message signs. Since 2000 there has been a complementary revolution in communications technology. The combination of modern wireless communications and ubiquitous computing power are having profound impacts on the supply and demand for transportation and on the culture of those using transportation. The planning process used in the latter part of the 20th century relied on fixed models of behavior and empirical but fixed data on demand and supply. As planning models became more sophisticated, integrating behavioral models of individuals and households, they were applied to test the adequacy of supply or the impacts of new policies (pricing, etc.) on supply issues. The combination of information technology and instantaneous communication of large and complex data sets has changed the approach to both planning models and actual system operations.

Most familiar to transportation planners are the advances from intelligent transportation systems (ITS). Smart vehicles (motor vehicles and transit vehicles), smart infrastructure, and smart materials all combine to provide real-time information on all aspects of supply. Remote and onsite computers make it possible to know:

- real-time traffic patterns
- performance characteristics of motor and transit vehicles
- actual vehicle locations and speeds
- vehicle-to-vehicle communication
- driver performance
- maintenance histories
- material behavior (fatigue of bridge structural elements, etc.)
- vehicle loads (goods and passengers)
- network traffic characteristics
- capacities and availability of parking lots and spaces

Transportation operators can adjust traffic flow or reschedule transit vehicles in real time, be more responsive to incidents, and be responsive to factors such as congestion and air quality. Vehicle-to-vehicle communications in congested area platoons can increase capacity to limits above those now set empirically. Transportation operators can relay this information to individual users through the Internet or text messages so they can plan their journeys. Archived data of this type will have a significant impact on planning supply models as changes in supply and changes in responses are documented and as new travel behavior is observed.

The availability of real-time data gives transportation systems users many more choices than they had in the past. Whether on mode choice—based on estimated arrival times, availability of parking, next bus coming,

The combination of information technology and instantaneous communication of large and complex data sets has changed the approach to both planning models and actual system operations.

The era of massive additions to supply is over; the new era will be one of maximizing the capacity of the supply, accommodating transportation systems to new societal objectives, and planning transportation as an integral part of land use, quality of life, and environmental constraints.

and other factors—or midjourney route choice, system users will derive new patterns of travel distinct from those based on past knowledge. Further, persons traveling in groups (household partners, carpoolers, colleagues attending a meeting but traveling separately) can make instantaneous changes in plans should there be some incident or congestion or transit breakdown. Such changes can be made by phone and Internet through texting or use of MySpace or other such communications. New activity choices can be made that are not predictable by traditional models. New applications of global positioning system (GPS) technology and mobile phones will make it possible for groups to know where each member is at all times, providing further input to both activity choice and transport choice to achieve the group's needs. The ability of both the system operator and the system user to have real-time comprehensive information about system characteristics and performance will necessitate changing traditional (static) supply-demand models. New models will use real-time information—meaning communications technology must be part of the operations and delivery planning process.

While much travel behavior theory has acknowledged that the 2009 household is not like the 1960 household, it has not addressed the fact that new types of travel models are needed. These new models must be activity- and lifestyle-driven, responsive to the flow of information individuals receive during the day and the decisions they make (and the influencers of those decisions) and the transportation choices they make (from those available) based on that information. In addition, the wired generation is also responsive to issues of environment and sustainability, and many choices will be based on values arising from those issues. Travel costs and utilities will have to account for emerging sensitivity to the impacts of travel choices on the environment.

Conclusion: A New Paradigm

The bodies of law and regulations that defined the highway and transit systems for most of the 20th century accomplished their tasks of building world-class infrastructure throughout the United States. But the era of massive additions to supply is over; the new era will be one of maximizing the capacity of the supply, accommodating transportation systems to new societal objectives, and planning transportation as an integral part of land use, quality of life, and environmental constraints. While the engineering methods of building, modernizing, and maintaining our infrastructure remain, and the tools to do so are part of our traditional DOTs and academic institutions, planning must take a new and more global perspective. This chapter is the start of a conversation on what that perspective should be—a conversation that must be joined by the generation of planners, policy analysts, economists, and engineers now being trained or entering the workforce.³

ENDNOTES

1. The New York State DOT held a major symposium, “After the Gas Tax,” in October 2008 to discuss this critical issue. The subject was innovative financing, a catchphrase for “How do we do it?” A conclusion was that there are many ad-hoc approaches and that new means have to be found to assure broad-based long-term capital and operating support for newly emerging programs.

2. See www.infrastructurereportcard.org.

3. Thanks to Richard Hanley, Cynthia Chen, and Todd Goldman for their review and comments.

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